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Hsieh et al.

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(54) **SEMICONDUCTOR ARRANGEMENT WITH ISOLATION STRUCTURE**

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Related U.S. Application Data

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(51) **Int. Cl.**
H10F 39/00 (2025.01)
H10F 39/18 (2025.01)

(52) **U.S. Cl.**
CPC **H10F 39/807** (2025.01); **H10F 39/011** (2025.01); **H10F 39/184** (2025.01)

(58) **Field of Classification Search**
CPC H01L 27/1463; H01L 27/14649; H01L 27/14683

See application file for complete search history.

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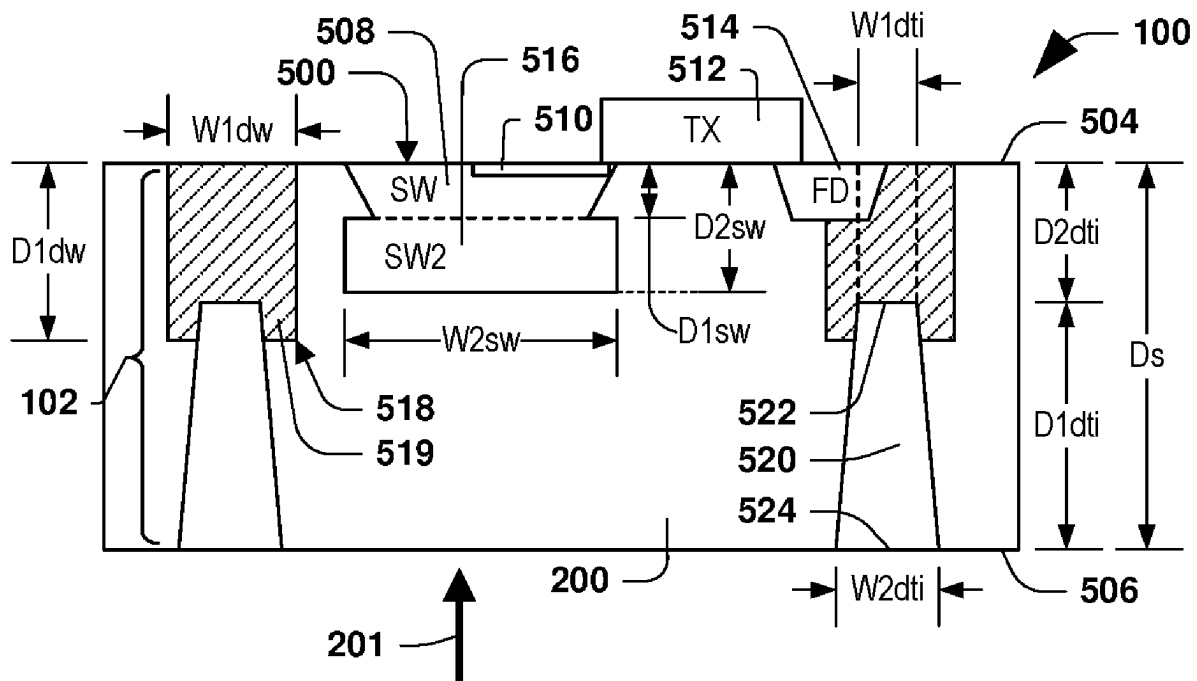
Primary Examiner — David Vu

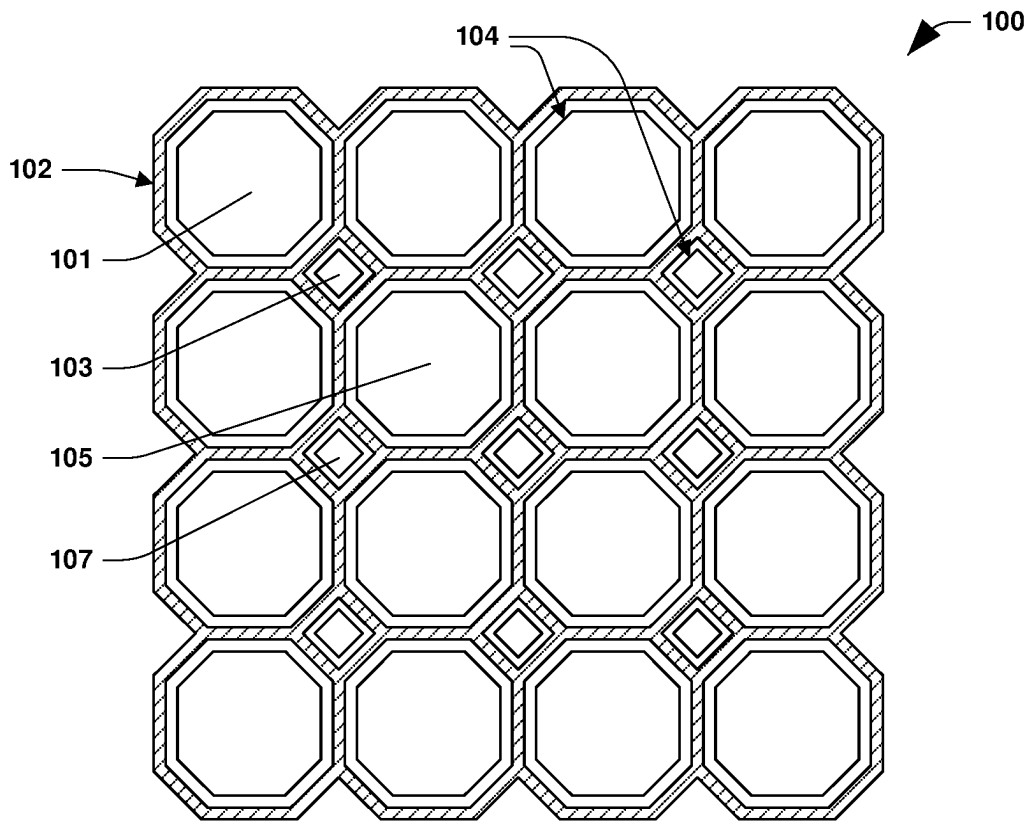
(74) *Attorney, Agent, or Firm* — Cooper Legal Group, LLC

(57) **ABSTRACT**

A semiconductor arrangement includes a photodiode extending to a first depth from a first side in a substrate. An isolation structure laterally surrounds the photodiode and includes a first well that extends into a first side of the substrate. A deep trench isolation extends into a second side of the substrate and at least a portion of the deep trench isolation underlies the first well.

20 Claims, 46 Drawing Sheets



**FIG. 1A**

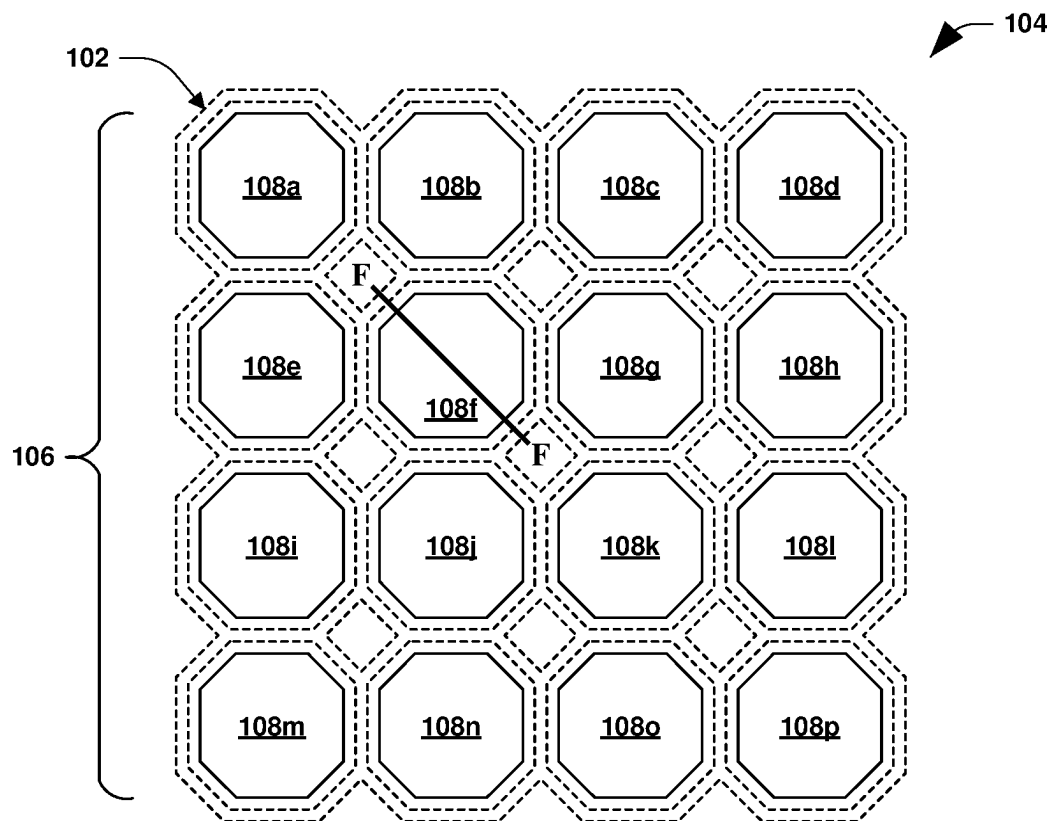


FIG. 1B

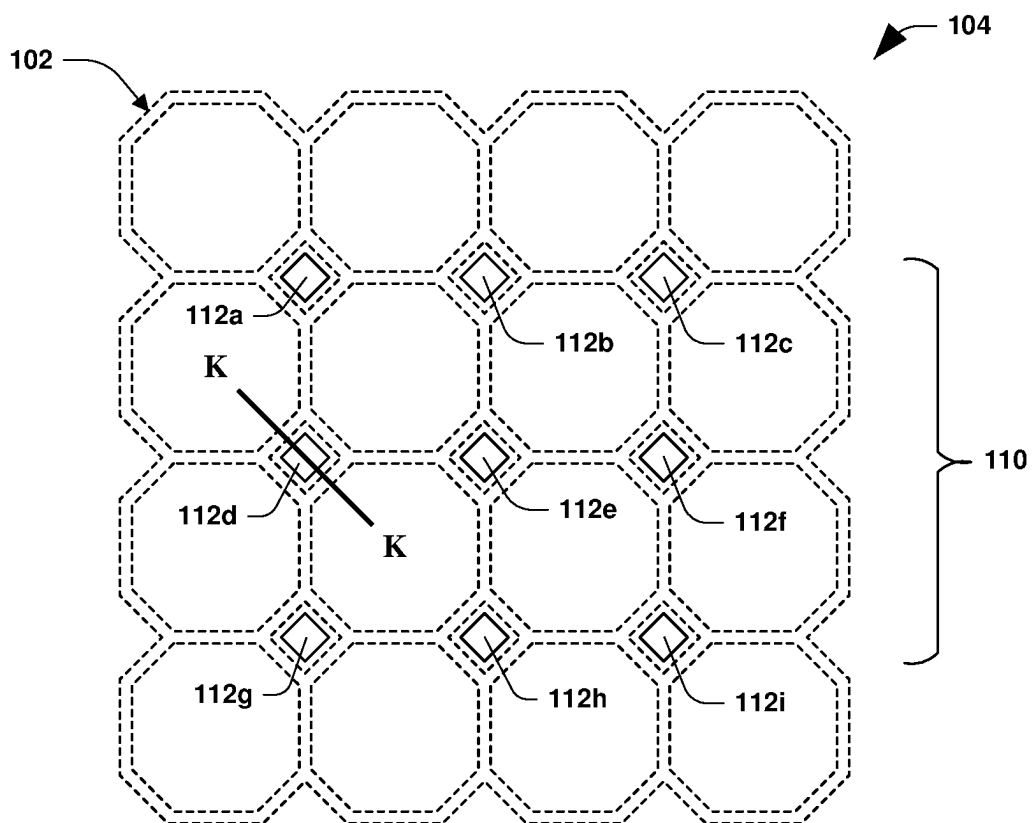


FIG. 1C

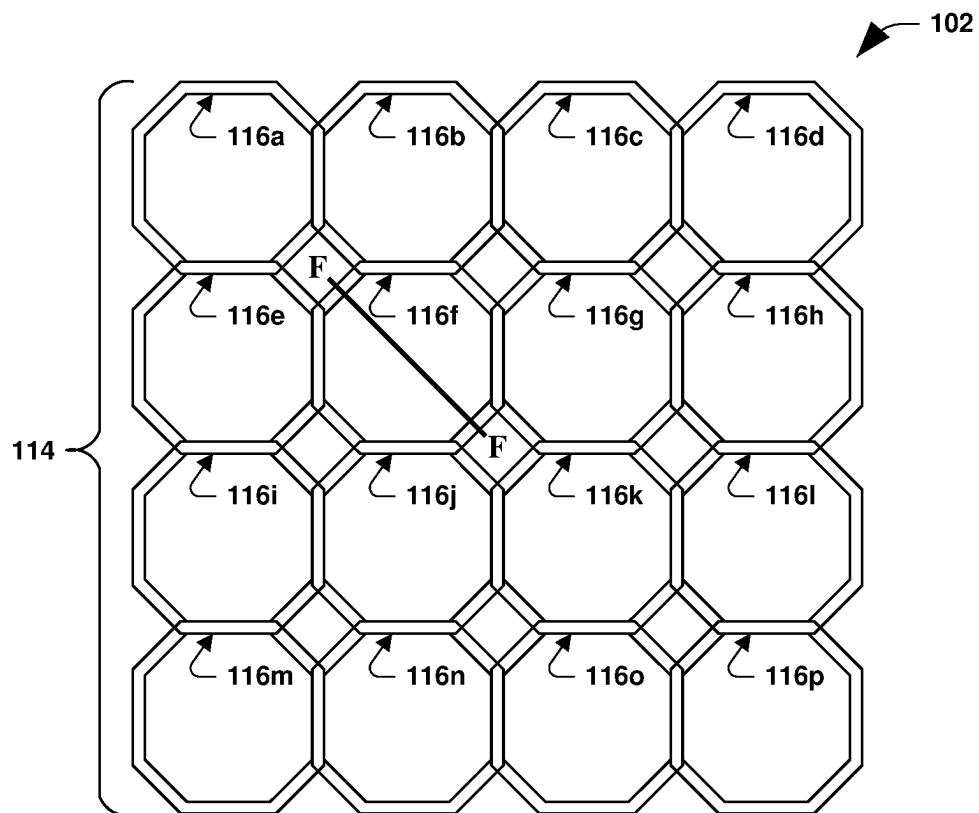


FIG. 1D

FIG. 1G

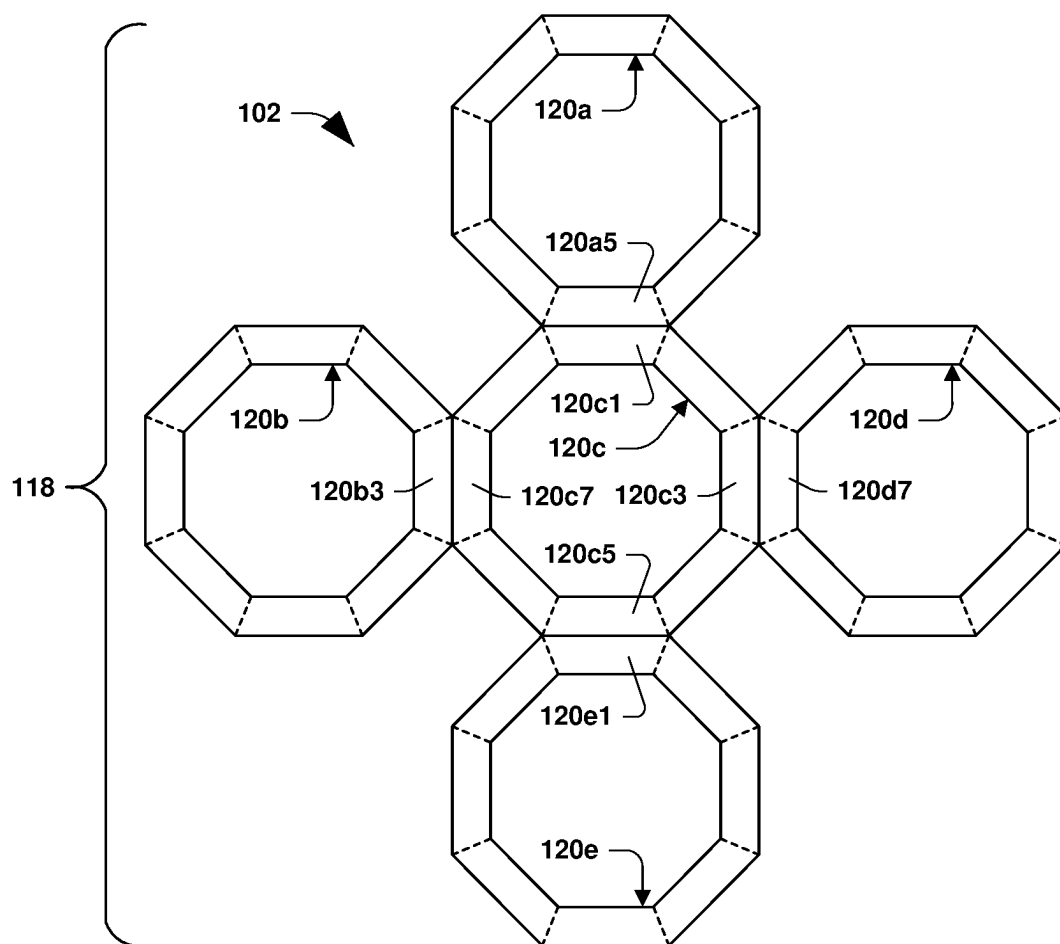


FIG. 1H

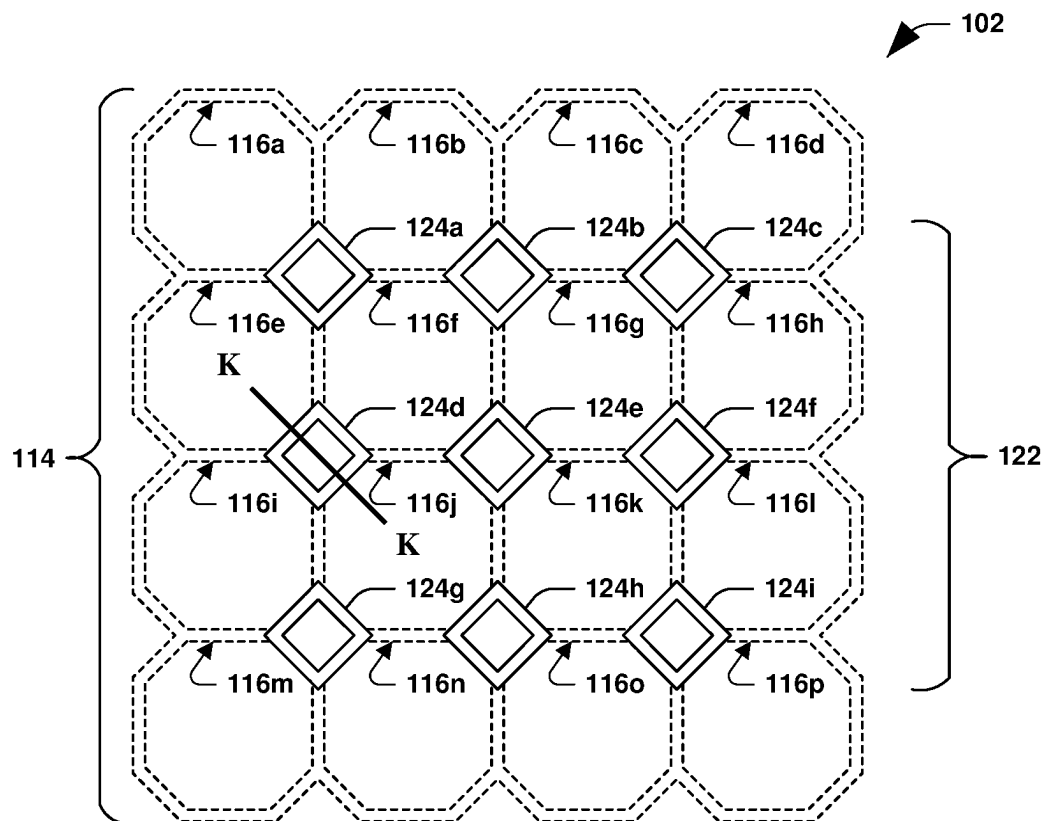


FIG. 11

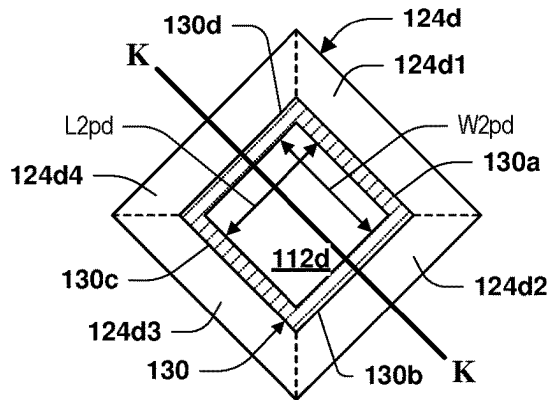


FIG. 1J

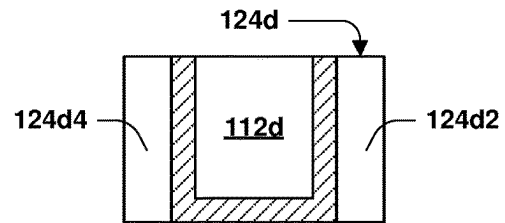


FIG. 1K

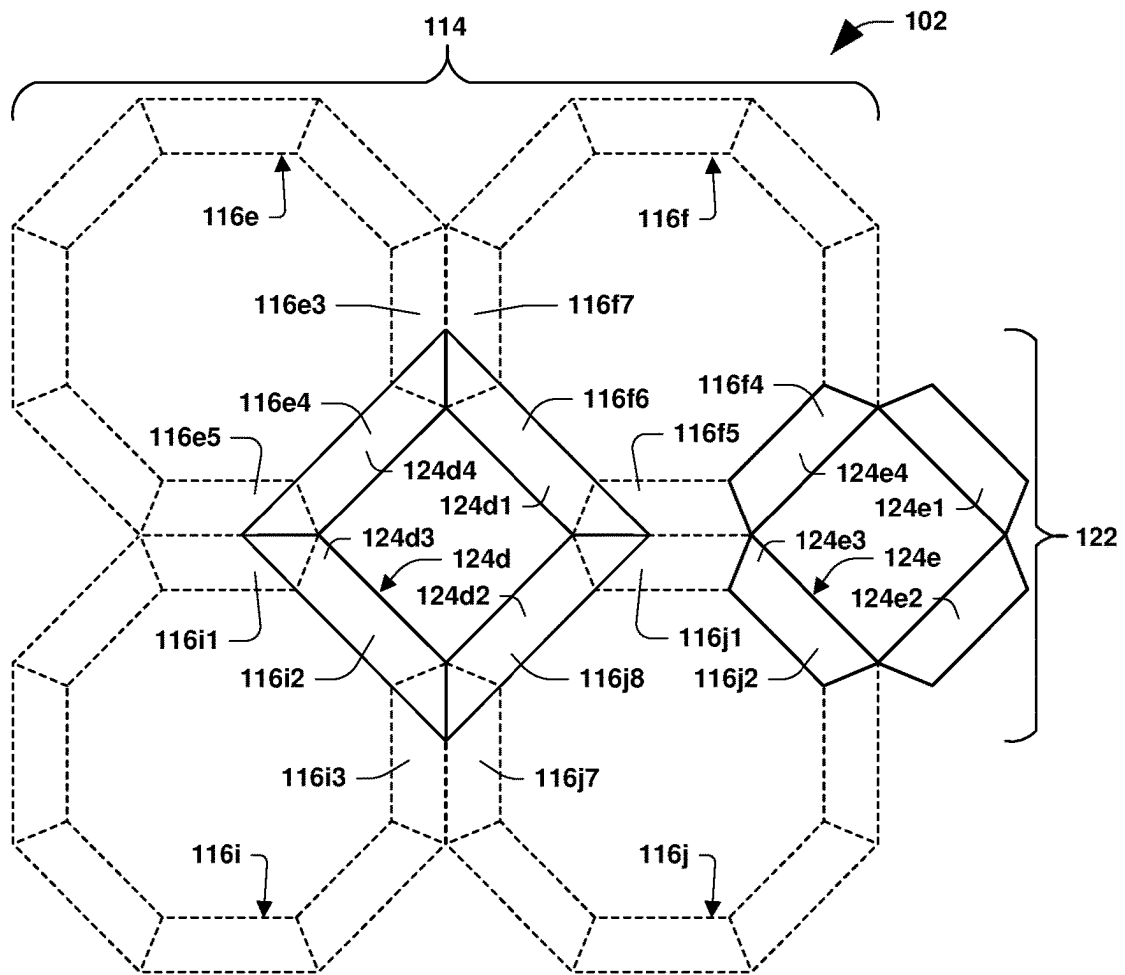
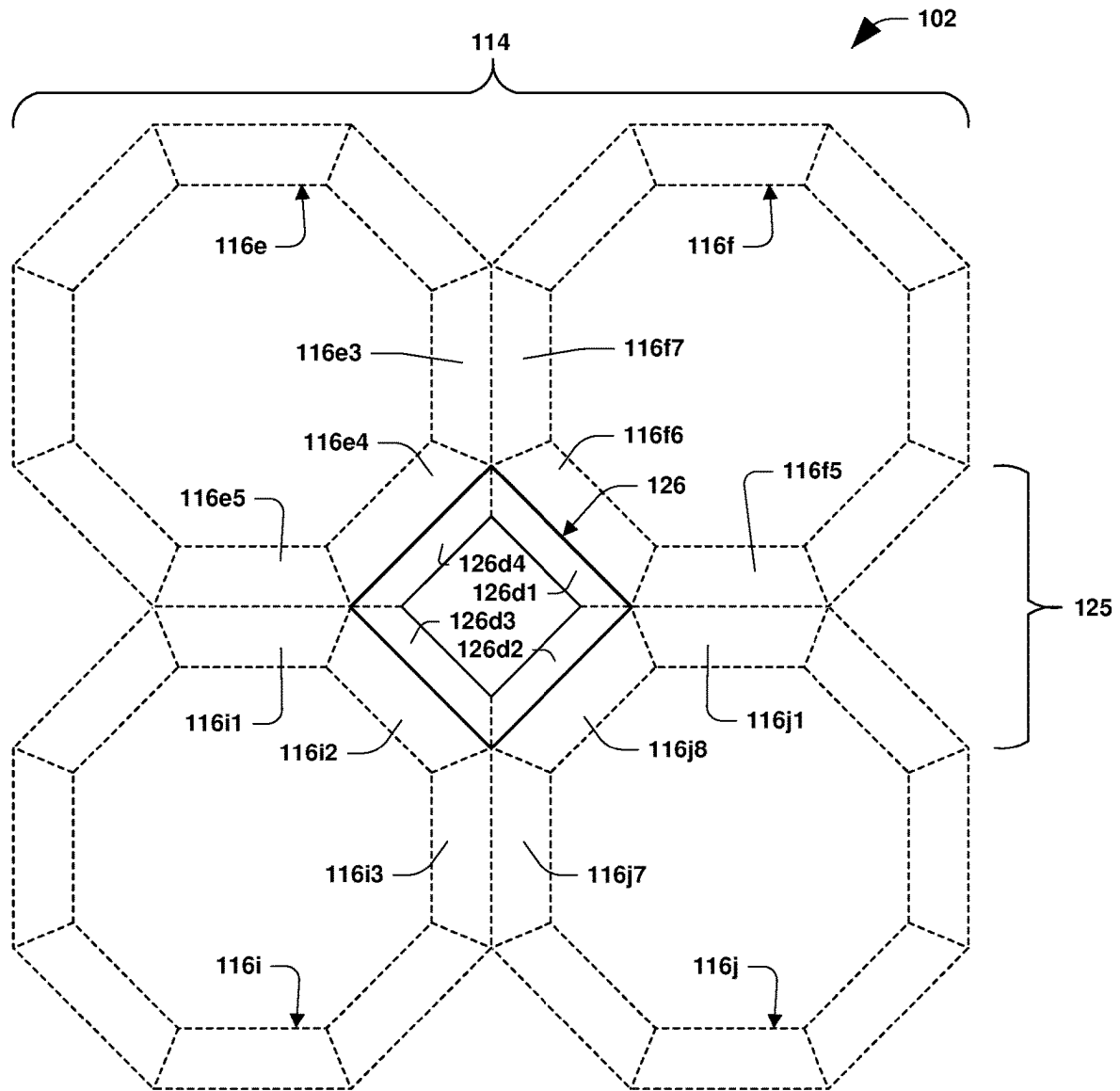


FIG. 1L

**FIG. 1M**

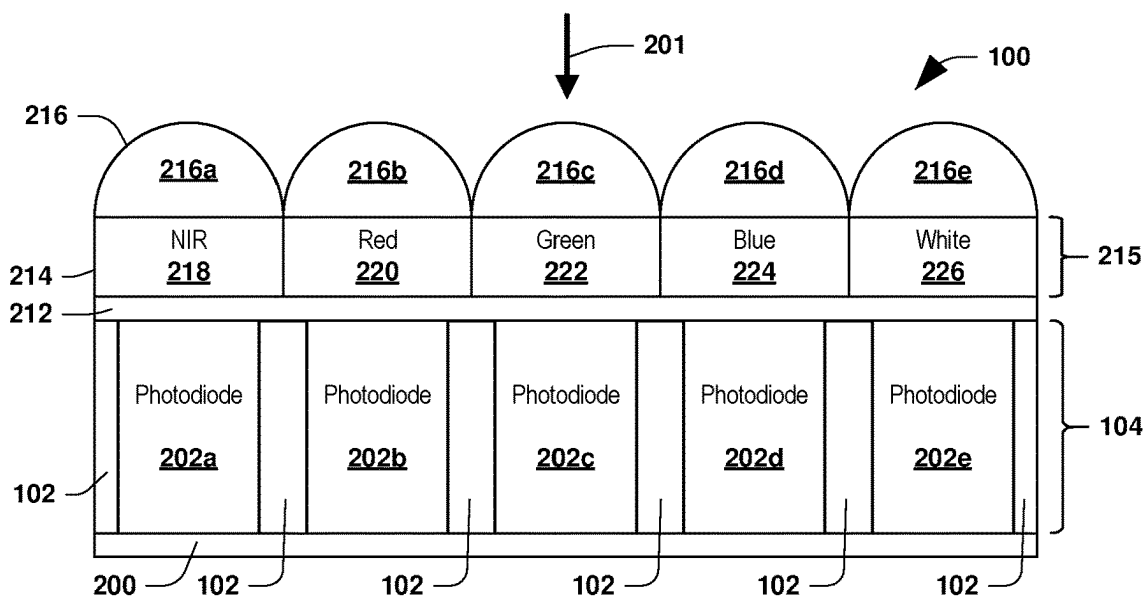


FIG. 2

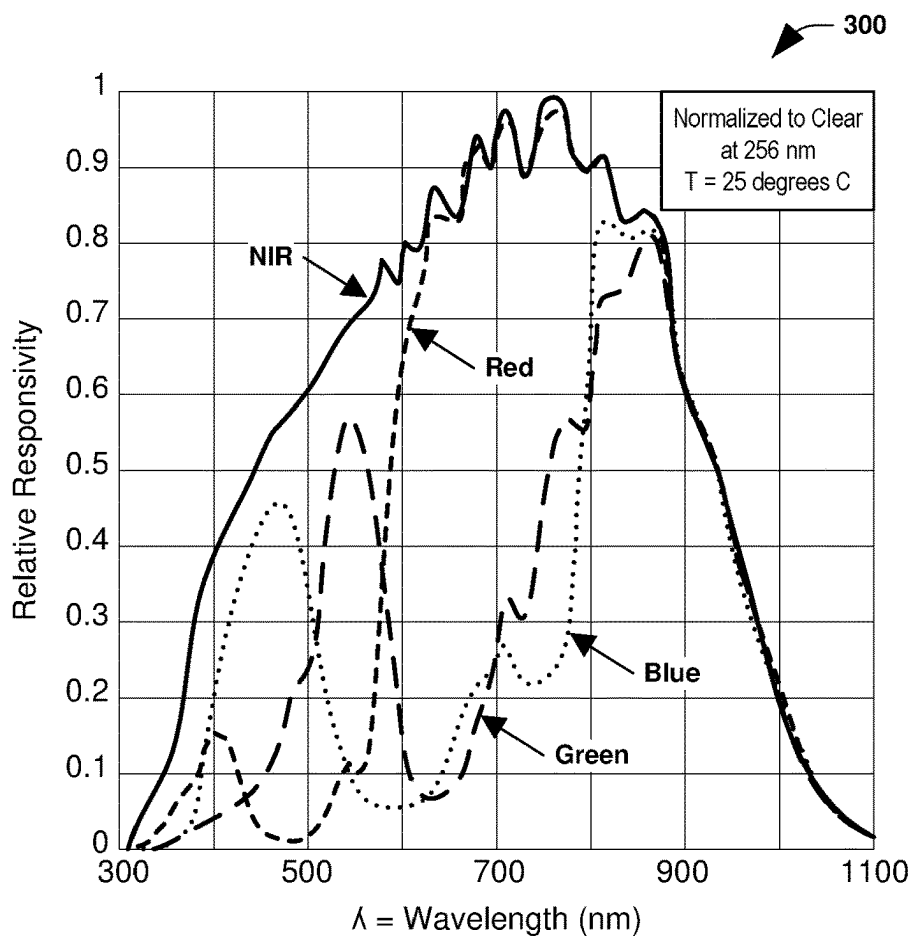


FIG. 3

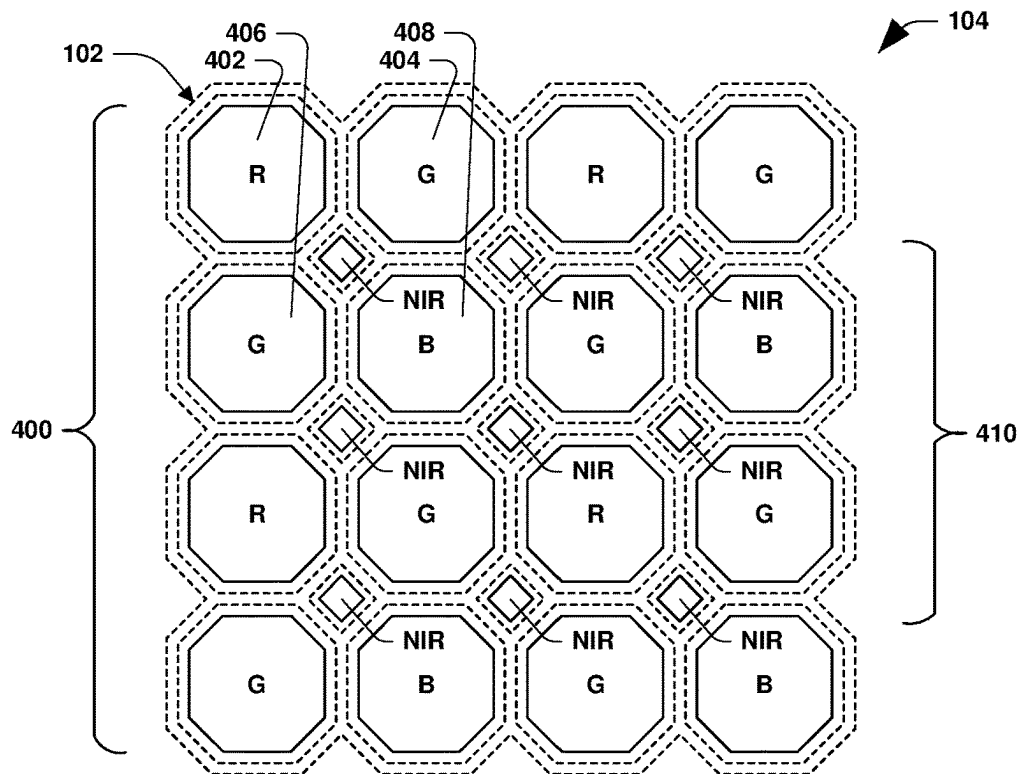


FIG. 4A

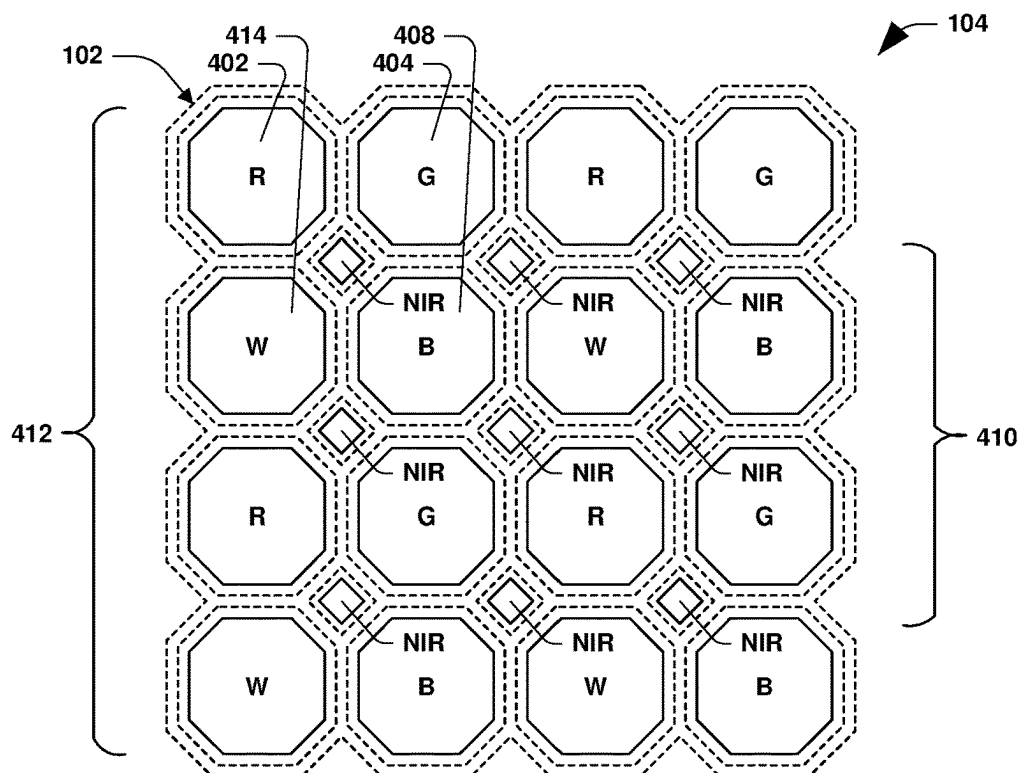


FIG. 4B

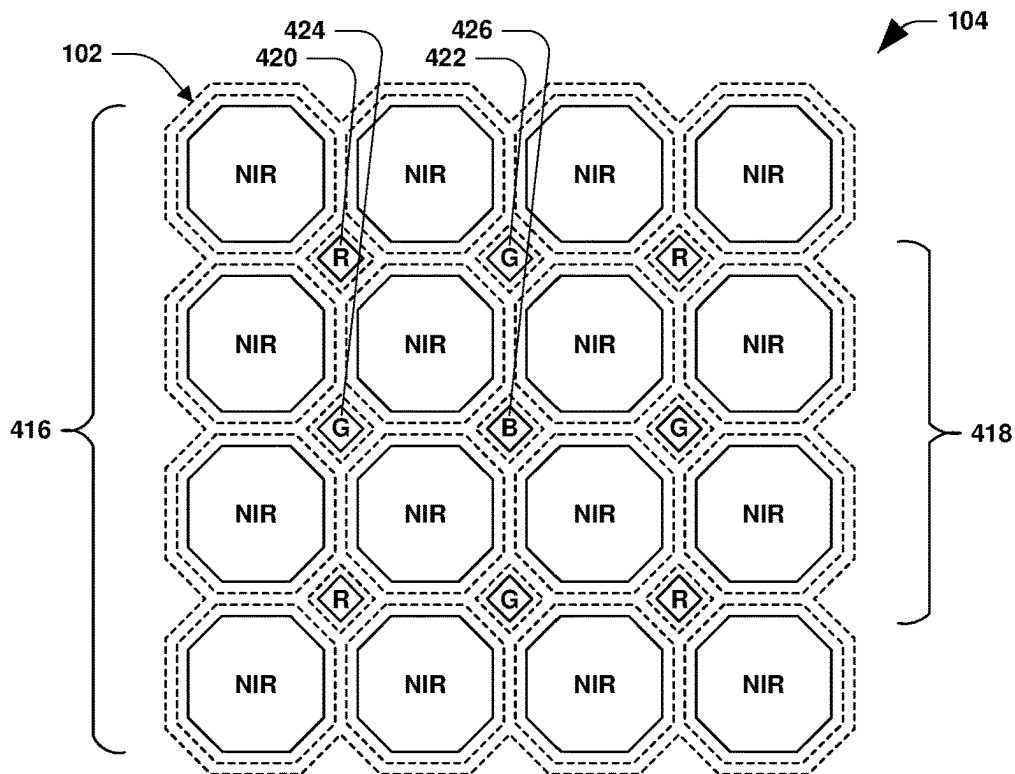


FIG. 4C

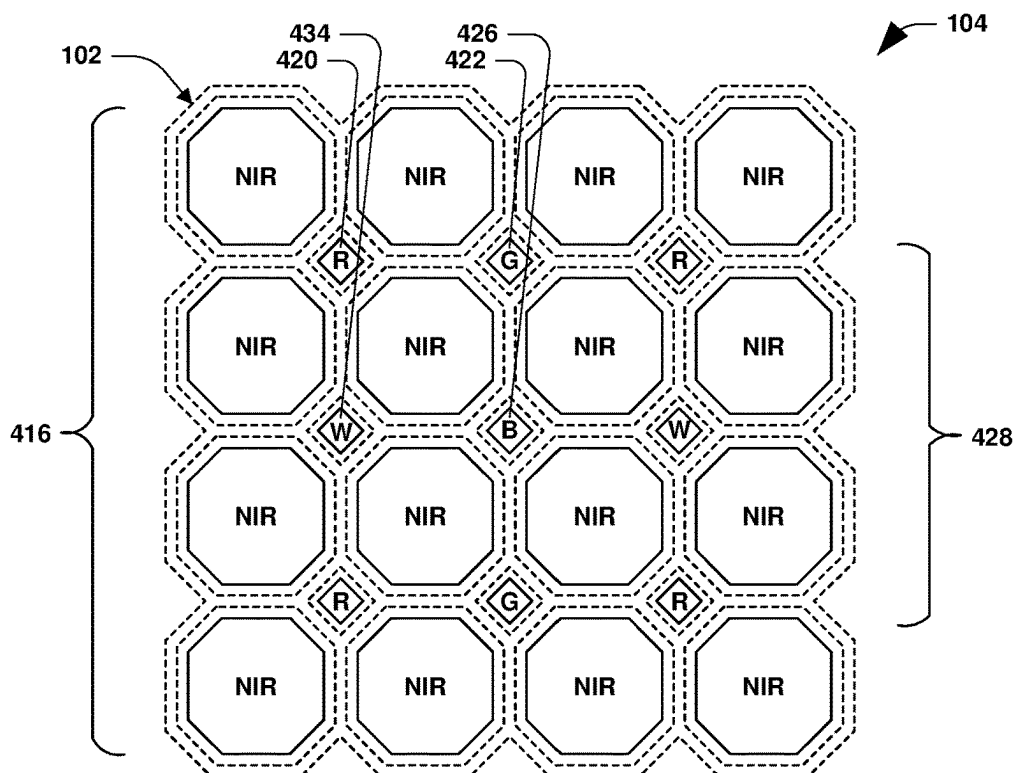
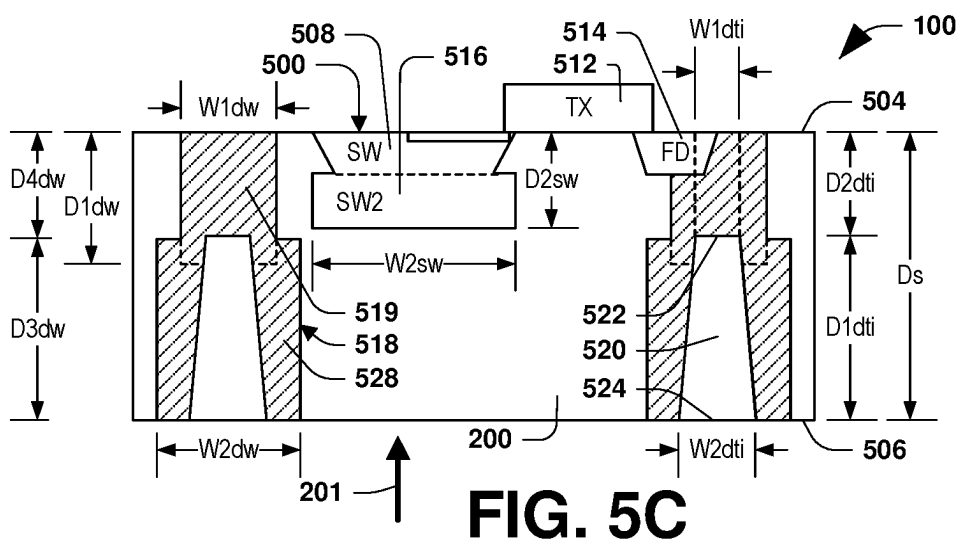
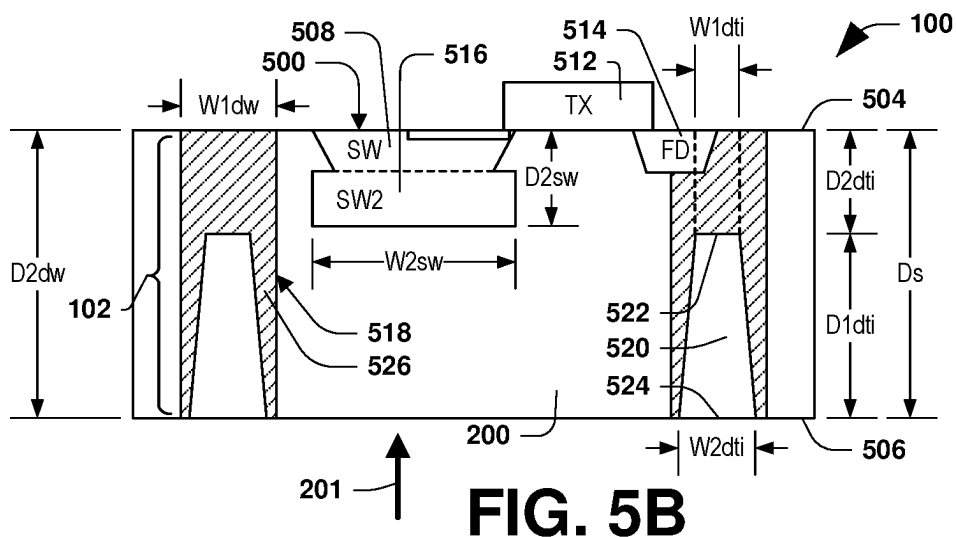
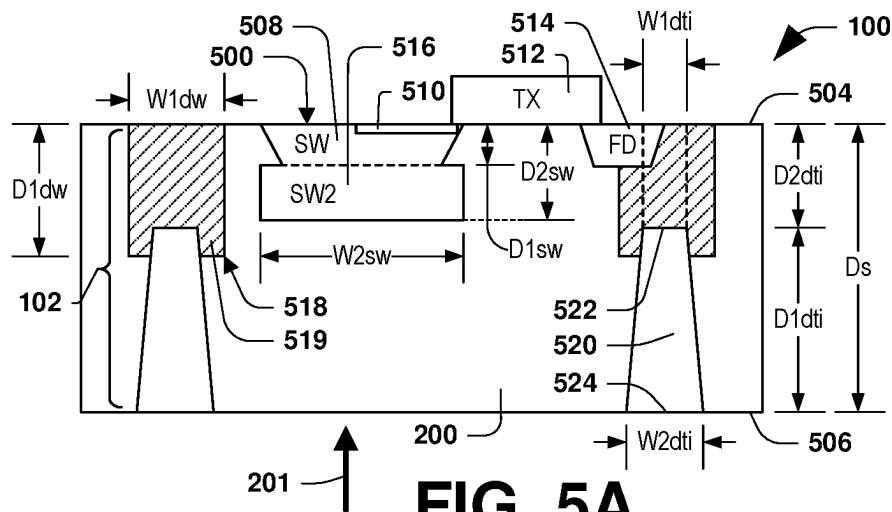


FIG. 4D



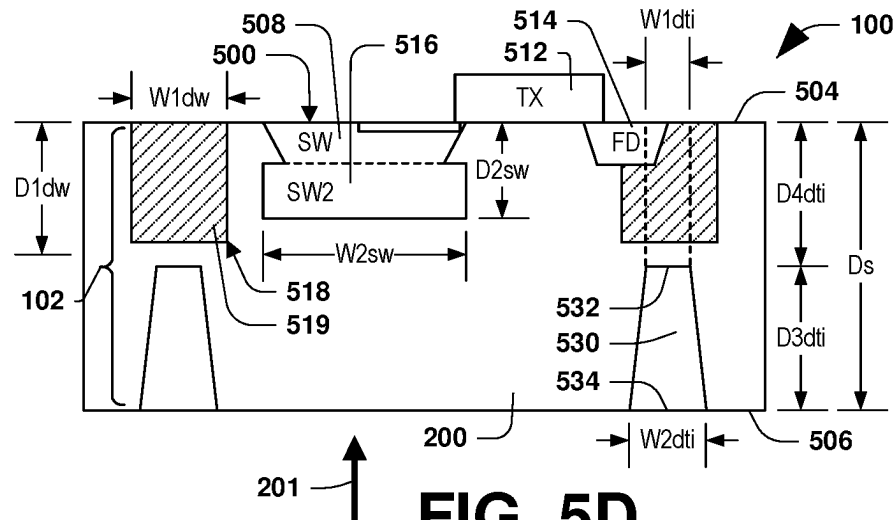


FIG. 5D

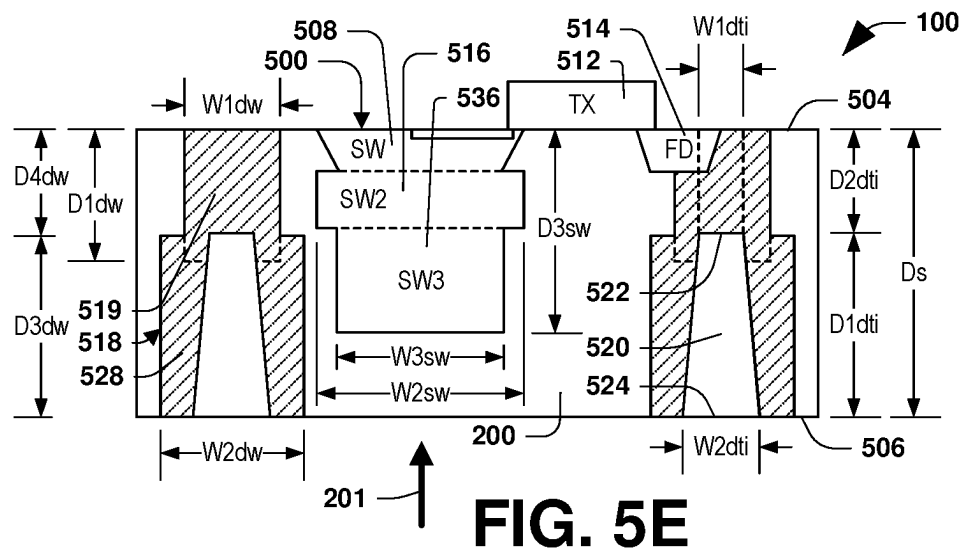


FIG. 5E

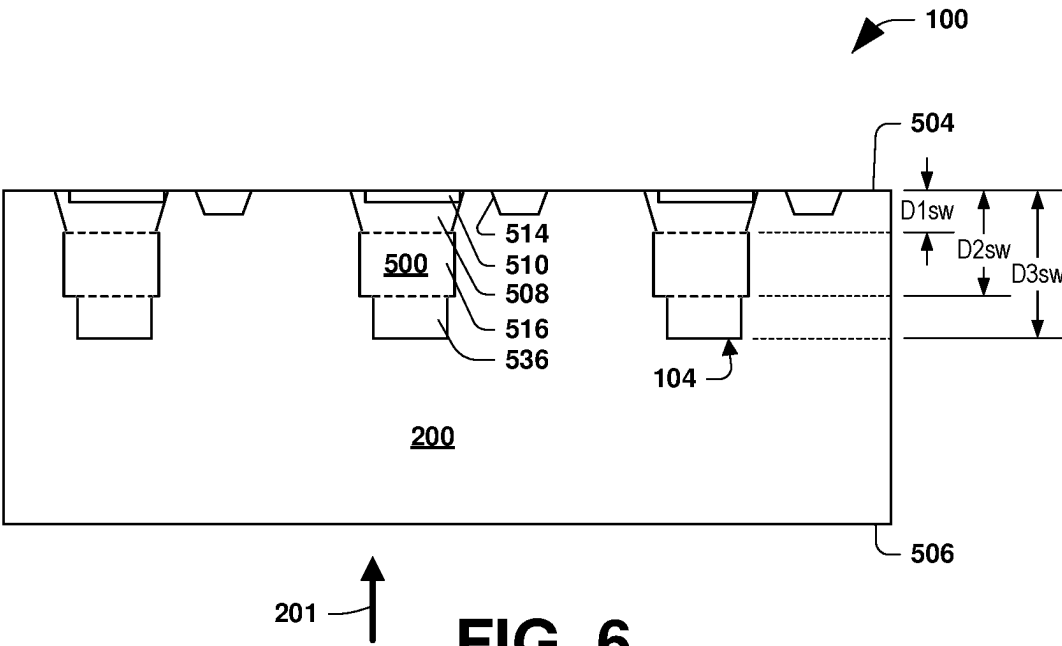
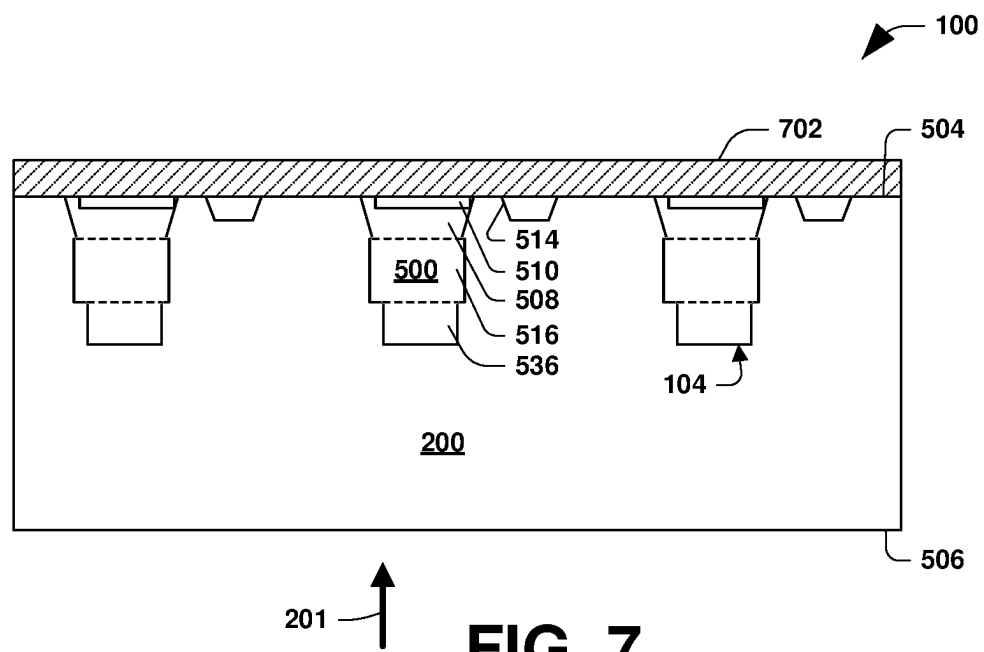
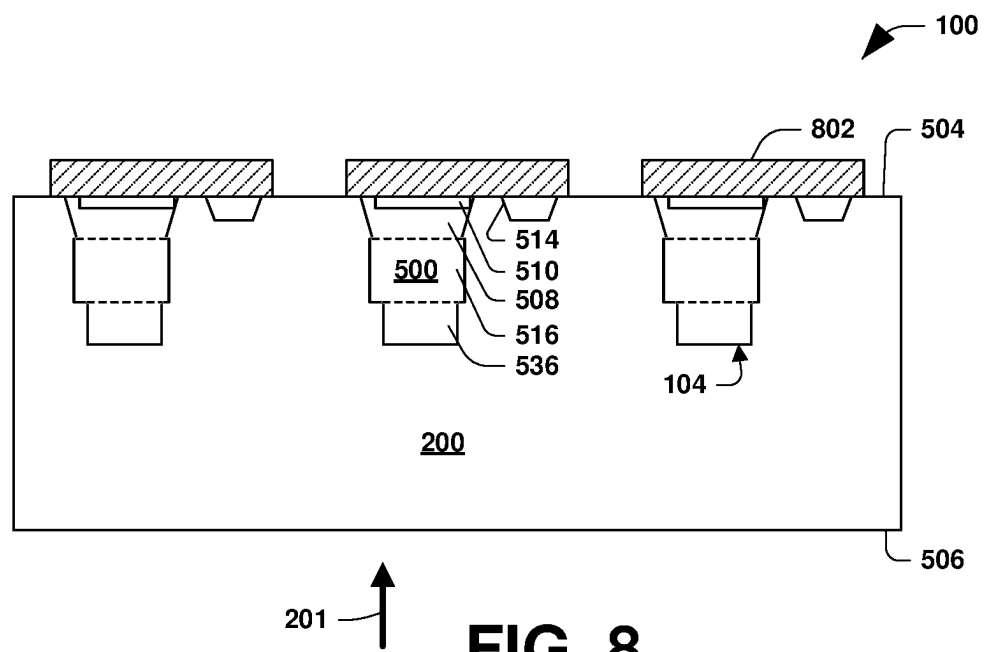


FIG. 6





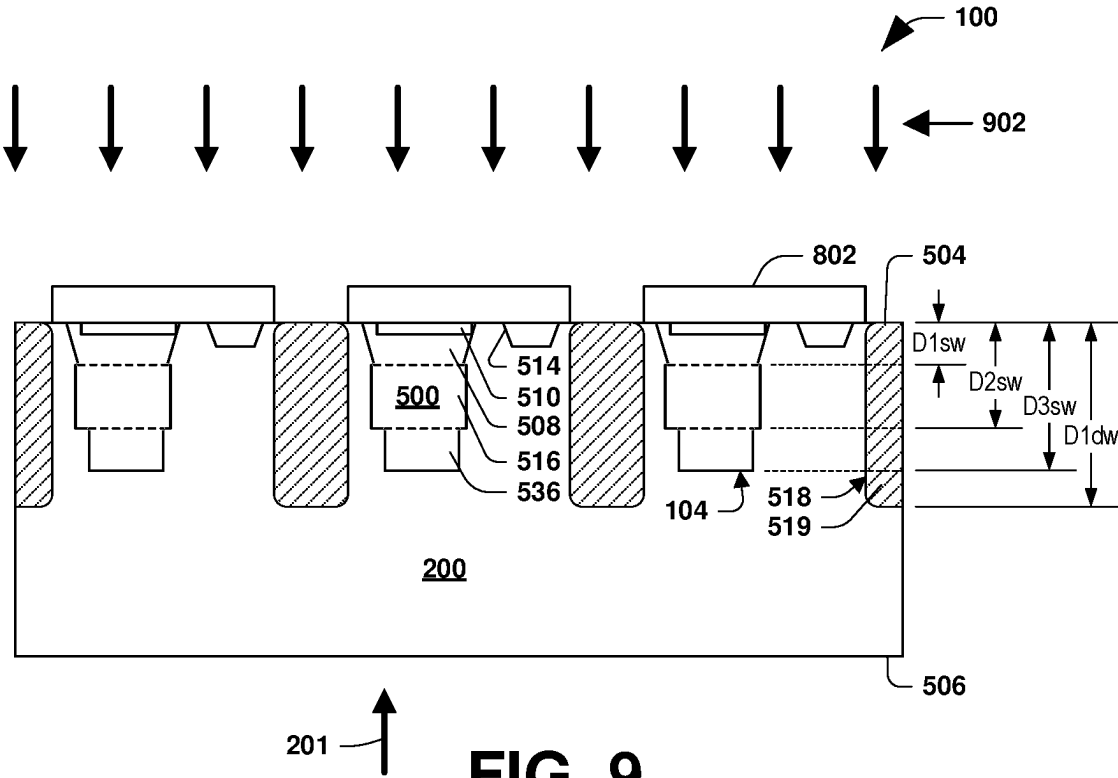
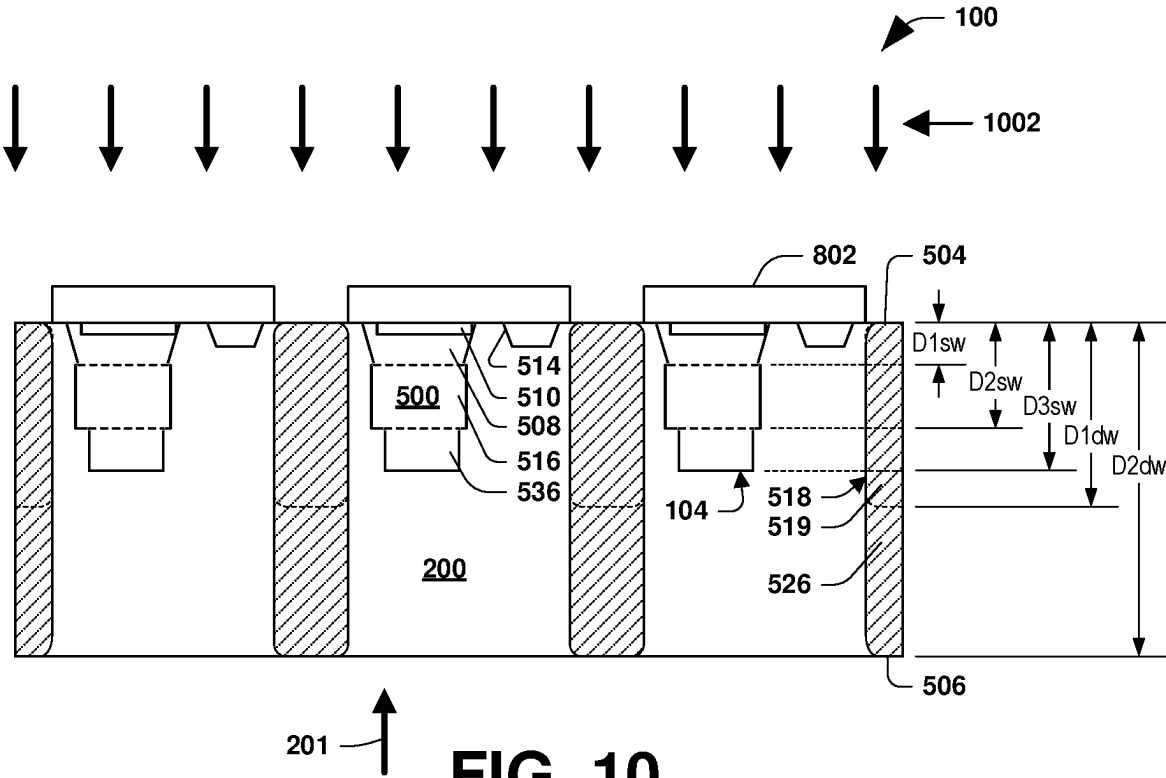
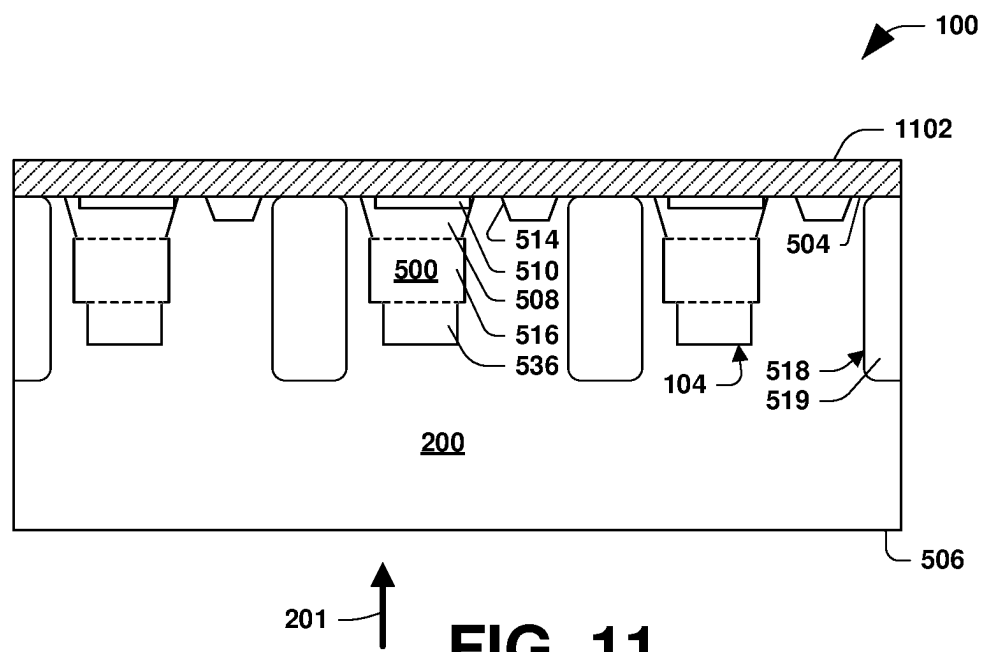


FIG. 9





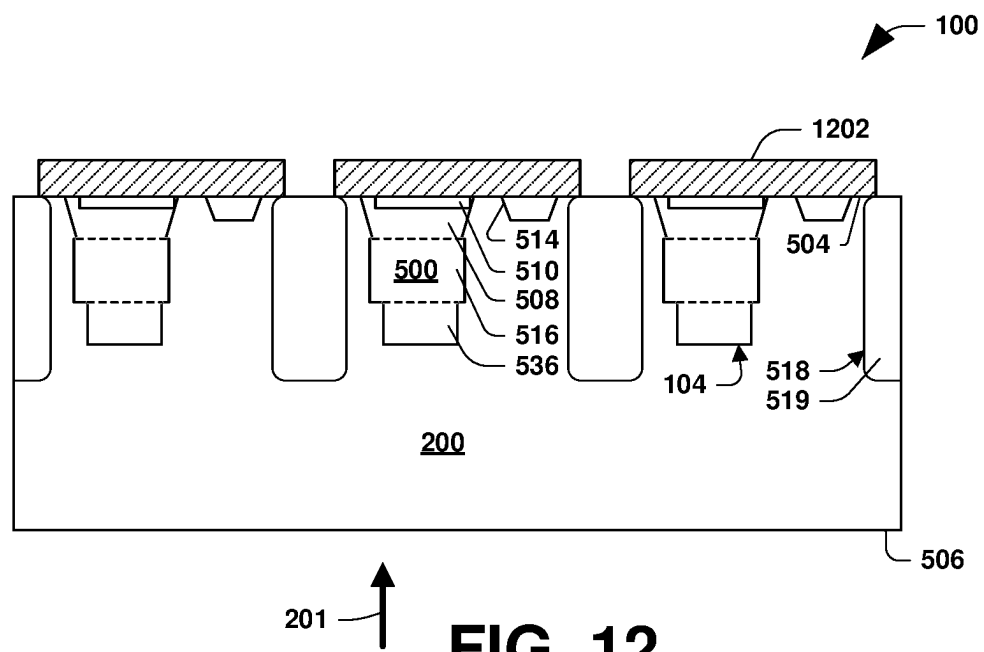
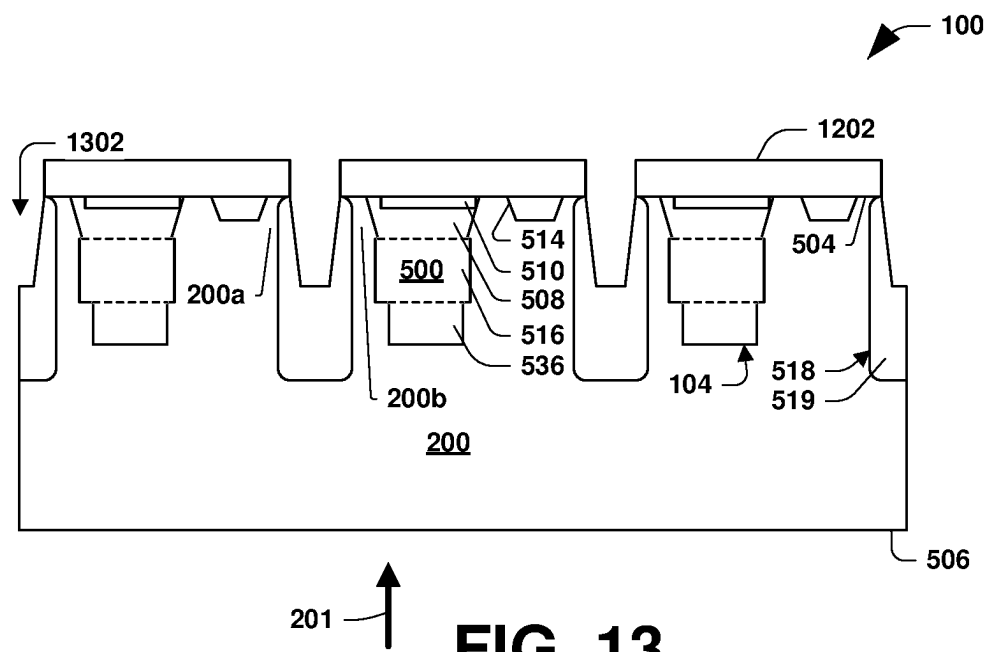
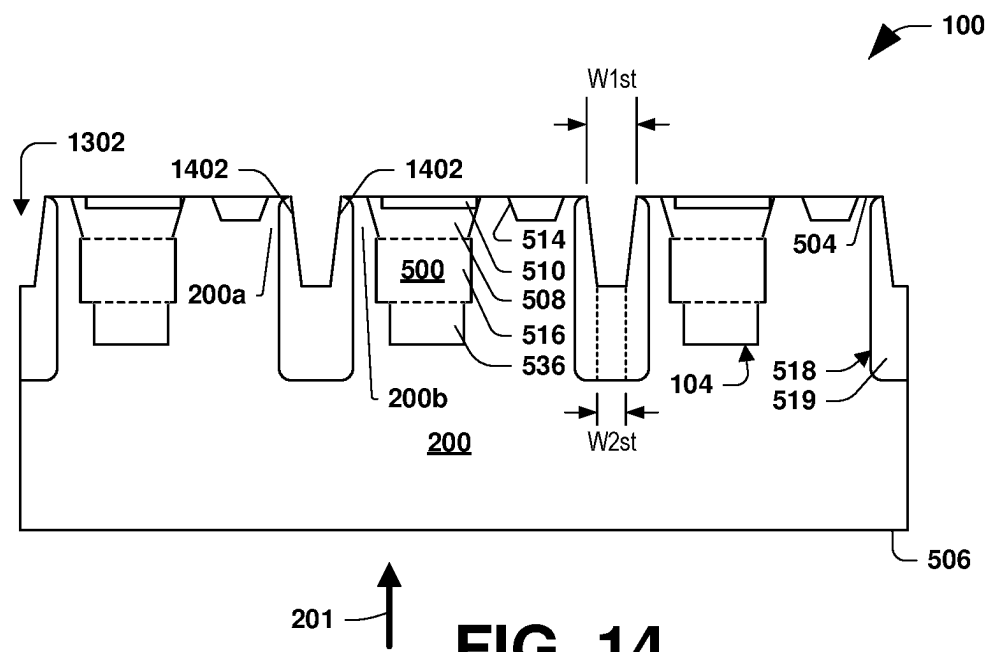
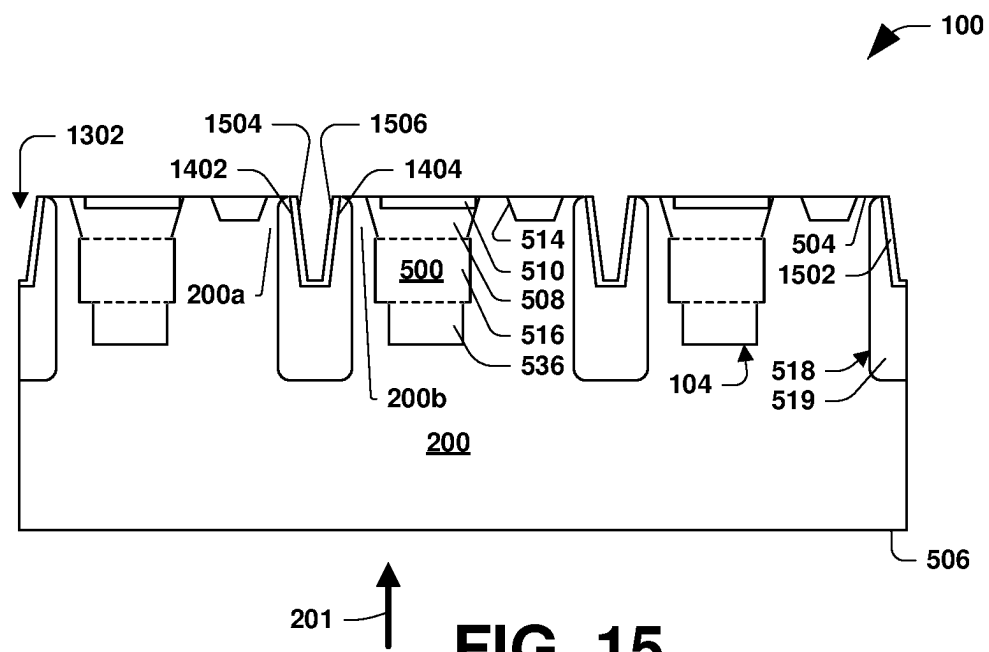
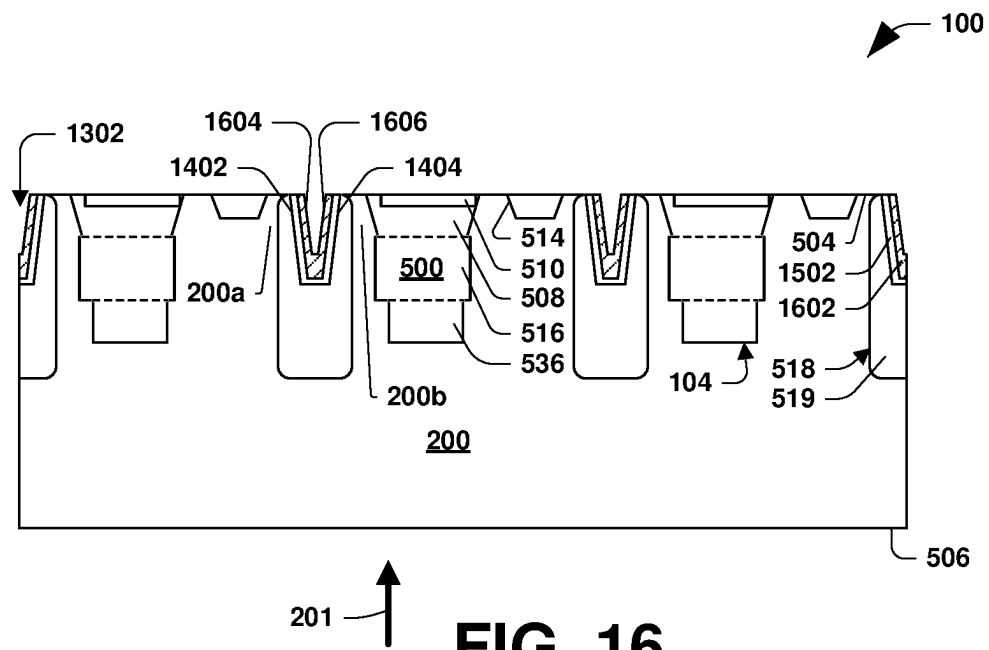


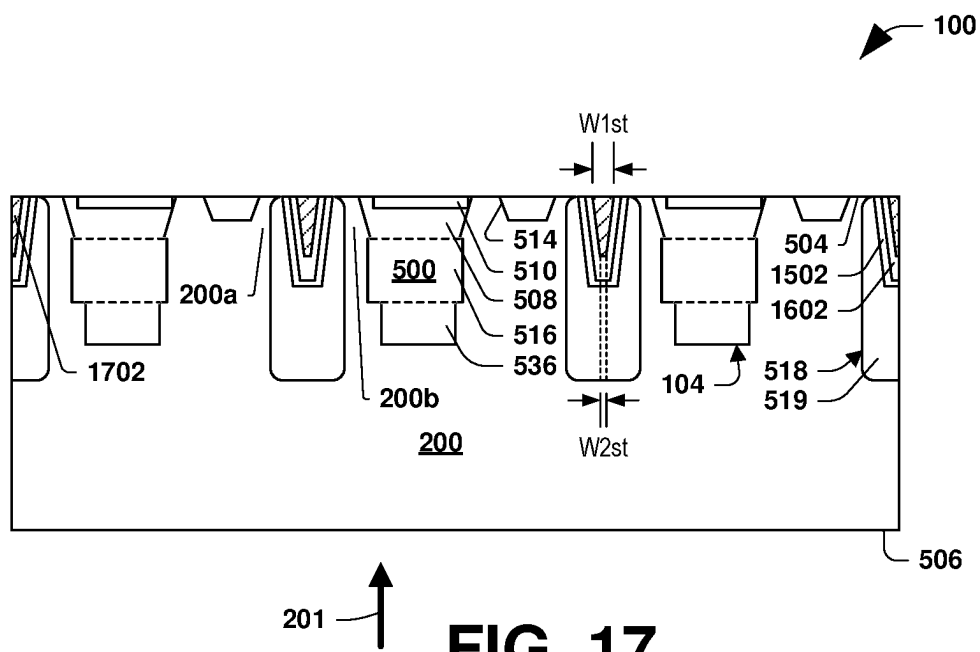
FIG. 12

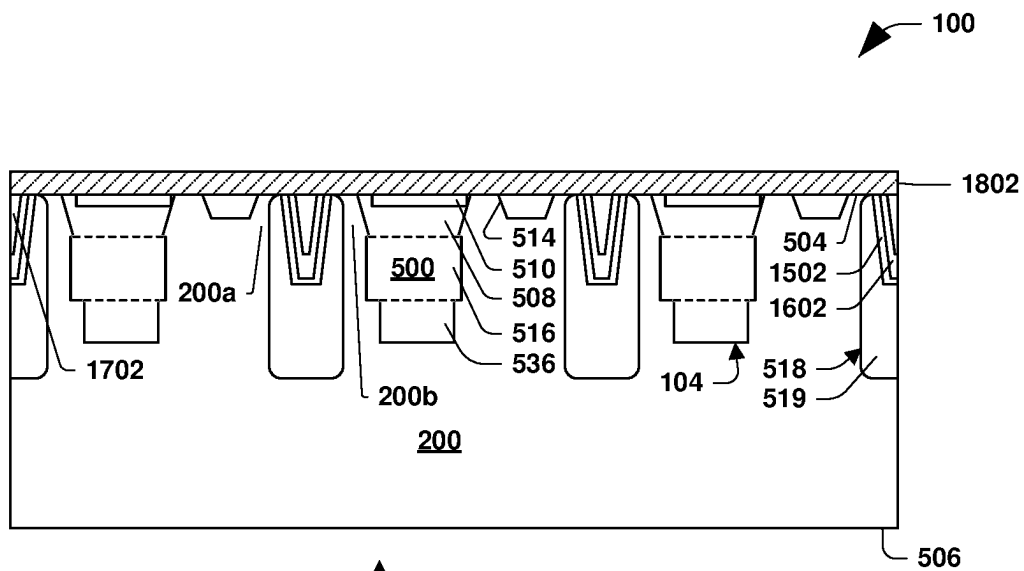




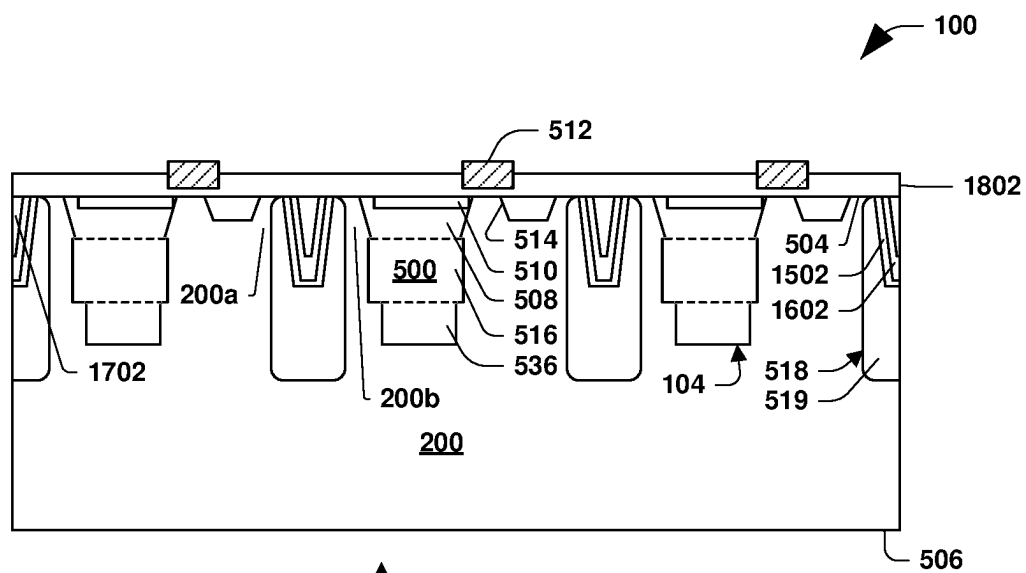








201 ↑ **FIG. 18**



201 ↑

FIG. 19

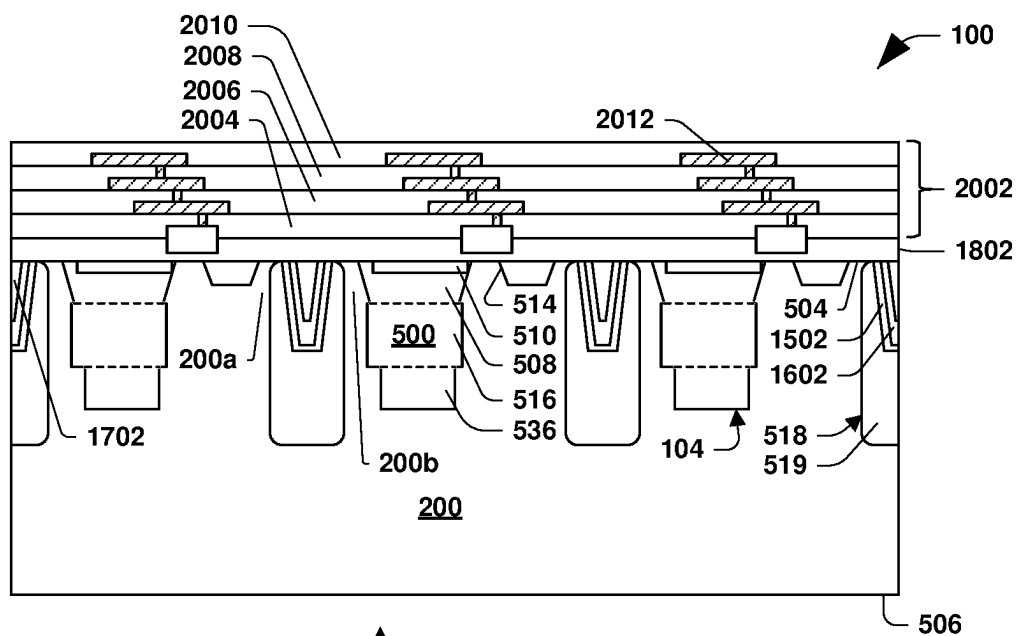
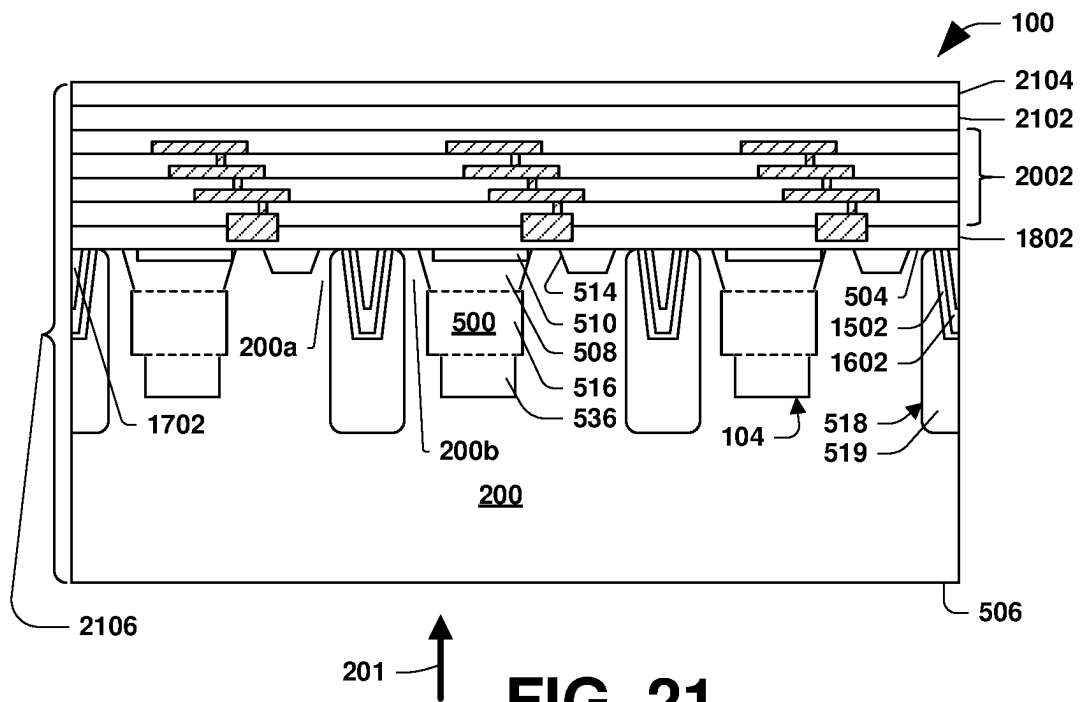
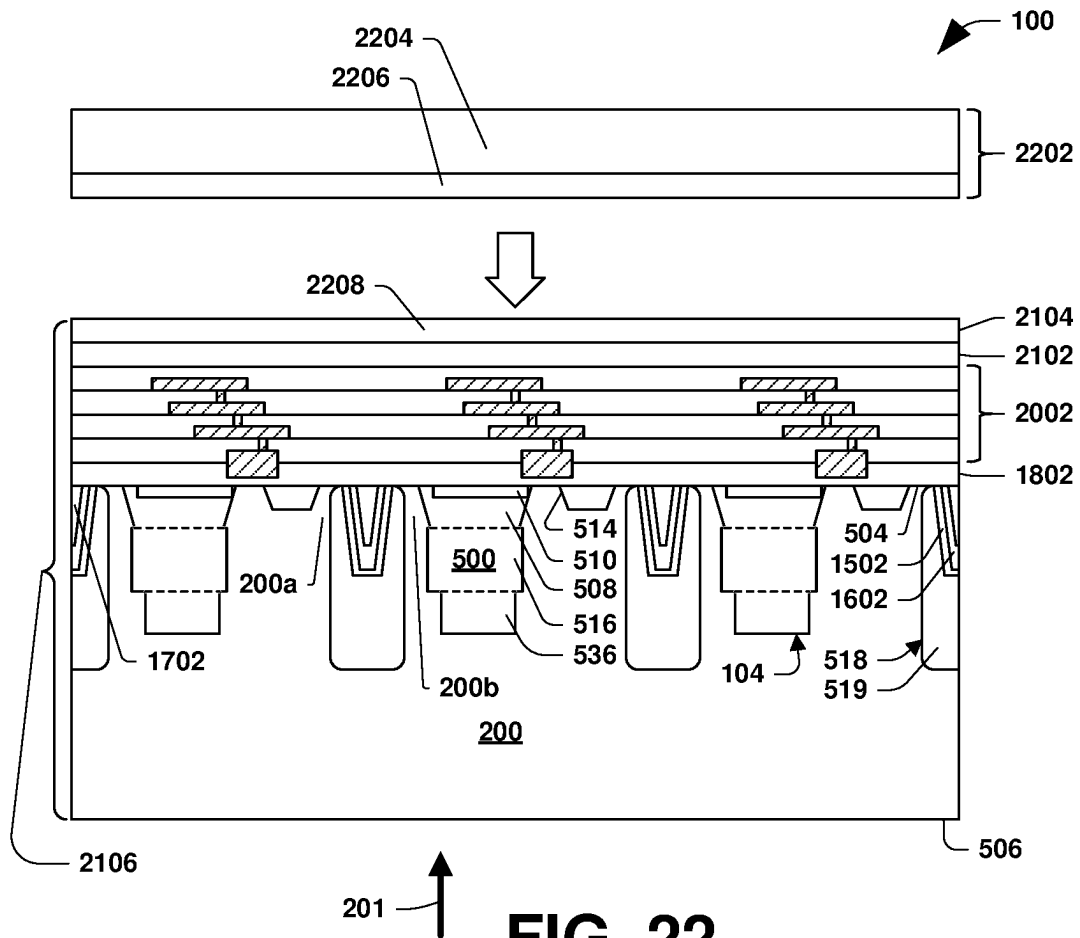


FIG. 20





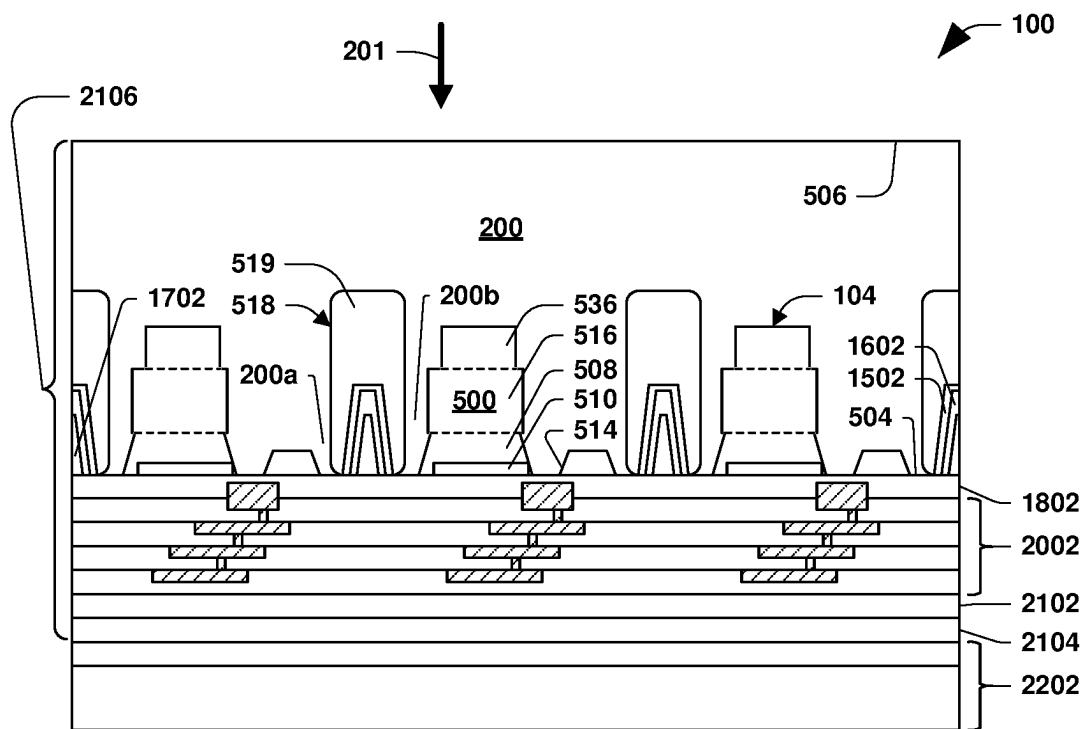


FIG. 23

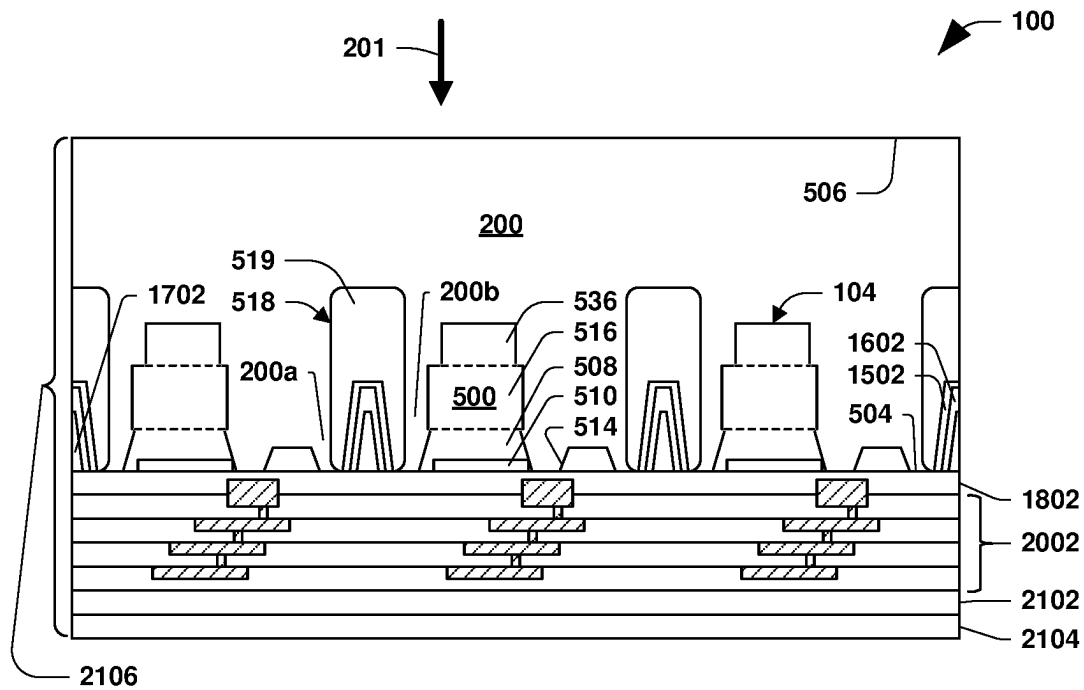
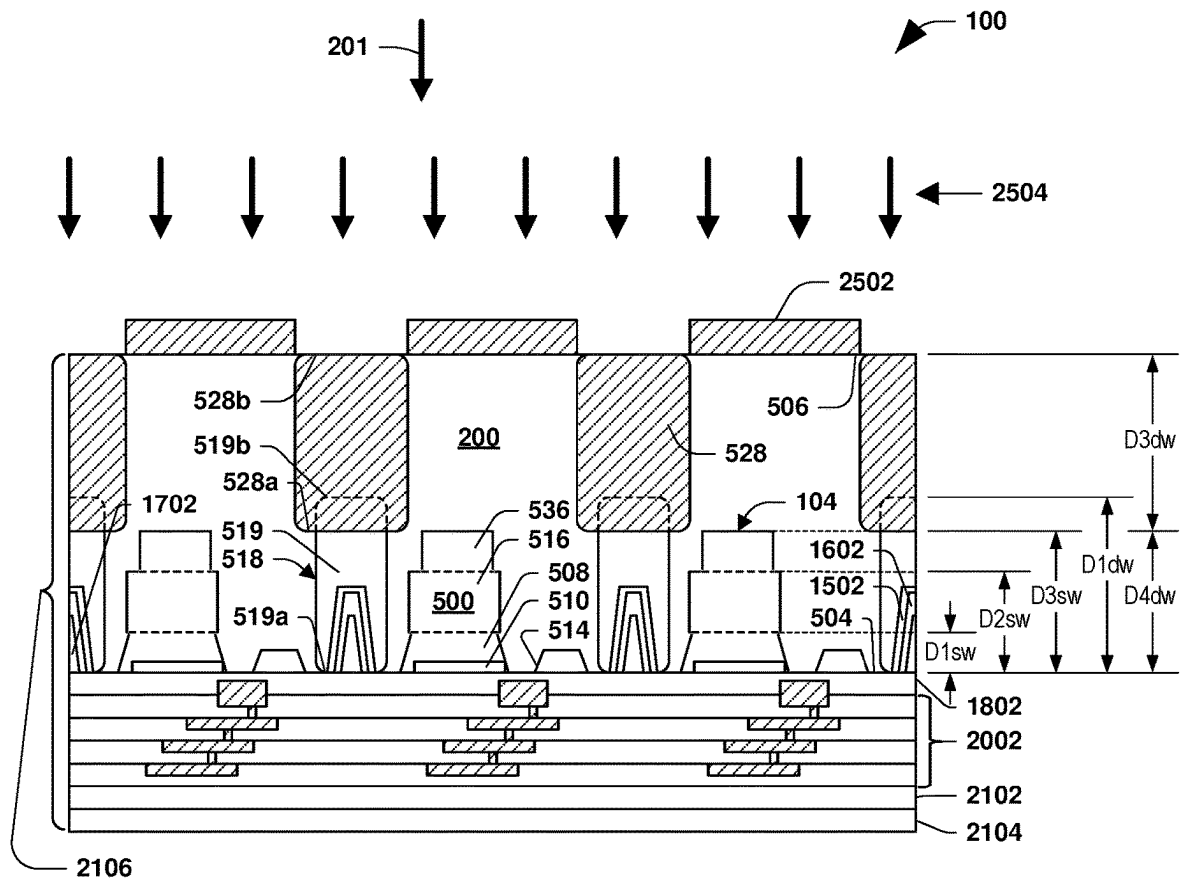


FIG. 24

**FIG. 25**

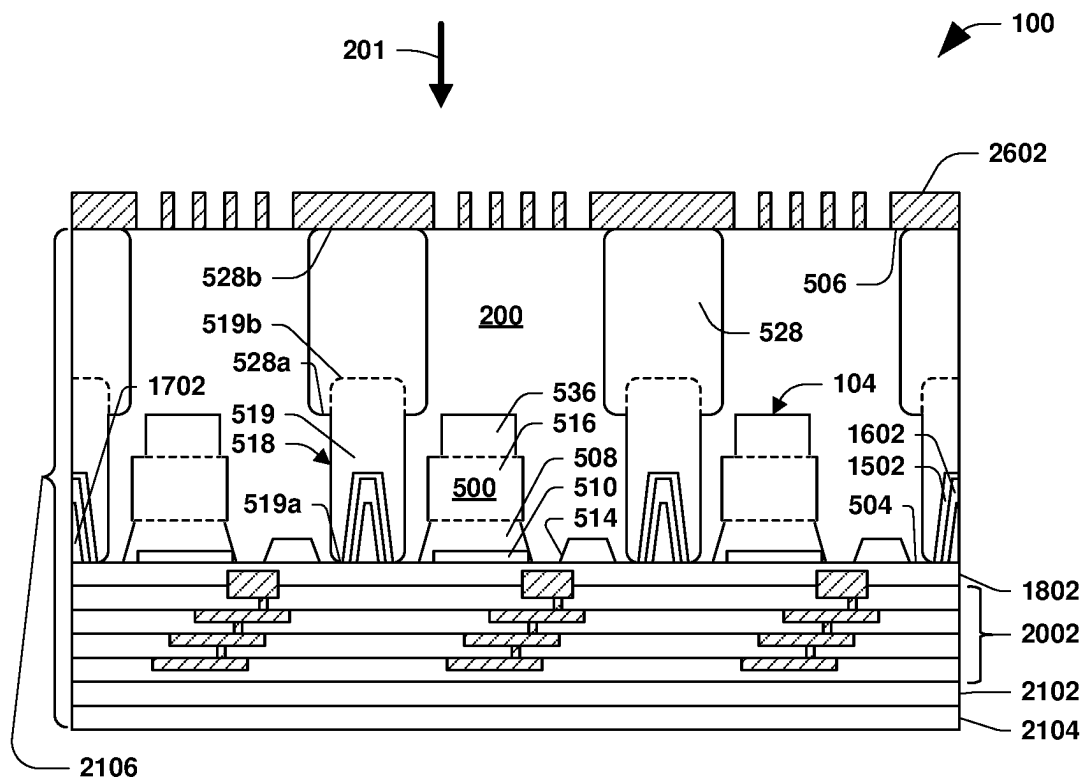


FIG. 26

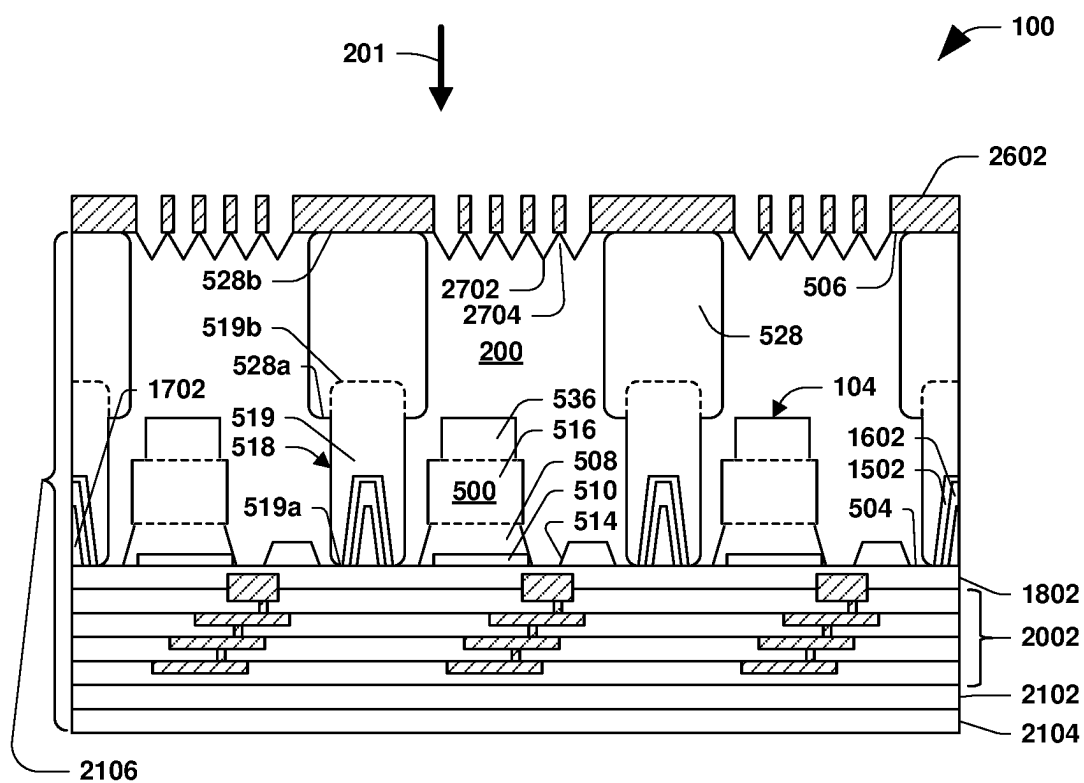


FIG. 27

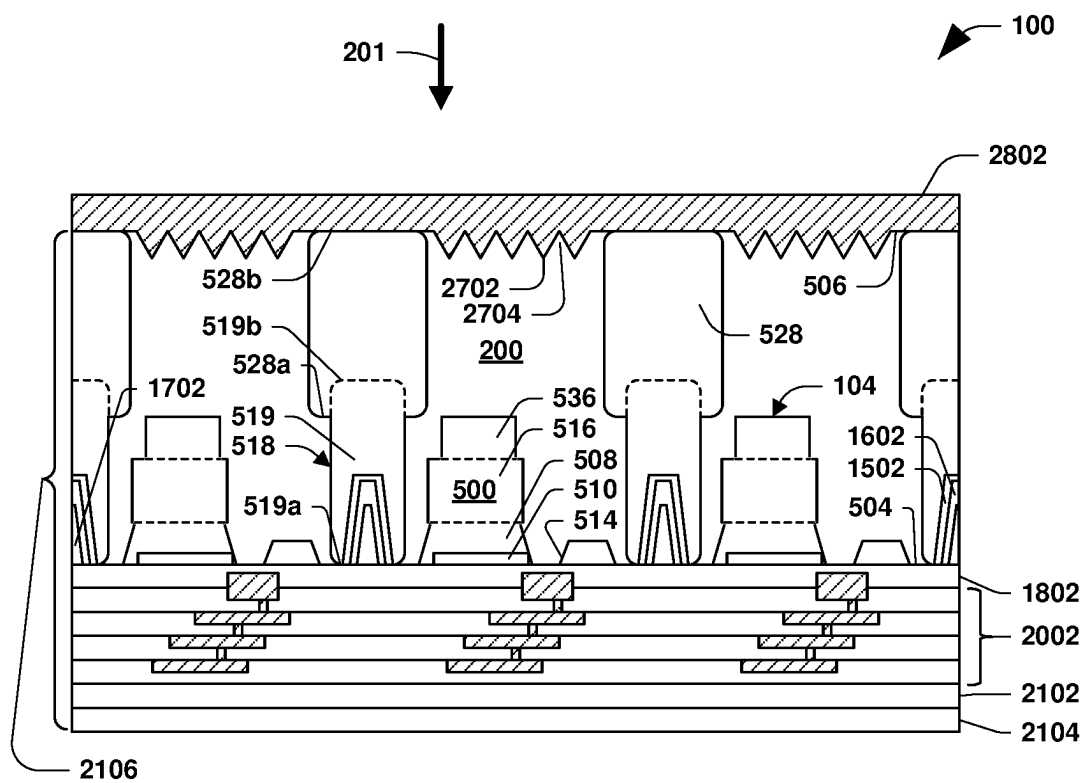


FIG. 28

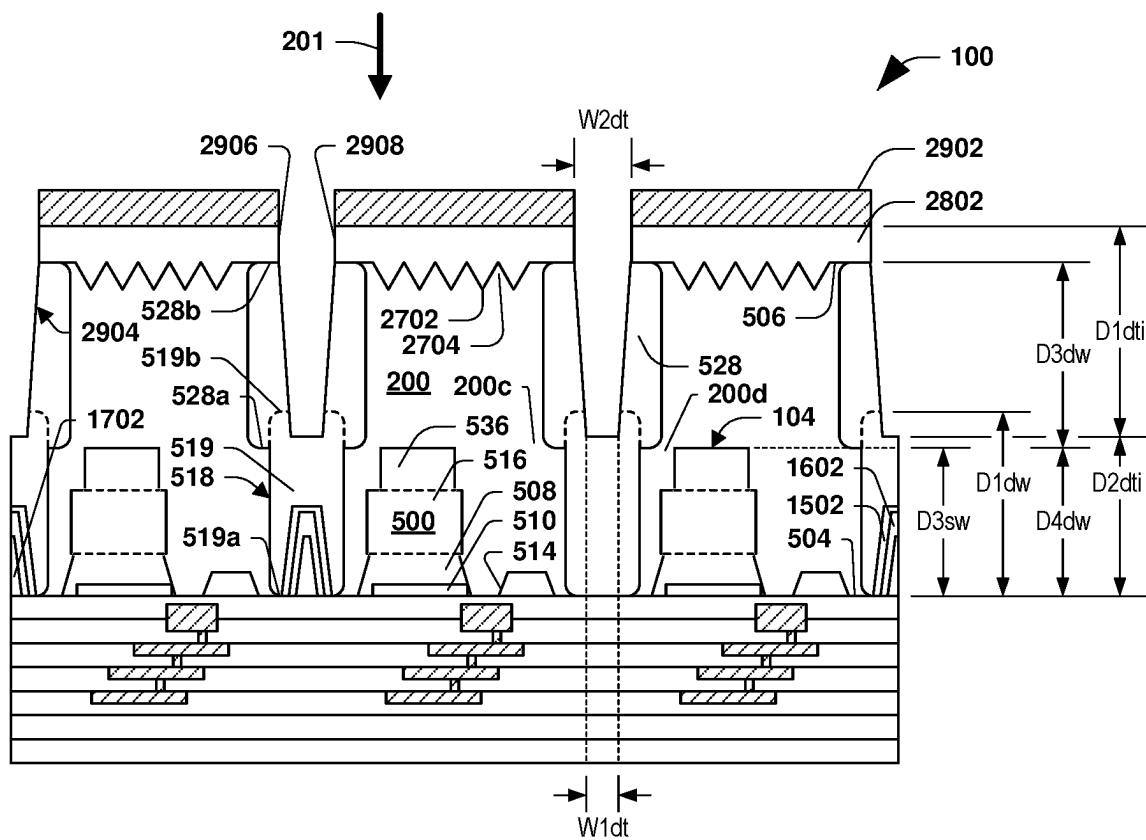


FIG. 29

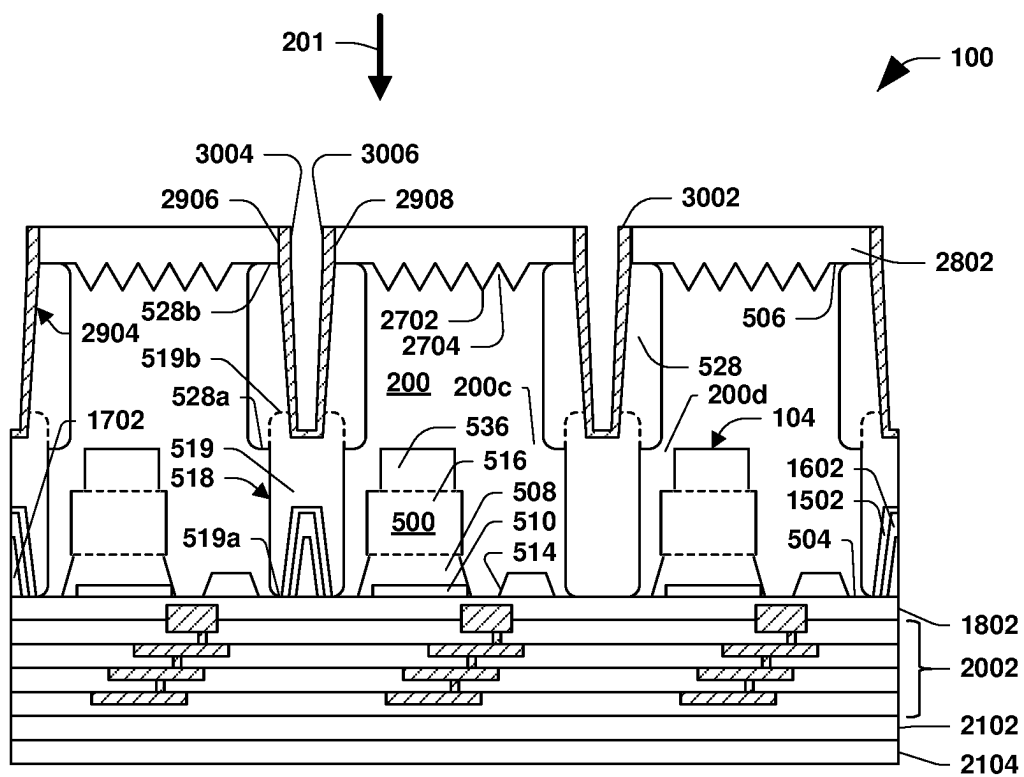


FIG. 30

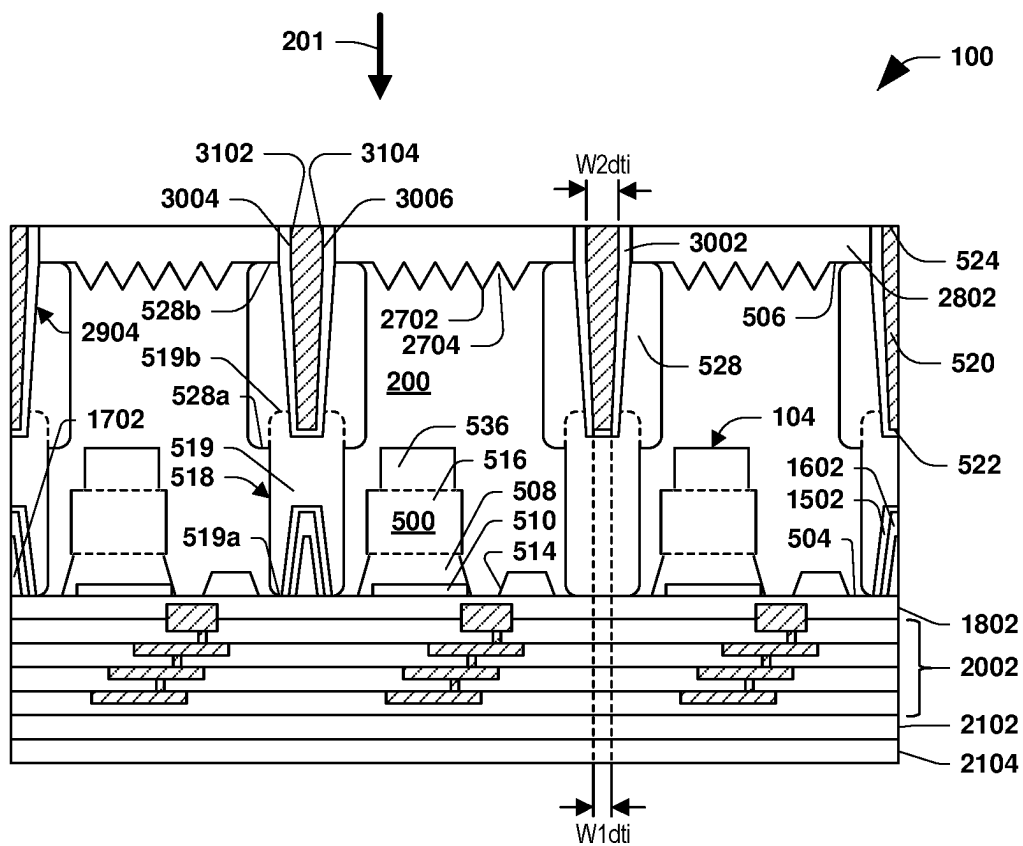


FIG. 31

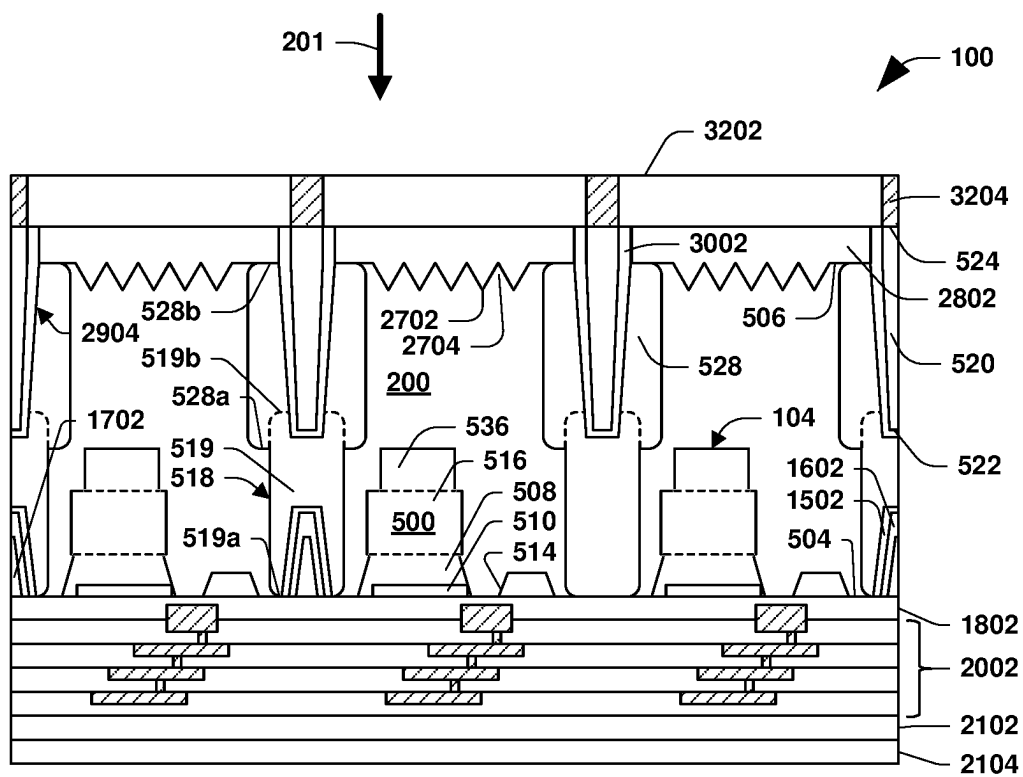


FIG. 32

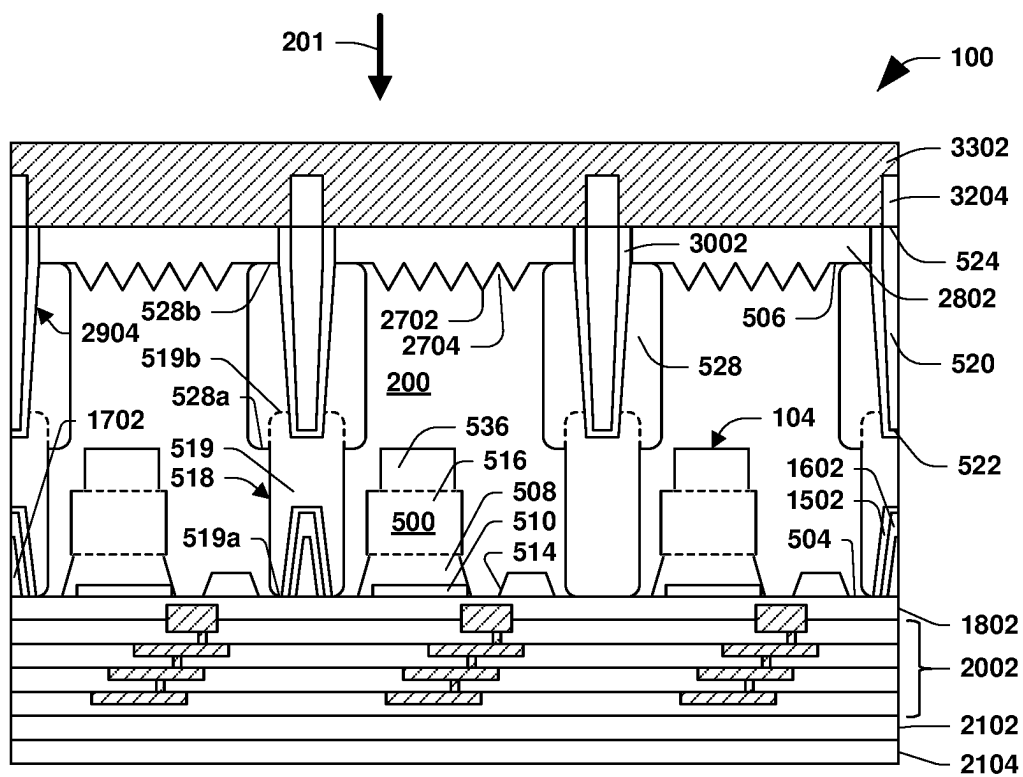


FIG. 33

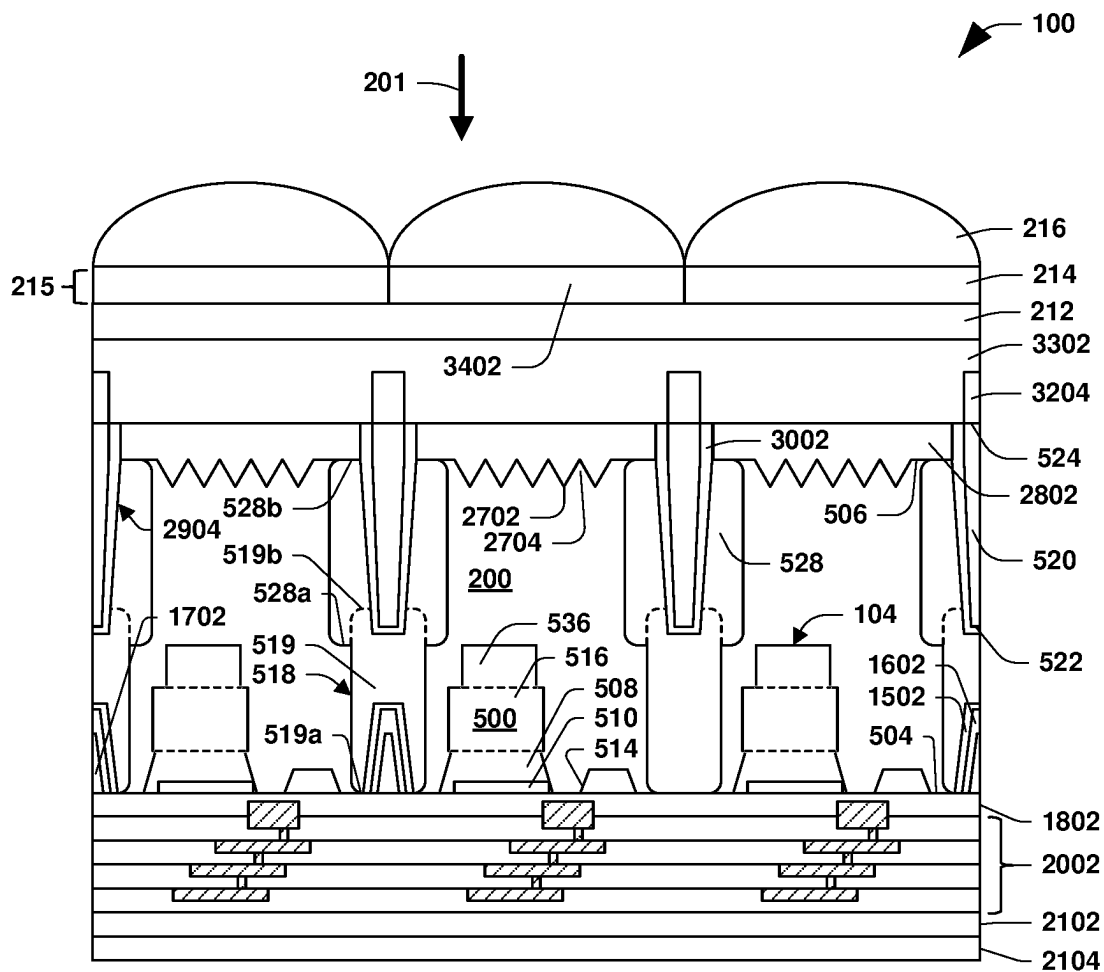
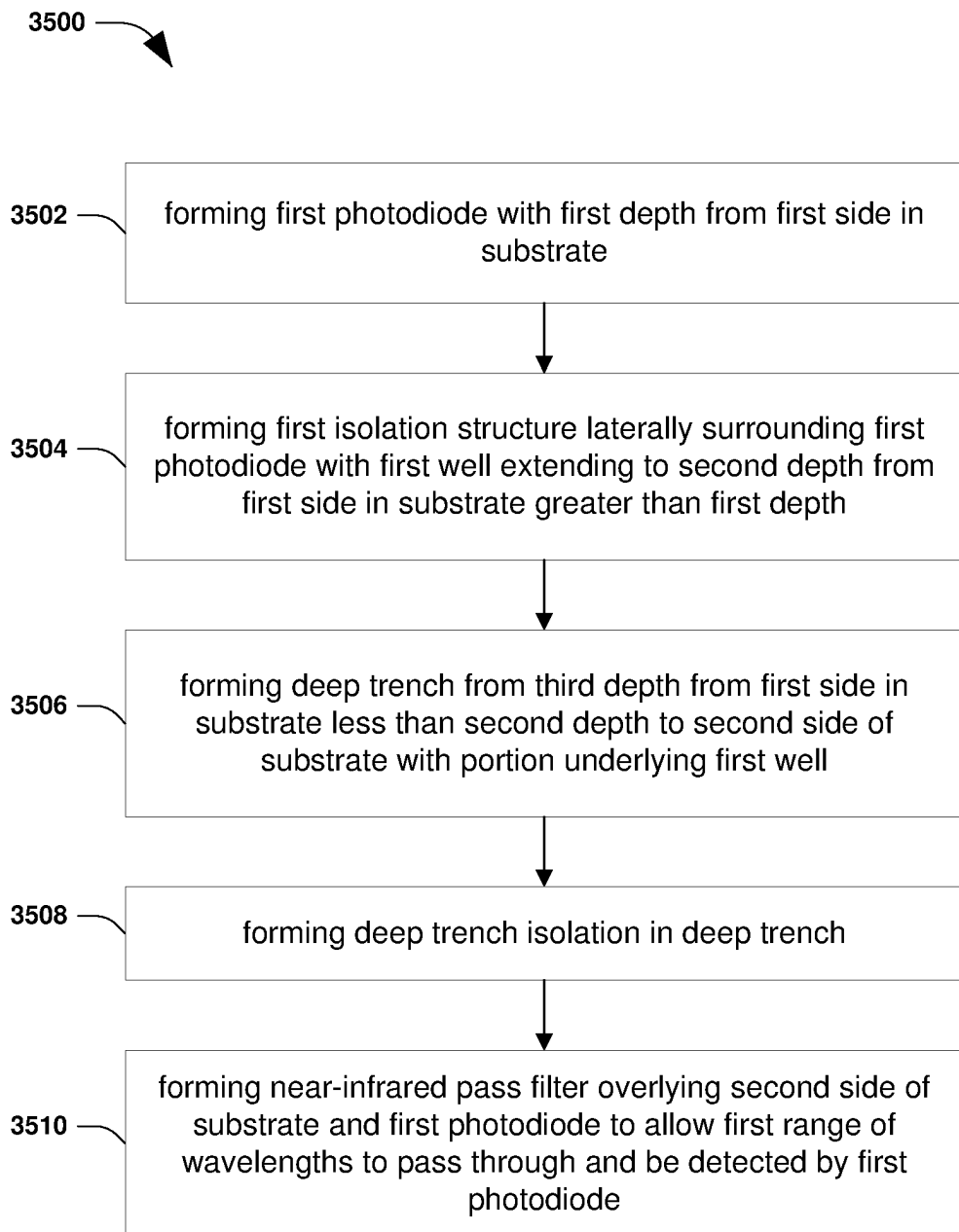
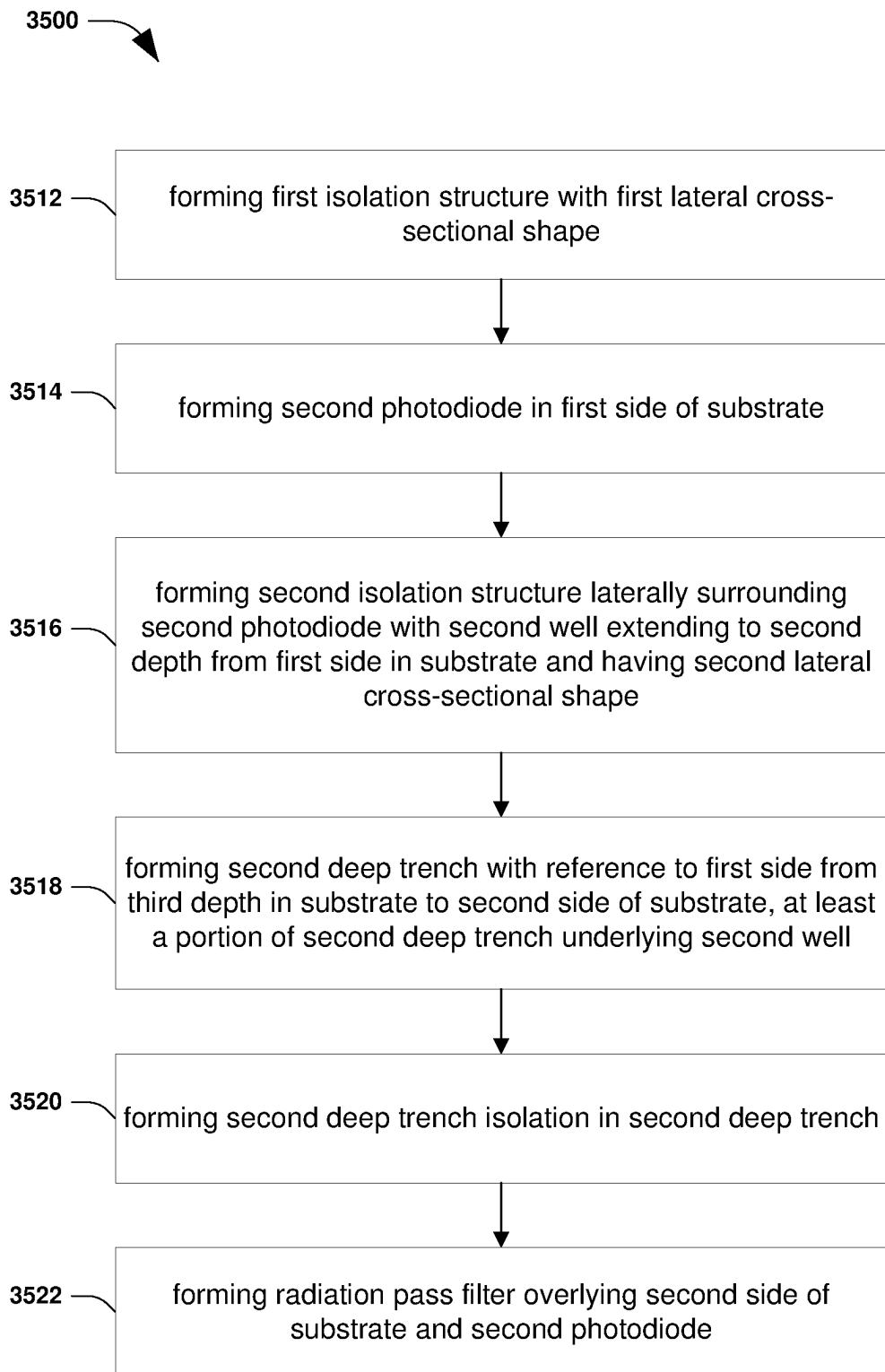


FIG. 34

**FIG. 35A**

**FIG. 35B**

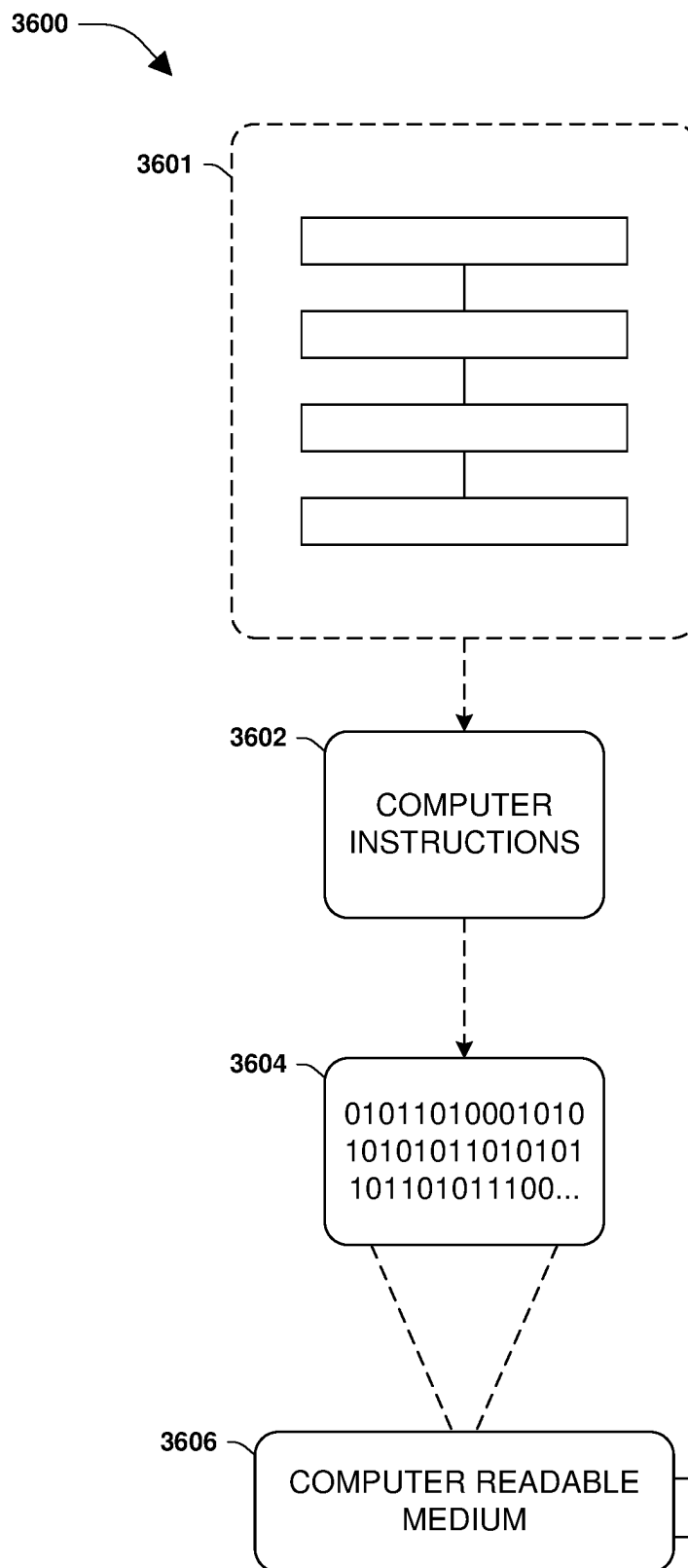


FIG. 36

SEMICONDUCTOR ARRANGEMENT WITH ISOLATION STRUCTURE

RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 17/458,720, titled “SEMICONDUCTOR ARRANGEMENT WITH ISOLATION STRUCTURE” and filed on Aug. 27, 2021, which is incorporated herein by reference.

BACKGROUND

A charge-coupled device (CCD), complementary metal-oxide-semiconductor (CMOS) radiation detecting elements, and other types of radiation detecting elements are used to convert an image focused on a radiation detecting element into an electrical signal. The device or element comprises an array of radiation detecting elements, such as photodiodes, configured to produce an electrical signal corresponding to an intensity of radiation impinging on the radiation detecting element. The electrical signal is used to display a corresponding image on a monitor or provide information about the optical image.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. 1A-1M illustrate a semiconductor arrangement including an isolation structure arrangement and a photodiode array, according to some embodiments.

FIG. 2 illustrates a semiconductor arrangement including a photodiode array, according to some embodiments.

FIG. 3 illustrates a graph of relative responsivity of a photodiode array with respect to wavelengths of detected radiation, according to some embodiments.

FIGS. 4A-4D illustrate a photodiode array, according to some embodiments.

FIGS. 5A-5E schematically illustrate a semiconductor arrangement including a photodiode and an isolation structure arrangement, according to some embodiments.

FIGS. 6-34 illustrate a semiconductor arrangement at various stages of fabrication, according to some embodiments.

FIGS. 35A and 35B illustrate a method of making a semiconductor arrangement, according to some embodiments.

FIG. 36 illustrates an exemplary computer-readable medium, according to some embodiments.

DETAILED DESCRIPTION

The following disclosure provides several different embodiments, or examples, for implementing different features of the provided subject matter with reference to the drawings, wherein like reference numerals are generally used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide an understanding of the claimed subject matter. It is evident, however, that the claimed subject matter can be practiced without these spe-

cific details. In other instances, structures and devices are illustrated in block diagram form in order to facilitate describing the claimed subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation illustrated in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. Also, relationship terms such as “connected to,” “adjacent to,” “coupled to,” and the like, may be used herein to describe both direct and indirect relationships. “Directly” connected, adjacent, or coupled may refer to a relationship in which there are no intervening components, devices, or structures. “Indirectly” connected, adjacent, or coupled may refer to a relationship in which there are intervening components, devices, or structures.

One or more semiconductor arrangements and one or more techniques for forming such semiconductor arrangements are provided. In an example, a semiconductor arrangement comprises a photodiode array formed over a substrate. The photodiode array comprises one or more photodiodes, such as image sensor pixels, configured to accumulate energy generated by optical radiation or near-infrared (NIR) radiation, such as from photons, of an optical image. A voltage of a photodiode can be read as an output for the optical image. In some embodiments, a photodiode is situated under one or more layers or components formed over a substrate. Because radiation travels along a path that comprises such layers or components before reaching the photodiode, signal strength of the radiation can decay before reaching the photodiode or the radiation can travel towards another photodiode. For example, the radiation can be detected by a neighboring or adjacent photodiode, which can result in crosstalk. Crosstalk can degrade performance of the semiconductor arrangement, increase noise, and decrease at least one of quality or intensity of signals produced by the semiconductor arrangement.

According to some embodiments, a semiconductor arrangement may have at least one of photodiodes, pinned layer photodiodes, reset transistors, source follower transistors, floating diffusions (also known as floating diodes), or transfer transistors. A CMOS semiconductor arrangement may be a CMOS active pixel image sensor (APS) with an intra-pixel charge transfer to a floating diffusion (FD). A pinned photodiode (PPD), also known as a pinned layer photodiode, is an example photodiode structure used in a CCD, a CMOS semiconductor arrangement, or a CMOS APS. A PPD provides, for example, at least one of low noise,

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high quantum efficiency, or low dark current. A CMOS semiconductor arrangement may be a front side illuminated (FSI) image sensor, detecting radiation from a front side, or a back side illuminated (BSI) image sensor, detecting radiation from a back side.

Accordingly, a semiconductor arrangement is provided herein. According to some embodiments, the semiconductor arrangement includes a photodiode that extends to a first depth from a first side in a substrate. An isolation structure laterally surrounds the photodiode, and the isolation structure has a first well extending to a second depth from the first side in the substrate greater than the first depth. A deep trench isolation extends from a third depth, from the first side in the substrate, to a fourth depth from the first side in the substrate. At least a portion of the deep trench isolation underlies the first well. Radiation enters the substrate from a first direction. The isolation structure has a shallow trench isolation tapered from a first maximum width to a first minimum width in a second direction opposite to the first direction. The isolation structure has the deep trench isolation tapered from a second minimum width to a second maximum width in the second direction. The semiconductor arrangement includes a first photodiode configured to detect a first range of wavelengths of radiation and a second photodiode configured to detect a second range of wavelengths of radiation.

A semiconductor arrangement including a photodiode array is provided herein. According to some embodiments, the semiconductor arrangement includes a photodiode array over a substrate, wherein the photodiode array comprises a first photodiode, a second photodiode, and a third photodiode. An isolation structure is disposed between the first photodiode and the second photodiode and surrounds the third photodiode. The isolation structure has a first well and a shallow trench isolation at least partially surrounded by the first well. A near-infrared pass filter overlies the second photodiode and is configured to allow a first range of wavelengths to pass through and be detected by the second photodiode. The photodiode array includes a first subset of photodiodes having a first lateral cross-sectional diameter and a second subset of photodiodes having a second lateral cross-sectional diameter. The first photodiode and the third photodiode are members of the first subset of photodiodes, and the second photodiode is a member of the second subset of photodiodes. Each of the first subset of photodiodes has a corresponding overlying radiation pass filter configured to allow a radiation pass range of wavelengths to pass through and be detected by a corresponding photodiode. Each of the second subset of photodiodes has a corresponding overlying radiation pass filter configured to allow a radiation pass range of wavelengths to pass through and be detected by a corresponding photodiode. The isolation structure includes a first subset of isolation structures surrounding the first subset of photodiodes with each having a first lateral cross-sectional shape. The isolation structure includes a second subset of isolation structures surrounding the second subset of photodiodes with each having a second lateral cross-sectional shape different from the first lateral cross-sectional shape. The first subset of isolation structures includes a plurality of laterally contiguous isolation structures laterally surrounding the third photodiode.

One or more methods of making a semiconductor arrangement are provided herein. According to some embodiments, a first photodiode with a first depth is formed from a first side in a substrate. A first isolation structure is formed laterally surrounding the first photodiode. The first isolation structure comprises a first well extending to a

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second depth from the first side in the substrate greater than the first depth. An etch into a second side in the substrate, opposite the first side, is performed to form a deep trench. The deep trench extends from a third depth from the first side in the substrate less than the second depth to the second side of the substrate. At least a portion of the deep trench underlies the first well. A deep trench isolation is formed in the deep trench. A near-infrared pass filter overlying the second side of the substrate and the first photodiode is formed. The near-infrared pass filter is configured to allow a first range of wavelengths to pass through and be detected by the first photodiode. Radiation enters the substrate from a first direction and from the second side of the substrate. The etch into the second side of the substrate may be performed such that the deep trench is tapered from a first maximum width to a first minimum width in the first direction. An etch into the first side of the substrate is performed to form a shallow trench. The shallow trench may be tapered from a first maximum width to a first minimum width in a second direction opposite to the first direction. A shallow trench isolation is formed in the shallow trench. The first isolation structure may be formed having a first lateral cross-sectional shape. A second photodiode is formed from the first side of the substrate. A second isolation structure is formed laterally surrounding the second photodiode. The second isolation structure comprises a second well extending to the second depth from the first side in the substrate and having a second lateral cross-sectional shape. A radiation pass filter is formed overlying the second side of the substrate and the second photodiode.

FIG. 1A illustrates a semiconductor arrangement 100, according to some embodiments. The semiconductor arrangement 100 comprises an isolation structure arrangement 102 at least one of surrounding or partially surrounding a photodiode array 104. The photodiode array 104 comprises a first photodiode 101, a second photodiode 103, a third photodiode 105, and a fourth photodiode 107, for example. FIG. 1B illustrates the photodiode array 104 comprising a first subset of photodiodes 106, including first photodiodes 108a-108p. According to some embodiments, the first photodiode 108a in FIG. 1B corresponds to the first photodiode 101 in FIG. 1A and the first photodiode 108p in FIG. 1B corresponds to the third photodiode 105 in FIG. 1A. Each photodiode of the first subset of photodiodes 106 has a corresponding overlying radiation pass filter configured to allow a radiation pass range of wavelengths to pass through and be detected by the corresponding photodiode, as set forth in greater detail below. FIG. 10 illustrates the photodiode array 104 comprising a second subset of photodiodes 110, including second photodiodes 112a-112i. According to some embodiments, the photodiode 112a in FIG. 10 corresponds to the second photodiode 103 in FIG. 1A, and the second photodiode 112d in FIG. 10 corresponds to the fourth photodiode 107 in FIG. 1A. According to some embodiments, the first subset of photodiodes 106 has a greater lateral cross-sectional diameter (as described in greater detail below) than the second subset of photodiodes 110. According to some embodiments, the first subset of photodiodes 106 has a greater lateral cross-sectional area (as described in greater detail below) than the second subset of photodiodes 110. Each photodiode of the second subset of photodiodes 110 has a corresponding overlying radiation pass filter configured to allow a radiation pass range of wavelengths to pass through and be detected by the corresponding photodiode, as set forth in greater detail below. FIG. 1D illustrates the isolation structure arrangement 102 comprising a first subset of isolation structures 114, includ-

ing first isolation structures **116a-116p**. According to some embodiments, each of the first isolation structures **116a-116p** laterally surrounds each of the first photodiodes **108a-108p** and is formed in an array.

FIG. 1E illustrates first isolation structure **116f** surrounding a first photodiode **108f**, according to some embodiments. FIG. 1F is a sectional view of the first isolation structure **116f** and the first photodiode **108f** taken along line F-F of FIGS. 1B, 1D, and 1E. The first isolation structure **116f** has a first lateral cross-sectional shape and includes a plurality of sections. The first isolation structure **116f** includes sections **116/1-116/8**. According to some embodiments, the first isolation structure **116f** has a piecewise-linear cross-sectional shape, such as a tetragon (e.g., a square, a rectangle, a diamond, a trapezoid, a parallelogram, or a rhombus), a pentagon, a hexagon, a heptagon, an octagon, or other cross-sectional shape. According to some embodiments, the first isolation structure **116f** has at least one of a regular piecewise-linear cross-sectional shape, where all sections have an equivalent length, or a non-regular piecewise-linear cross-sectional shape, where all section do not have an equivalent length. For example, sections **116/8**, **116/1**, **116/2**, **116/4**, **116/5**, and **116/6** may have an equivalent length and sections **116/7** and **116/3** may have an equivalent length longer than sections **116/8**, **116/1**, **116/2**, **116/4**, **116/5**, and **116/6**. In another example, sections **116/8**, **116/1**, **116/2**, **116/4**, **116/5**, and **116/6** may have an equivalent length and sections **116/7** and **116/3** may have an equivalent length shorter than sections **116/8**, **116/1**, **116/2**, **116/4**, **116/5**, and **116/6**. Other arrangements and/or configurations of first isolation structure **116f** are within the scope of the present disclosure.

According to some embodiments, the first isolation structure **116f** has sections that are not piecewise-linear. For example, sections **116/8**, **116/1**, **116/2**, **116/4**, **116/5**, and **116/6** are curved and sections **116/7** and **116/3** are piecewise-linear. In another example, sections **116/8**, **116/1**, **116/2**, **116/4**, **116/5**, and **116/6** are curved with a first degree of curvature and sections **116/7** and **116/3** are curved with a second degree of curvature. According to some embodiments, the first isolation structure **116f** is one of circular, elliptical, or approximating a circle. According to some embodiments, a layout of the first isolation structure **116f** is an octagon with variations due to fabrication resulting in sections approximating a circle. Other structures and/or configurations of the first isolation structure **116f** are within the scope of the present disclosure.

The first photodiode **108f** has a first lateral cross-sectional diameter **W1pd** and a first lateral cross-sectional length **L1pd**. According to some embodiments, the first photodiode **108f** has a first lateral cross-sectional shape. According to some embodiments, the first photodiode **108f** has a first lateral cross-sectional area bounded by the first isolation structure **116f**. According to some embodiments, the first photodiode **108f** has a first lateral cross-sectional shape approximating a lateral cross-sectional shape of the first isolation structure **116f**. According to some embodiments, the first photodiode **108f** has a portion approximating a first lateral cross-sectional shape of the first isolation structure **116f**. According to some embodiments, a layout of the first photodiode **108f** has a portion approximating an octagon, or other lateral cross-sectional shape set forth above, with variations due to fabrication resulting in the portion approximating a circle. First photodiode **108f** may transfer charge to a floating diffusion, as described in greater detail below. According to some embodiments, the floating diffusion is bounded by the first isolation structure **116f**. According to

some embodiments, the floating diffusion has a portion approximating a lateral cross-sectional shape of the first isolation structure **116f**. According to some embodiments, a layout of the floating diffusion has a portion approximating an octagon, or other lateral cross-sectional shape set forth above, with variations due to fabrication resulting in the portion approximating a circle. Other structures and/or configurations of the first photodiode **108f** are within the scope of the present disclosure.

FIG. 1G illustrates the isolation structure arrangement **102** comprising a first subset of isolation structures **114**, according to some embodiments. Each of the first subset of isolation structures **114** laterally surrounds a photodiode from the first subset of photodiodes **106** and is formed in an array. The first subset of isolation structures **114** includes a plurality of first isolation structures adjacent to the first isolation structure **116f**. The plurality of first isolation structures includes first isolation structures **116b**, **116g**, **116j**, and **116e**. According to some embodiments, each of the first isolation structures **116a-116p** illustrated in FIG. 1D (e.g., the first isolation structure **116f**) is at least one of adjacent to, contiguous with, overlapping, or overlapping a portion of at least one first isolation structure (e.g., one of the first isolation structures **116a-116p**). According to some embodiments, the first isolation structure **116f** includes a section **116/1** that is contiguous with a section **116b5** of the first isolation structure **116b**. According to some embodiments, each of the first isolation structures **116a-116p** is at least one of adjacent to, contiguous with, overlapping, or overlapping a portion of at least two first isolation structures (e.g., two of the first isolation structures **116a-116p**). According to some embodiments, the first isolation structure **116f** includes a section **116/3** that is contiguous with a section **116g7** of the first isolation structure **116g**. According to some embodiments, each of the first isolation structures **116a-116p** is at least one of adjacent to, contiguous with, overlapping, or overlapping a portion of at least three first isolation structures (e.g., three of the first isolation structures **116a-116p**). According to some embodiments, the first isolation structure **116f** includes a section **116/5** that is contiguous with a section **116/1** of the first isolation structure **116j**. According to some embodiments, each of the first isolation structures **116a-116p** is at least one of adjacent to, contiguous with, overlapping, or overlapping a portion of at least four first isolation structures (e.g., four of the first isolation structures **116a-116p**). According to some embodiments, the first isolation structure **116f** includes a section **116/7** that is contiguous with a section **116e3** of the first isolation structure **116e**. Other structures and/or configurations of the first isolation structures **116a-116p** are within the scope of the present disclosure.

According to some embodiments, each of the first isolation structures **116a-116p** (e.g., the first isolation structure **116f**) includes a section that is the same as a section of at least one first isolation structure. For example, the first isolation structure **116f** may include the section **116/1** that is the same as the section **116b5** of the first isolation structure **116b**. According to some embodiments, each of the first isolation structures **116a-116p** includes a section that is the same as a section of a second first isolation structure. For example, the first isolation structure **116f** may also include the section **116/3** that is the same as the section **116g7** of the first isolation structure **116g**. According to some embodiments, each of the first isolation structures **116a-116p** includes a third section that is the same as a section of a third first isolation structure. For example, the first isolation structure **116f** may also include the section **116/5** that is the

same as the section **116f** of the first isolation structure **116j**. According to some embodiments, each of the first isolation structures **116a-116p** includes a fourth section that is the same as a section of a fourth first isolation structure. For example, the first isolation structure **116f** may include the section **116f7** that is the same as the section **116e3** of the first isolation structure **116e**. Other structures and/or configurations of the first isolation structures **116a-116p** are within the scope of the present disclosure.

FIG. 1H illustrates the isolation structure arrangement **102** comprising a first subset of isolation structures **118**, including first isolation structures **120a-120e**, according to some embodiments. The first subset of isolation structures **118** may replace the first subset of isolation structures **114** in FIG. 1G. Each of the first isolation structures **120a-120e** laterally surrounds a photodiode from the first subset of photodiodes **106** and is formed in an array. According to some embodiments, each of the first isolation structures **120a-120e** includes a plurality of sections as set forth with reference to first isolation structure **116f**, illustrated in FIG. 1E (e.g., sections **116f1-116f8**). In the example illustrated in FIG. 1F, each of the first isolation structures **120a-120e** laterally surrounds a corresponding photodiode (e.g., a photodiode **112a-e** of the first subset of photodiodes **106**). In other words adjacent photodiodes may be laterally separated by at least two first isolation structures (e.g., two of the first isolation structures **120a-120e**). For example, a first photodiode surrounded by the first isolation structure **120c** is separated from a first photodiode surrounded by the first isolation structure **120b** by both the first isolation structure **120c** and the first isolation structure **120b**.

According to some embodiments, each of the first isolation structures **120a-120e** (e.g., the first isolation structure **120c**) is adjacent to and contiguous with at least one of the first isolation structures **120a-120e**. For example, the first isolation structure **120c** may include a section **120c1** that is contiguous with a section **120a5** of the first isolation structure **120a**. According to some embodiments, each of the first isolation structures **120a-120e** is adjacent to and contiguous with at least two first isolation structures (e.g., two of the first isolation structures **120a-120e**). For example, the first isolation structure **120c** may also include a section **120c3** that is contiguous with a section **120d7** of the first isolation structure **120d**. According to some embodiments, each of the first isolation structures **120a-120e** is adjacent to and contiguous with at least three first isolation structures (e.g., three of the first isolation structures **120a-120e**). For example, the first isolation structure **120c** may also include a section **120c5** that is contiguous with a section **120e1** of the first isolation structure **120e**. According to some embodiments, each of the first isolation structures **120a-120e** is adjacent to and contiguous with at least four first isolation structures (e.g., four of the first isolation structures **120a-120e**). For example, the first isolation structure **120c** may also include a section **120c7** that is contiguous with a section **120b3** of the first isolation structure **120b**. Other structures and/or configurations of the first isolation structures **120a-120e** are within the scope of the present disclosure.

FIG. 1I illustrates the isolation structure arrangement **102** comprising a second subset of isolation structures **122**, including second isolation structures **124a-124i**. According to some embodiments, each of the second isolation structures **124a-124i** laterally surrounds each of the second photodiodes **112a-112i** and is formed in an array. According to some embodiments, the second subset of isolation structures **122**, including the second isolation structures **124a-124i**, are interleaved between the first subset of isolation

structures **114**, including the first isolation structures **116a-116p**. Other structures and/or configurations of the second isolation structures **124a-124i** are within the scope of the present disclosure.

FIG. 1J illustrates the second isolation structure **124d** surrounding the second photodiode **112d**, according to some embodiments. FIG. 1K is a sectional view of the second isolation structure **124d** and the second photodiode **112d** taken along line K-K of FIGS. 1C, 1I, and 1J. The second isolation structure **124d** has a second lateral cross-sectional shape and includes a plurality of sections. The second isolation structure **124d** includes sections **124d1-124d4**. According to some embodiments, the sections **124d1-124d4** of the second isolation structure **124d** are at least one of sections or portions of sections of adjacent second isolation structures. The second isolation structure **124d** includes a border **130**. According to some embodiments, the border **130** is a p-n junction border and the sections **124d1-124d4** are sections of first isolation structures (e.g., the first isolation structures **116a-116p**). The border **130** includes border sections **130a-130d**. According to some embodiments, the border sections **130a-130d** are p-n junction border sections and the sections **124d1-124d4** are isolation structures of first isolation structures (e.g., the first isolation structures **116a-116p**).

According to some embodiments, the second isolation structure **124d** has a second piecewise-linear cross-sectional shape, such as a tetragon (e.g., a square, a rectangle, a diamond, a trapezoid, a parallelogram, or a rhombus). According to some embodiments, the second isolation structure **124d** has a regular piecewise-linear cross-sectional shape, where all sections have an equivalent length. According to some embodiments, the second isolation structure **124d** has a non-regular piecewise-linear cross-sectional shape, where all sections do not have an equivalent length. For example, the sections **124d4** and **124d1** may have an equivalent length and the sections **124d2** and **124d3** may have an equivalent length longer than the sections **124d4** and **124d1**. In another example, the sections **124d4** and **124d1** may have an equivalent length and the sections **124d2** and **124d3** may have an equivalent length shorter than the sections **124d4** and **124d1**. Other structures and/or configurations of the second isolation structure **124d** are within the scope of the present disclosure.

According to some embodiments, the second isolation structure **124d** has sections that are not piecewise-linear. For example, the sections **124d4** and **124d1** may be curved and the sections **124d2** and **124d3** may be piecewise-linear. In another example, the sections **124d4** and **124d1** may be curved with a first degree of curvature and the sections **124d2** and **124d3** may be curved with a second degree of curvature. According to some embodiments, the second isolation structure **124d** may be at least one of circular, approximating a circle, elliptical, or approximating an ellipse. According to some embodiments, the layout of the second isolation structure **124d** is a diamond with variations due to fabrication resulting in sections approximating a circle or an ellipse. Other structures and/or configurations of the second isolation structure **124d** are within the scope of the present disclosure.

The second photodiode **112d** has a second lateral cross-sectional diameter **W2pd** and a second lateral cross-sectional length **L2pd**. According to some embodiments, the lateral cross-sectional diameter **W2pd** of second photodiode **112d** is less than the first lateral cross-sectional diameter **W1pd** of the first photodiode **108f**. According to some embodiments, the lateral cross-sectional length **L2pd** of the second photo-

diode **112d** is less than the first lateral cross-sectional length **L1pd** of the first photodiode **108f**. According to some embodiments, the second photodiode **112d** has a second lateral cross-sectional shape. According to some embodiments, the second photodiode **112d** has a second lateral cross-sectional area bounded by the second isolation structure **124d**. According to some embodiments, the second lateral cross-sectional shape of the second photodiode **112d** is less than the first lateral cross-sectional shape of the first photodiode **108f**. According to some embodiments, the second lateral cross-sectional area of the second photodiode **112d** is less than the first lateral cross-sectional area of the first photodiode **108f**. According to some embodiments, the second photodiode **112d** has a second lateral cross-sectional shape approximating the shape of the second isolation structure **124d**. For example, a layout of the second photodiode **112d** may have a portion approximating a diamond, or other second lateral cross-sectional shape set forth above, with variations due to fabrication resulting in a portion approximating a circle or an ellipse. The second photodiode **112d** may transfer charge to a floating diffusion, as described in greater detail below. According to some embodiments, the floating diffusion is bounded by the second isolation structure **124d**. According to some embodiments, the floating diffusion has a portion approximating a diamond, or other second lateral cross-sectional shape set forth above, with variations due to fabrication resulting in the portion approximating a circle or an ellipse. Other structures and/or configurations of the second photodiode **112d** are within the scope of the present disclosure.

FIG. 1L illustrates the isolation structure arrangement **102** comprising the second subset of isolation structures **122**, according to some embodiments. Each of the second subset of isolation structures **122** laterally surrounds a photodiode from the second subset of photodiodes **110** and is formed in an array. The second subset of isolation structures **122**, including a plurality of second isolation structures, is adjacent to the first subset of isolation structures **114**, including the plurality of first isolation structures. For example, the second isolation structure **124d** is adjacent to the first isolation structures **116e**, **116f**, **116i**, and **116j**, and the second isolation structure **124e** is adjacent to the first isolation structures **116f** and **116j**. According to some embodiments, each of the second isolation structures **124d** and **124e** includes a plurality of sections. For example, the second isolation structure **124d** includes sections **124d1**-**124d4** and the second isolation structure **124e** includes sections **124e1**-**124e4**. According to some embodiments, each of the first isolation structures **116e**, **116f**, **116i**, and **116j** includes a plurality of sections. The first isolation structure **116e** includes sections **116e1**-**116e8** (e.g., sections **116e3**, **116e4**, and **116e5**); the first isolation structure **116f** includes sections **116f1**-**116f8** (e.g., sections **116f4**, **116f5**, **116f6**, and **116f7**); the first isolation structure **116i** includes sections **116i1**-**116i8** (e.g., sections **116i1**, **116i2**, and **116i3**); and the first isolation structure **116j** includes sections **116j1**-**116j8** (e.g., sections **116j1**, **116j2**, **116j7**, and **116j8**). According to some embodiments, each of the second subset of isolation structures **122** (e.g., the second isolation structures **124d** and **124e**) is at least one of adjacent to or contiguous with at least one first isolation structure in the first subset of isolation structures **114** (e.g., first isolation structures **116e**, **116f**, **116i**, **116j**).

According to some embodiments, each of the second subset of isolation structures **122** includes a section that is the same as a section of one of the first subset of isolation structures **114**. For example, the second isolation structure

124e includes the section **124e4** that is the same as the section **116f4** of the first isolation structure **116f**. In another example, the second isolation structure **124e** includes the section **124e3** that is the same as the section **116j2** of the first isolation structure **116j**. According to some embodiments, each of the second subset of isolation structures **122** includes a section that is the same as at least one of a section, a portion of a section, or a plurality of portions of sections of a first isolation structure. For example, the second isolation structure **124d** includes the section **124d1** that is at least one of the same as the section **116f6**, the same as a portion of the section **116f7**, or the same as a portion of the section **116f5**. The second isolation structure **124d** includes the section **124d2** that is at least one of the same as the section **116f8**, the same as a portion of the section **116f1**, or the same as a portion of the section **116f7**. The second isolation structure **124d** includes the section **124d3** that is at least one of the same as the section **116i2**, the same as a portion of the section **116i1**, or the same as a portion of the section **116i3**. Likewise, the second isolation structure **124d** includes the section **124d4** that is at least one of the same as the section **116e4**, the same as a portion of the section **116e3**, or the same as a portion of the section **116e5**. Other structures and/or configurations of the plurality of second isolation structures **124d** and **124e** are within the scope of the present disclosure.

FIG. 1M illustrates the isolation structure arrangement **102** comprising a second subset of isolation structures **125**, according to some embodiments. The second subset of isolation structures **125** may replace the second subset of isolation structures **122** in FIG. 1L. Each of the second subset of isolation structures **125** laterally surrounds a photodiode from the second subset of photodiodes **110** and is formed in an array. The second subset of isolation structures **125** comprises a plurality of second isolation structures (e.g., second isolation structure **126**). According to some embodiments, second isolation structure **126** may be formed in an array such as the array of the second subset of isolation structures **122** of FIG. 1L. The second subset of isolation structures **125** laterally surrounds second photodiodes (e.g., the second photodiodes **112a**-**112i** of FIG. 10). According to some embodiments, the second isolation structure **126** includes a plurality of sections **126d1**, **126d2**, **126d3**, **126d4**, similar to sections of the second isolation structure **124d** of FIG. 1J. In the example illustrated in FIG. 1M, the second isolation structure **126** laterally surrounds a corresponding second photodiode (e.g., the second photodiodes **112a**-**112i** illustrated in FIG. 10) of the second subset of photodiodes **110**. According to some embodiments, diagonally adjacent photodiodes (e.g., the second photodiodes **112a**-**112i** illustrated in FIG. 10 that are diagonally adjacent to the first photodiodes **108a**-**108p** illustrated in FIG. 1D) are diagonally separated by at least two isolation structures. For example, a second photodiode disposed within the second isolation structure **126** is diagonally adjacent to a first photodiode disposed within the first isolation structure **116e**, and is separated by the second isolation structure **126** and the first isolation structure **116e**. Other structures and/or configurations of the second subset of the isolation structures **125** are within the scope of the present disclosure.

According to some embodiments, the second isolation structure **126** is at least one of adjacent to, contiguous with, overlapping, or overlapping a portion of at least one first isolation structure. For example, the second isolation structure **126** may include a section **126d1** that is adjacent to and contiguous with a section **116f6** of the first isolation structure **116f**. According to some embodiments, the second

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isolation structure **126** is at least one of adjacent to, contiguous with, overlapping, or overlapping a portion of at least two first isolation structures (e.g., two of the first isolation structures **116e**, **116f**, **116i**, **116j**). For example, the second isolation structure **126** may also include a section **126d2** that is adjacent to and contiguous with a section **116j8** of the first isolation structure **116j**. According to some embodiments, the second isolation structure **126** is at least one of adjacent to, contiguous with, overlapping, or overlapping a portion of at least three first isolation structures (e.g., three of the first isolation structures **116e**, **116f**, **116i**, **116j**). For example, the second isolation structure **126** may also include a section **126d3** that is adjacent to and contiguous with a section **116i2** of the first isolation structure **116i**. According to some examples, the second isolation structure **126** is at least one of adjacent to, contiguous with, overlapping, or overlapping a portion of at least four first isolation structures (e.g., four of the first isolation structures **116e**, **116f**, **116i**, **116j**). For example, the second isolation structure **126** may also include a section **126d4** that is adjacent to and contiguous with a section **116e4** of the first isolation structure **116e**. Other structures and/or configurations of the second isolation structure **126** are within the scope of the present disclosure.

According to some embodiments, the first isolation structures **116a-116p** of the first subset of isolation structures **114** are larger than the second isolation structures **124a-124i** of the second subset of isolation structures **122**. According to some embodiments, the first isolation structures **116a-116p** have at least one of a larger lateral cross-sectional diameter or a larger cross-sectional area. According to some embodiments, the first isolation structures **116a-116p** have a different lateral cross-sectional shape than the second isolation structures **124a-124i**. According to some embodiments, each of the first photodiodes **108a-108p** of the first subset of photodiodes **106** are larger than each of the second photodiodes **112a-112i** of the second subset of photodiodes **110**. According to some embodiments, each of the first photodiodes **108a-108p** has at least one of a larger lateral cross-sectional diameter or a larger cross-sectional area than each of the second photodiodes **112a-112i**. According to some embodiments, each of the first photodiodes **108a-108p** has a different lateral cross-sectional shape than each of the second photodiodes **112a-112i**.

According to some embodiments, the first subset of photodiodes **106** and the second subset of photodiodes **110** are image sensors, such as at least one of optical image sensors, proximity image sensors, motion image sensors, infrared image sensors, or near-infrared (NIR) image sensors. NIR image sensors may be used for security, personal authentication, range finding applications, to enhance a color optical image, etc. Optical image sensors may use an array of photodiodes to detect an optical image in ranges of wavelengths of color optical radiation with a color filter pattern, for example. The color filter pattern may be, for example, a Bayer filter pattern of red-green-blue (RGB), with a 2x2 color unit cell of two green filters in the diagonal positions and blue and red in the off-diagonal positions. The color filter pattern may be, for example, a Bayer filter pattern of red-green-blue-white (RGBW), with a 2x2 color unit cell of one green filter and one white filter in the diagonal positions and blue and red in the off-diagonal positions. NIR image sensors may use an array of photodiodes to detect an NIR image in a range of wavelengths of NIR radiation. The image detected by the NIR image sensors may be used to digitally enhance the detected optical image, for example.

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According to some embodiments, the semiconductor arrangement **100** may be used to detect an optical image and a corresponding NIR image. The first subset of photodiodes **106** may be laid out with a color filter pattern to detect an optical image and the second subset of photodiodes **110** may be laid out with an NIR filter to detect a corresponding NIR image. Alternatively, the first subset of photodiodes **106** may be laid out with an NIR filter pattern to detect an NIR image and the second subset of photodiodes **110** may be laid out with a color filter pattern to detect a corresponding optical image.

According to some embodiments, the first subset of photodiodes **106** may be resized with respect to the second subset of photodiodes **110**. For example, the first subset of photodiodes **106** may be resized to capture an optical image with greater quality than an NIR image captured by the second subset of photodiodes **110** when the NIR image is used to enhance the optical image. The first subset of photodiodes **106** may be resized corresponding to a degree of quality with respect to the NIR image. For example, the first subset of photodiodes **106** may be resized to have a larger lateral cross-sectional diameter than the second subset of photodiodes **110**. The first subset of photodiodes **106** may be resized to have a lateral cross-sectional diameter that is 3 times larger than the second subset of photodiodes **110**. The first subset of photodiodes **106** may be resized to have a lateral cross-sectional diameter that is 2.5 times larger than the second subset of photodiodes **110**. The first subset of photodiodes **106** may be resized to have a lateral cross-sectional diameter that is 2 times larger than the second subset of photodiodes **110**. The first subset of photodiodes **106** may be resized to have a lateral cross-sectional diameter that is 1.5 times larger than the second subset of photodiodes **110**. Other resizing of the lateral cross-sectional diameter of first subset of photodiodes **106** with respect to the second subset of photodiodes **110** is within the scope of the present disclosure.

The first subset of photodiodes **106** may be resized to have a larger lateral cross-sectional area or a larger lateral cross-sectional shape than the second subset of photodiodes **110**. For example, the first subset of photodiodes **106** may be resized to have a lateral cross-sectional area or a larger lateral cross-sectional shape that is 3 times larger than the second subset of photodiodes **110**. The first subset of photodiodes **106** may be resized to have a lateral cross-sectional area or a larger lateral cross-sectional shape that is 2.5 times larger than the second subset of photodiodes **110**. The first subset of photodiodes **106** may be resized to have a lateral cross-sectional area or a larger lateral cross-sectional shape that is 2 times larger than the second subset of photodiodes **110**. The first subset of photodiodes **106** may be resized to have a lateral cross-sectional area or a larger lateral cross-sectional shape that is 1.5 times larger than the second subset of photodiodes **110**. Other resizing of the lateral cross-sectional area or the larger lateral cross-sectional shape of first subset of photodiodes **106** with respect to the second subset of photodiodes **110** is within the scope of the present disclosure.

According to some embodiments, the first subset of photodiodes **106** may be resized with respect to the first subset of isolation structures **114** and the second subset of photodiodes **110** may be resized with respect to the second subset of isolation structures **122**. The first subset of isolation structures **114** may be resized to have a larger lateral cross-sectional diameter than the second subset of isolation structures **122**. For example, the first subset of isolation structures **114** may be resized to have a lateral cross-

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sectional diameter that is 3 times larger than the second subset of isolation structures 122. The first subset of isolation structures 114 may be resized to have a lateral cross-sectional diameter that is 2.5 times larger than the second subset of isolation structures 122. The first subset of isolation structures 114 may be resized to have a lateral cross-sectional diameter that is 2 times larger than the second subset of isolation structures 122. The first subset of isolation structures 114 may be resized to have a lateral cross-sectional diameter that is 1.5 times larger than the second subset of isolation structures 122. Other resizing of the lateral cross-sectional diameter of first subset of isolation structures 114 with respect to the second subset of isolation structures 122 is within the scope of the present disclosure.

The first subset of isolation structures 114 may be resized to have at least one of a larger lateral cross-sectional area or a larger lateral cross-sectional shape than the second subset of isolation structures 122. For example, the first subset of isolation structures 114 may be resized to have at least one of a lateral cross-sectional area or a larger lateral cross-sectional shape that is 3 times larger than the second subset of isolation structures 122. The first subset of isolation structures 114 may be resized to have at least one of a lateral cross-sectional area or a larger lateral cross-sectional shape that is 2.5 times larger than the second subset of isolation structures 122. The first subset of isolation structures 114 may be resized to have at least one of a lateral cross-sectional area or a larger lateral cross-sectional shape that is 2 times larger than the second subset of isolation structures 122. The first subset of isolation structures 114 may be resized to have at least one of a lateral cross-sectional area or a larger lateral cross-sectional shape that is 1.5 times larger than the second subset of isolation structures 122. Other resizing of the lateral cross-sectional area or the larger lateral cross-sectional shape of first subset of isolation structures 114 with respect to the second subset of isolation structures 122 is within the scope of the present disclosure.

According to some embodiments, the first subset of isolation structures 114 may be resized with an octagon shape and the second subset of isolation structures 122 may be resized with a diamond shape. In this configuration, the first subset of isolation structures 114 may form an array and the second subset of isolation structures 122 may form an array. According to some embodiments, each of the first subset of isolation structures 114 may be at least one of contiguous with, overlapping, or overlapping a portion of four adjacent second isolation structures (e.g., the second isolation structures 124a-124i) of the second subset of isolation structures 122. According to some embodiments, each of the second subset of isolation structures 122 may be at least one of contiguous with, overlapping, or overlapping a portion of four adjacent first isolation structures (e.g., the first isolation structures 116a-116p) of the first subset of isolation structures 114. Other structures and/or configurations of the first subset of isolation structures 114 with respect to the second subset of isolation structures 122 are within the scope of the present disclosure.

FIG. 2 illustrates the semiconductor arrangement 100 including the photodiode array 104, according to some embodiments. The photodiode array 104 comprises a substrate 200 and photodiodes 202a-202e over the substrate 200. The photodiode array 104 is configured to sense radiation, such as incident light, which is projected towards the substrate 200 along a direction of projected radiation 201. The photodiodes (e.g. the photodiodes 202a-202e) of the photodiode array 104 are separated by the isolation structure arrangement 102. An ARC layer 212 (i.e., an

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antireflective coating layer) is arranged over the photodiodes 202a-202e and over the substrate 200. A radiation filter layer 214 is arranged over the ARC layer 212, over the photodiodes 202a-202e, and over the substrate 200. The radiation filter layer 214 comprises a plurality of radiation pass filters 215, each configured to allow a radiation pass range of wavelengths to pass through and be detected by the corresponding photodiode. The plurality of radiation pass filters 215 comprises an NIR pass filter 218 and a plurality of color pass filters. The plurality of color pass filters comprises a red pass filter 220, a green pass filter 222, a blue pass filter 224, and a white pass filter 226. The NIR pass filter 218 is configured to allow an NIR range of wavelengths to pass through and be detected by the photodiode 202a. The color pass filters are configured to allow a color range of wavelengths to pass through and be detected by a corresponding photodiode. The red pass filter 220 is configured to allow a red range of wavelengths to pass through and be detected by the photodiode 202b. The green pass filter 222 is configured to allow a green range of wavelengths to pass through and be detected by the photodiode 202c. The blue pass filter 224 is configured to allow a blue range of wavelengths to pass through and be detected by the photodiode 202d. The white pass filter 226 is configured to allow a white range of wavelengths to pass through and be detected by the photodiode 202e. Other arrangements and/or configurations of the radiation filter layer 214 or the plurality of the radiation pass filters 215 are within the scope of the present disclosure.

A micro-lens array 216 is arranged over the ARC layer 212 and the radiation filter layer 214, according to some embodiments. The micro-lens array 216 is arranged to steer radiation towards the photodiode array 104. The micro-lens array 216 includes a plurality of micro-lenses (e.g. micro-lenses 216a-216e) configured to transmit radiation to a corresponding photodiode (e.g., the photodiodes 202a-202e) of the photodiode array 104. A micro-lens 216a is configured to transmit radiation through the NIR pass filter 218 to the photodiode 202a. A micro-lens 216b is configured to transmit radiation through the red pass filter 220 to the photodiode 202b. A micro-lens 216c is configured to transmit radiation through the green pass filter 222 to the photodiode 202c. A micro-lens 216d is configured to transmit radiation through the blue pass filter 224 to the photodiode 202d. A micro-lens 216e is configured to transmit radiation through the white pass filter 226 to the photodiode 202e. Other arrangements and/or configurations of the micro-lens array 216 are within the scope of the present disclosure.

FIG. 3 illustrates a graph 300 of relative responsivity of the photodiode array 104 with respect to wavelengths of detected radiation, according to some embodiments. Relative responsivity, also known as spectral responsivity, may be expressed as a ratio of generated photocurrent to incident radiation power, expressed in Amps (A)/Watts (W). Relative responsivity may also be known as wavelength-dependence, and may be expressed as quantum efficiency or a ratio of a number of photo-generated carriers to incident photons, which is a dimensionless quantity. Graph 300 is normalized to clear at a wavelength of 256 nanometers (nm) and a temperature (T) of 25 degrees Celsius (° C.). According to some embodiments, optical radiation, also known as visible radiation, may be narrowly set forth from about 420 nm to about 680 nm and may be broadly set forth from about 380 nm to about 800 nm. White radiation is generally a combination of all optical radiation. According to some embodiments, optical radiation comprises red radiation (e.g., generally a red range of wavelengths from about 620 nm to about 700 nm), green radiation (e.g., generally a green range

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of wavelengths from about 492 nm to about 577 nm), blue radiation (e.g., generally a blue range of wavelengths from about 455 nm to about 492 nm), and white radiation (e.g., generally a white range of wavelengths from about 380 nm to about 800 nm). According to some embodiments, NIR

radiation may be set forth as an NIR range of wavelengths from about 750 nm to about 1400 nm or may be set forth as an NIR range of wavelengths from about 780 nm to about 2500 nm. Graph 300 illustrates a relative responsivity of blue radiation over 0.4 for wavelengths between about 450 nm to about 500 nm and wavelengths between about 780 nm to about 950 nm. A relative responsivity of green radiation over 0.4 is provided for wavelengths between about 750 nm to about 950 nm. A relative responsivity of red radiation over 0.4 is provided for wavelengths between about 550 nm to about 950 nm. A relative responsivity of NIR radiation over 0.4 is provided for NIR wavelengths between about 400 nm to about 950 nm. Accordingly, with detection of NIR wavelengths in addition to detection of optical wavelengths, the photodiode array 104 may detect a greater number of photo-generated carriers in response to incident photons than detection of optical wavelengths alone. The detected NIR radiation may be at least one of translated into a digital image for display or used to enhance a detected optical image.

FIGS. 4A-4D illustrate photodiode array 104, according to some embodiments. FIG. 4A illustrates a first subset of photodiodes 400 comprising a plurality of color photodiodes and a second subset of photodiodes 410 comprising a plurality of NIR photodiodes. According to some embodiments, the first subset of photodiodes 400 has a color filter pattern according to a Bayer filter pattern of RGB. The first subset of photodiodes 400 comprises a red photodiode 402 configured to detect the red range of wavelengths, green photodiodes 404 and 406 configured to detect the green range of wavelengths, and a blue photodiode 408 configured to detect the blue range of wavelengths. According to some embodiments, the red photodiode 402 corresponds to the photodiode 202b that is configured to receive the red range of wavelengths from the red pass filter 220, the green photodiodes 404 and 406 correspond to the photodiode 202c that is configured to receive the green range of wavelengths from the green pass filter 222, and the blue photodiode 408 corresponds to the photodiode 202d that is configured to receive the blue range of wavelengths from the blue pass filter 224. Together, the red photodiode 402, the green photodiodes 404 and 406, and the blue photodiode 408 form a 2x2 color unit cell of two green filters in the diagonal positions and blue and red filters in the off-diagonal positions. According to some embodiments, the red photodiode 402, the green photodiodes 404 and 406, and the blue photodiode 408 are formed in an array that repeats through the first subset of photodiodes 400. According to some embodiments, each NIR photodiode of the second subset of photodiodes 410 corresponds to the photodiode 202a that is configured to detect and receive the NIR range of wavelengths from the NIR pass filter 218. According to some embodiments, each NIR photodiode of the second subset of photodiodes 410 is formed in an array that repeats through the second subset of photodiodes 410. Other arrangements and/or configurations of the first subset of photodiodes 400 and the second subset of photodiodes 410 are within the scope of the present disclosure.

FIG. 4B illustrates a first subset of photodiodes 412 comprising a plurality of color photodiodes and the second subset of photodiodes 410 comprising the plurality of NIR

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photodiodes. According to some embodiments, the first subset of photodiodes 412 has a color filter pattern according to a Bayer filter pattern of RGBW. The first subset of photodiodes 412 comprises the red photodiode 402 configured to detect the red range of wavelengths, the green photodiode 404 configured to detect the green range of wavelengths, the blue photodiode 408 configured to detect the blue range of wavelengths, and a white photodiode 414 configured to detect the white range of wavelengths. According to some embodiments, the white photodiode 414 corresponds to the photodiode 202e that is configured to receive the white range of wavelengths from the white pass filter 226. According to some embodiments, the red photodiode 402, the green photodiode 404, the blue photodiode 408, and the white photodiode 414 are formed in an array that repeats through the first subset of photodiodes 400. Other arrangements and/or configurations of the first subset of photodiodes 412 and the second subset of photodiodes 410 are within the scope of the present disclosure.

FIG. 4C illustrates a first subset of photodiodes 416 comprising a plurality of NIR photodiodes and a second subset of photodiodes 418 comprising a plurality of color photodiodes. According to some embodiments, each NIR photodiode of the first subset of photodiodes 416 corresponds to the photodiode 202a that is configured to detect and receive the NIR range of wavelengths from the NIR pass filter 218. According to some embodiments, each NIR photodiode of the first subset of photodiodes 416 is formed in an array that repeats through the first subset of photodiodes 416. According to some embodiments, the second subset of photodiodes 418 comprises a color filter pattern according to the Bayer filter pattern of RGB. The second subset of photodiodes 418 comprises a red photodiode 420 configured to detect the red range of wavelengths, green photodiodes 422 and 424 configured to detect the green range of wavelengths, and a blue photodiode 426 configured to detect the blue range of wavelengths. According to some embodiments, the red photodiode 420 corresponds to the photodiode 202b that is configured to receive the red range of wavelengths from the red pass filter 220, the green photodiodes 422 and 424 correspond to the photodiode 202c that is configured to receive the green range of wavelengths from the green pass filter 222, and the blue photodiode 426 corresponds to the photodiode 202d that is configured to receive the blue range of wavelengths from the blue pass filter 224. Together, the red photodiode 420, the green photodiodes 422 and 424, and the blue photodiode 426 form a 2x2 color unit cell of two green filters in the diagonal positions and blue and red filters in the off-diagonal positions. According to some embodiments, the red photodiode 420, the green photodiodes 422 and 424, and the blue photodiode 426 are formed in an array that repeats through the second subset of photodiodes 418. Other arrangements and/or configurations of the first subset of photodiodes 416 and the second subset of photodiodes 418 are within the scope of the present disclosure.

FIG. 4D illustrates the first subset of photodiodes 416 comprising the plurality of NIR photodiodes and a second subset of photodiodes 428 comprising a plurality of color photodiodes. According to some embodiments, the second subset of photodiodes 428 comprises a color filter pattern according to the Bayer filter pattern of RGBW. The second subset of photodiodes 428 comprises the red photodiode 420 configured to detect the red range of wavelengths, the green photodiode 422 configured to detect the green range of wavelengths, the blue photodiode 426 configured to detect the blue range of wavelengths, and a white photodiode 434

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configured to detect the white range of wavelengths. According to some embodiments, the white photodiode 434 corresponds to the photodiode 202e that is configured to receive the white range of wavelengths from the white pass filter 226. According to some embodiments, the red photodiode 420, the green photodiode 422, the blue photodiode 426, and the white photodiode 434 are formed in an array that repeats through the second subset of photodiodes 428. Other arrangements and/or configurations of the first subset of photodiodes 416 and the second subset of photodiodes 428 are within the scope of the present disclosure.

FIGS. 5A-5E schematically illustrate the semiconductor arrangement 100 including a photodiode 500 and the isolation structure arrangement 102, according to some embodiments. FIG. 5A illustrates the photodiode 500 formed over the substrate 200. The substrate 200 comprises a first side 504 and a second side 506 opposite to the first side 504. The substrate 200 has a substrate depth D_s . The semiconductor arrangement 100 is illustrated as inverted compared to the example illustrated in FIG. 3, such that the direction of projected radiation 201 is projected towards the second side 506 of the substrate 200. The photodiode 500 comprises a charge storage well 508, a pinning layer 510, a transfer gate 512, and a floating diffusion 514. The charge storage well 508 is arranged between the substrate 200 and the pinning layer 510.

According to some embodiments, the charge storage well 508 may extend from the first side 504 of the substrate 200 into the substrate 200 by a storage well depth $D1_{sw}$. The pinning layer 510 may be over twenty times thinner than the charge storage well 508. For example, in a 180 nm process, the pinning layer 510 may be about 100 nm thick, while the charge storage well 508 may be about 2,500-nm thick. According to some embodiments, the storage well depth $D1_{sw}$ of the charge storage well 508 may be considered the depth of photodiode 500 when compared to other components of semiconductor arrangement 100. The charge storage well 508 comprises a first deep storage well section 516. The first deep storage well section 516 is contiguous with and forms a part of charge storage well 508. According to an example, when the first deep storage well section 516 forms a part of charge storage well 508, the first deep storage well section 516 may extend into the first side 504 of the substrate 200 by a deep storage well depth $D2_{sw}$. The first deep storage well section 516 has a deep storage well width $W2_{sw}$. According to some embodiments, the deep storage well depth $D2_{sw}$ of the charge storage well 508 may be considered the depth of photodiode 500 when compared to other components of semiconductor arrangement 100.

According to some embodiments, the isolation structure arrangement 102 comprises an isolation structure 518 and a deep trench isolation 520. The isolation structure 518 laterally surrounds the photodiode 500. The isolation structure 518 comprises a first well 519, which is also known as a deep well. The first well 519 extends into the first side 504 of the substrate 200 by a deep well depth $D1_{dw}$ and has a deep well width $W1_{dw}$. According to some embodiments, the deep well depth $D1_{dw}$ is a maximum deep well depth and the deep well width $W1_{dw}$ is a maximum deep well width. According to some embodiments, the deep well depth $D1_{dw}$ of the first well 519 is greater than the storage well depth $D1_{sw}$ of the charge storage well 508 of the photodiode 500. According to some embodiments, the deep well depth $D1_{dw}$ of the first well 519 is greater than the deep storage well depth $D2_{sw}$ of the first deep storage well section 516 of the photodiode 500. Other arrangements and/or configura-

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tions of the deep well depth $D1_{dw}$ and the deep well width $W1_{dw}$ are within the scope of the present disclosure.

The deep trench isolation 520 surrounds the photodiode 500 and has a first side 522 and a second side 524. The deep trench isolation 520 extends into the substrate 200 from the second side 506 of the substrate 200 by a deep trench isolation depth $D1_{dti}$. The deep trench isolation 520 extends into the substrate 200 from the second side 506 of the substrate 200 by a deep trench isolation depth $D2_{dti}$ measured from the first side 504 of the substrate 200. Hence, when measured from the first side 504 of the substrate 200, the deep trench isolation 520 extends into the substrate 200 from the deep trench isolation depth $D2_{dti}$ to the deep trench isolation depth $D1_{dti}$. According to some embodiments, the second side 524 of the deep trench isolation 520 is the same as the second side 506 of the substrate 200. According to some embodiments, the second side 524 of the deep trench isolation 520 is not the same as the second side 506 of the substrate 200. For example, when measured from the first side 504 of the substrate 200, the second side 524 of the deep trench isolation 520 may be less than the second side 506 of the substrate 200. In other words, $D2_{dti} + D1_{dti} < D_s$. The deep trench isolation 520 extends from the first side 504 of the substrate 200 from deep trench isolation depth $D2_{dti}$ to deep trench isolation depth $D1_{dti}$. According to some embodiments, the deep trench isolation 520 is not tapered. According to some embodiments, the deep trench isolation 520 is tapered. For example, the deep trench isolation 520 may be tapered from a minimum width $W1_{dti}$ to a maximum width $W2_{dti}$ in a direction opposite to the direction of projected radiation 201. According to some embodiments and with respect to the first side 504 of the substrate 200, a portion of the deep trench isolation 520 underlies the first well 519. According to some embodiments and with respect to the first side 504 of the substrate 200, the deep trench isolation 520 completely underlies the first well 519. Other arrangements and/or configurations of the first well 519 and the deep trench isolation 520 are within the scope of the present disclosure.

According to some embodiments, the first well 519 laterally surrounds the photodiode 500 and at least a portion of the first well 519 laterally surrounds the deep trench isolation 520. A portion of the first well 519 laterally surrounds the deep trench isolation 520 such that an outer periphery of the portion of the first well 519 laterally surrounds an outer periphery of a portion of the deep trench isolation 520. According to some embodiments, the first well 519 overlaps a portion of the deep trench isolation 520. With respect to the first side 504 of the substrate 200, the deep trench isolation depth $D2_{dti}$ of the deep trench isolation 520 is less than the deep well depth $D1_{dw}$ of the first well 519. According to some embodiments, the deep trench isolation 520 has a maximum width $W2_{dti}$ and the isolation structure 518 has a maximum width of the deep well width $W1_{dw}$, and the maximum width $W2_{dti}$ is less than the deep well width $W1_{dw}$. Other arrangements and/or configurations of the first well 519 and the deep trench isolation 520 are within the scope of the present disclosure.

In some embodiments, the deep trench isolation depth $D1_{dti}$ of the deep trench isolation 520 is formed with dimensions compared to the substrate depth D_s of the substrate 200. According to an example, a greater deep trench isolation depth $D1_{dti}$ may provide at least one of greater electrical isolation or greater optical isolation of photodiode 500. According to an example, the deep trench isolation depth $D1_{dti}$ may be limited by vertical overlap in the direction of projected radiation 201 with a portion of

photodiode 500, such as floating diffusion 514. The deep trench isolation depth $D1dti$ may provide increased isolation of photodiode 500, depending upon a deep storage well depth $D2sw$ of a first deep storage well section 516 of the photodiode 500. According to an example, the deep storage well depth $D2sw$ may correspond to the substrate depth Ds of the substrate 200 and is described in greater detail below. In some embodiments, the deep trench isolation depth $D1dti$ is between about 45% and 80% of the substrate depth Ds of the substrate 200. In some embodiments, the deep trench isolation depth $D1dti$ is between about 50% and 75% of the substrate depth Ds of the substrate 200. In some embodiments, the deep trench isolation depth $D1dti$ is between about 60% and 70% of the substrate depth Ds of the substrate 200. In some embodiments, the deep storage well width $W2sw$ of the photodiode 500 is between 40% and 45% of the substrate depth Ds of substrate 200. In some embodiments, the deep storage well width $W2sw$ of the photodiode 500 is between 35% and 40% of the substrate depth Ds of substrate 200. Other arrangements and/or configurations of deep trench isolation 520 are within the scope of the present disclosure.

In some embodiments, the deep storage well width $W2sw$ of the photodiode 500 may be between about 2.0 micrometers (μm) and 2.5 μm , with a deep well width $W1dw$ of the first well 519 being between about 0.19 μm and 0.41 μm and a maximum width $W2dti$ of the deep trench isolation 520 being less than the deep well width $W1dw$. In some embodiments, the deep storage well width $W2sw$ of the photodiode 500 may be between about 2.1 μm and 2.4 μm , with a deep well width $W1dw$ of the first well 519 being between about 0.33 μm and 0.41 μm and a maximum width $W2dti$ of deep trench isolation 520 being less than deep well width $W1dw$. In some embodiments, the deep storage well width $W2sw$ of the photodiode may be between about 2.2 μm and 2.3 μm , with a deep well width $W1dw$ of the first well 519 being between about 0.35 μm and 0.39 μm and a maximum width $W2dti$ of the deep trench isolation 520 being less than the deep well width $W1dw$. Other arrangements and/or configurations of photodiode 500 and deep trench isolation 520 are within the scope of the present disclosure.

FIG. 5B illustrates the photodiode 500 formed over the substrate 200, according to some embodiments. The isolation structure 518 laterally surrounds the photodiode 500. The isolation structure 518 comprises a first well 526, which is also known as a deep well. The first well 526 extends into the first side 504 of the substrate 200 by a deep well depth $D2dw$ and has a deep well width $W1dw$. According to some embodiments, the deep well depth $D2dw$ is a maximum deep well depth and the deep well width $W1dw$ is a maximum deep well width. According to some embodiments, the deep well depth $D2dw$ of the first well 526 is greater than the deep storage well depth $D2sw$ of the first deep storage well section 516 of the photodiode 500. According to some embodiments, the deep well depth $D2dw$ of the first well 526 extends from the first side 504 of the substrate 200 to the second side 506 of the substrate 200. According to some embodiments, the deep well depth $D2dw$ of the first well 526 extends from the first side 504 of the substrate 200 to a depth greater than a depth of the deep trench isolation 520 measured from the first side 504 of the substrate 200. In other words, $D2dw > D2dti + D1dti$. According to some embodiments, the deep well width $W1dw$ of the first well 526 is greater than a minimum width $W1dti$ of deep trench isolation 520. According to some embodiments, the deep well width $W1dw$ of the first well 526 is greater than a maximum width $W2dti$ of the deep trench isolation 520. According to

some embodiments, the first well 526 completely surrounds the deep trench isolation 520. Other arrangements and/or configurations of the first well 526 and the deep trench isolation 520 are within the scope of the present disclosure.

FIG. 5C illustrates the photodiode 500 formed over the substrate 200, according to some embodiments. The isolation structure 518 laterally surrounds the photodiode 500 and comprises the first well 519 and a second well 528, which is also known as a deep well. The second well 528 extends into the substrate 200 from the second side 506 of the substrate 200 by a deep well depth $D3dw$ and has a deep well width $W2dw$. As measured from the first side 504 of the substrate 200, the second well 528 extends from a deep well depth $D4dw$ to the deep well depth $D3dw$. In this case, isolation structure 518 has a total depth from the first side 504 of the substrate 200 of $D4dw + D3dw$. According to some embodiments, the first well 519 overlaps the second well 528. According to some embodiments, the first well 519 and the second well 528 are contiguous and formed with the same dopant type. According to some embodiments, a combination of the first well 519 and the second well 528 completely surrounds the deep trench isolation 520. According to some embodiments, a combination of the first well 519 and the second well 528 extends from the first side 504 of the substrate 200 to the second side 506 of the substrate 200. According to some embodiments, when measured from the first side 504 of the substrate 200, the deep well depth $D4dw$ is greater than the deep storage well depth $D2sw$ of the first deep storage well section 516 of the photodiode 500. The deep well width $W1dw$ of the first well 519 is less than the deep well width $W2dw$ of the second well 528 and completely surrounds the deep trench isolation 520. Because the deep well width $W1dw$ of the first well 519 is less than the deep well width $W2dw$ of the second well 528, the photodiode 500 may be configured with a greater area. Other arrangements and/or configurations of the first well 519, the second well 528, and the deep trench isolation 520 are within the scope of the present disclosure.

FIG. 5D illustrates the photodiode 500 formed over the substrate 200, according to some embodiments. The isolation structure 518 laterally surrounds the photodiode 500 and comprises the first well 519. A deep trench isolation 530 surrounds the photodiode 500 and has a first side 532 and a second side 534. The deep trench isolation 530 extends into the substrate 200 from the second side 506 of the substrate 200 by a deep trench isolation depth $D3dti$. The deep trench isolation 530 extends into the substrate 200 from the second side 506 of the substrate 200 by a deep trench isolation depth $D4dti$ measured from the first side 504 of the substrate 200. Hence, when measured from the first side 504 of the substrate 200, the deep trench isolation 530 extends into the substrate 200 from the deep trench isolation depth $D4dti$ to the deep trench isolation depth $D3dti$. The isolation structure 518 and the first well 519 do not overlap the deep trench isolation 530. The deep well width $W1dw$ of the first well 519 is greater than a maximum width $W2dti$ of the deep trench isolation 530. Other arrangements and/or configurations of the isolation structure 518 and the deep trench isolation 530 are within the scope of the present disclosure.

FIG. 5E illustrates the photodiode 500 formed over the substrate 200, according to some embodiments. The isolation structure 518 laterally surrounds the photodiode 500 and comprises the first well 519 and the second well 528. The second well 528 extends into the substrate 200 from the second side 506 of the substrate 200 by the deep well depth $D3dw$ and has the deep well width $W2dw$. As measured from the first side 504 of the substrate 200, the second well 528

extends from the deep well depth $D4dw$ to the deep well depth $D3dw$. In this case, the isolation structure **518** has a total depth from the first side **504** of the substrate **200** of $D4dw+D3dw$. The charge storage well **508** comprises a first deep storage well section **516** and a second deep storage well section **536**. The first deep storage well section **516** and second deep storage well section **536** are contiguous with and form a part of charge storage well **508**. According to some embodiments, the second deep storage well section **536** extends into the substrate **200** by a deep storage well depth $D3sw$ from the first side **504** of the substrate **200**.

According to some embodiments, the second deep storage well depth $D3sw$ of charge storage well **508** may be considered the depth of the photodiode **500** when compared to other components of the semiconductor arrangement **100**. The first deep storage well section **516** has a deep storage well width $W2sw$ and the second deep storage well section **536** has a deep storage well width $W3sw$. According to some embodiments, the width $W2sw$ of first deep storage well section **516** is wider than the width $W3sw$ of second deep storage well section **536**. The first deep storage well section **516** is laterally surrounded by the first well **519**, and the second deep storage well section **536** is laterally surrounded by the second well **528**. When measured from the first side **504** of the substrate **200**, the deep storage well depth $D3sw$ of the second deep storage well section **536** is greater than the deep trench isolation depth $D2dti$ of the deep trench isolation **520**.

In some embodiments, the deep storage well width $W3sw$ of the photodiode **500** may be configured in relation to a deep well width $W2dw$ of the second well **528**. A greater deep storage well width $W3sw$ may provide a greater amount of material to receive charge carriers for photodiode **500** and a greater deep well width $W2dw$ may provide at least one of a greater electrical isolation or a greater optical isolation. The deep storage well width $W3sw$ and the deep well width $W2dw$ may also limit the density of photodiodes in the photodiode array **104**. According to some embodiments, a range of deep storage well widths $W3sw$ and a range of deep well widths $W2dw$ may provide charge carriers for detection by the photodiode **500** and at least one of an electrical isolation or an optical isolation. According to some embodiments, the deep storage well width $W3sw$ of the photodiode **500** may be between about 1.0 μm and 1.4 μm , with a deep well width $W2dw$ of the second well **528** being between about 1.0 μm and 1.4 μm and a maximum width $W2dti$ of the deep trench isolation **520** being less than deep well width $W2dw$. In some embodiments, the deep storage well width $W3sw$ of the photodiode **500** may be between about 1.1 μm and 1.3 μm , with a deep well width $W2dw$ of the second well **528** being between about 1.1 μm and 1.3 μm and a maximum width $W2dti$ of deep trench isolation **520** being less than deep well width $W2dw$. In some embodiments, the deep storage well width $W3sw$ of the photodiode **500** may be between about 1.15 μm and 1.25 μm , with a deep well width $W2dw$ of the second well **528** being between about 1.15 μm and 1.25 μm and a maximum width $W2dti$ of the deep trench isolation **520** being less than deep well width $W2dw$. Other arrangements and/or configurations of deep trench isolation **520** are within the scope of the present disclosure.

FIG. 6 illustrates the semiconductor arrangement **100** at a stage of fabrication, according to some embodiments. The semiconductor arrangement **100** comprises the substrate **200**. In some embodiments, the substrate **200** corresponds to a device wafer of the semiconductor arrangement **100**. The first side **504** of the substrate **200** corresponds to a front side

of the substrate **200** and the second side **506** of the substrate **200** corresponds to a back side of the substrate **200**. The semiconductor arrangement **100** is configured to sense radiation, such as incident light, which is projected towards the substrate **200** along the direction of projected radiation **201**.

The substrate **200** comprises at least one of an epitaxial layer, a silicon-on-insulator (SOI) structure, a wafer, or a die formed from a wafer. The substrate **200** comprises at least one of silicon, germanium, carbide, arsenide, gallium, arsenic, phosphide, indium, antimonide, SiGe, SiC, GaAs, GaN, GaP, InGaP, InP, InAs, InSb, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP, or other suitable material. The substrate **200** comprises at least one of monocrystalline silicon, crystalline silicon with a $\langle 100 \rangle$ crystallographic orientation, crystalline silicon with a $\langle 110 \rangle$ crystallographic orientation, crystalline silicon with a $\langle 111 \rangle$ crystallographic orientation, or other suitable material. The substrate **200** has at least one doped region. Other structures and/or configurations of the substrate **200** are within the scope of the present disclosure.

In some embodiments, the semiconductor arrangement **100** comprises at least one photodiode of the photodiode array **104** in the substrate **200**. In some embodiments, the photodiode **500** corresponds to at least one of the first photodiode **101**, the second photodiode **103**, the third photodiode **105**, or the fourth photodiode **107** of the photodiode array **104**. In some embodiments, the photodiode **500** corresponds to at least one photodiode of the first subset of photodiodes **106** (e.g., the first photodiodes **108a-108p**) or the second subset of photodiodes **110** (e.g., the second photodiodes **112a-112i**). The photodiode **500** is formed by at least one of doping, ion implantation, molecular diffusion, or other suitable techniques. In some embodiments, the photodiode **500** comprises at least one of a pinned layer photodiode, a phototransistor, a photogate, or other suitable component. At least some of the photodiodes of the photodiode array **104** can vary from one another to have at least one of different heights, thicknesses, widths, material compositions, etc. The substrate **200** may comprise any number of photodiodes or other components within the scope of the present disclosure.

At least some of the photodiodes of the photodiode array **104** or other components comprise at least one of boron, germanium, indium, phosphorous, BF_2 , arsenic, antimony, fluorine, InAs, InSb, GaSb, GaAs, InP, a silicide, silicon, doped silicon, or other suitable material. The photodiode array **104** is configured to sense radiation, such as incident light, which is projected towards the substrate **200** along the direction of projected radiation **201**. At least some of the photodiodes of the photodiode array **104** comprise a material that is relatively highly absorptive to NIR wavelengths or optical wavelengths. Other structures and/or configurations of the photodiode **500**, the photodiode array **104**, or other components, are within the scope of the present disclosure.

Photodiode **500** comprises the charge storage well **508**, arranged between the substrate **200** and the pinning layer **510**, and the floating diffusion **514**. According to some embodiments, the photodiode **500** comprises the first deep storage well section **516** and the second deep storage well section **536** are contiguous with and form a part of the charge storage well **508**. Other structures and/or configurations of the first deep storage well section **516** and the second deep storage well section **536** are within the scope of the present disclosure.

In some embodiments, the substrate **200** comprises an epitaxial layer. In some embodiments, the substrate **200** comprises a first-type epitaxial layer with a first-type dopant,

which is at least one of a p-type or an n-type. A second-type dopant is the other dopant, of the p-type or the n-type, from the first-type dopant. In some embodiments, the charge storage well **508**, the floating diffusion **514**, the first deep storage well section **516**, and the second deep storage well section **536** comprise the second-type dopant. In some embodiments, the floating diffusion **514** comprises the second-type dopant in a higher concentration than the charge storage well **508**, the first deep storage well section **516**, and the second deep storage well section **536**. The substrate **200** is formed with the first-type dopant to form the at least one doped region by at least one of ion implantation, molecular diffusion, or other suitable techniques. A number or an amount of dopants into the substrate **200** is controlled, such as to control a concentration of dopants. In some embodiments, an energy of dopants implanted into the substrate **200** is controlled, such as to control a depth to which dopants are implanted. In some embodiments, a depth of dopants in the substrate **200** is controlled by increasing or decreasing a voltage used to direct the dopants into the substrate **200**. In some embodiments, a depth of dopants in the substrate **200** is controlled by increasing or decreasing a time of exposure of dopants to the substrate **200**. In some embodiments, a depth of dopants in the substrate **200** is controlled by increasing or decreasing a concentration of dopants exposed to the substrate **200**. Other processes and/or procedures for controlling a depth of dopants in the substrate **200** are within the scope of the present disclosure.

In some embodiments, the charge storage well **508** of the photodiode **500** comprises a second type dopant, which is the other type of the substrate **200**. The charge storage well **508** forms a p-n junction with the substrate **200**. The charge storage well **508** is also known as a buried signal charge storage well. The floating diffusion **514** comprises the second type dopant and has a higher concentration of dopants than charge storage well **508**. The floating diffusion **514** forms a p-n junction with the substrate **200**. The substrate **200** may comprise a more heavily doped region under floating diffusion **514** to repel photoelectrons so they may be collected by the charge storage well **508**. The pinning layer **510** comprises the first-type dopant and has a higher concentration than the substrate **200**. Other configurations and/or compositions of the substrate **200** and the photodiode **500** are within the scope of the present disclosure.

FIG. 7 illustrates the semiconductor arrangement **100** at a stage of fabrication with a first mask layer **702** formed over the substrate **200**, according to some embodiments. The first mask layer **702** at least one of overlies the substrate **200**, is in direct contact with the substrate **200**, or is in indirect contact with the substrate **200**. In some embodiments, the first mask layer **702** is a hard mask layer. The first mask layer **702** comprises at least one of an oxide, a nitride, a metal, or other suitable material. The first mask layer **702** is formed by at least one of physical vapor deposition (PVD), sputtering, chemical vapor deposition (CVD), low pressure CVD (LPCVD), atomic layer chemical vapor deposition (ALCVD), ultrahigh vacuum CVD (UHVCVD), reduced pressure CVD (RPCVD), atomic layer deposition (ALD), molecular beam epitaxy (MBE), liquid phase epitaxy (LPE), spin on, growth, or other suitable techniques. Other structures and/or configurations of the first mask layer **702** are within the scope of the present disclosure.

FIG. 8 illustrates the semiconductor arrangement **100** at a stage of fabrication with the first mask layer **702** patterned to form a first patterned mask layer **802** over the substrate **200**, according to some embodiments. According to some embodiments, a photoresist (not shown) is used to form the

first patterned mask layer **802**. The photoresist is formed over the first mask layer **702** by at least one of PVD, sputtering, CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. The photoresist comprises a light-sensitive material, where properties, such as solubility, of the photoresist are affected by light. The photoresist is a negative photoresist or a positive photoresist. With respect to a negative photoresist, regions of the negative photoresist become insoluble when illuminated by a light source, such that application of a solvent to the negative photoresist during a subsequent development stage removes non-illuminated regions of the negative photoresist. A pattern formed in the negative photoresist is thus a negative image of a pattern defined by opaque regions of a template, such as a mask, between the light source and the negative photoresist. In a positive photoresist, illuminated regions of the positive photoresist become soluble and are removed via application of a solvent during development. Thus, a pattern formed in the positive photoresist is a positive image of opaque regions of the template, such as a mask, between the light source and the positive photoresist. One or more etchants have a selectivity such that the one or more etchants remove or etch away one or more layers exposed or not covered by the photoresist at a greater rate than the one or more etchants remove or etch away the photoresist. Accordingly, an opening in the photoresist allows the one or more etchants to form a corresponding opening in the one or more layers under the photoresist, and thereby transfer a pattern in the photoresist to the one or more layers under the photoresist. The photoresist is stripped or washed away after the pattern transfer.

An etching process used to remove portions of the first mask layer **702** to form the first patterned mask layer **802** is at least one of a dry etching process, a wet etching process, an anisotropic etching process, an isotropic etching process, a plasma etching process, or another suitable etching process. The etching process uses at least one of HF, diluted HF, HCl₂, H₂S, or other suitable material. In some embodiments, the etching process performed to remove portions of the first mask layer **702** and form the first patterned mask layer **802** also removes at least some of the substrate **200**, such as portions of the substrate **200** underlying openings in the first patterned mask layer **802**. Other processes and/or techniques for forming the first patterned mask layer **802** are within the scope of the present disclosure.

FIG. 9 illustrates the semiconductor arrangement **100** at a stage of fabrication with the isolation structure **518** comprising the first well **519**, according to some embodiments. The first well **519** is formed in the substrate **200**. In some embodiments, the first well **519** is formed via a deep implantation **902** and comprises the first-type dopant. In some embodiments, the first well **519** comprises the first-type dopant in a higher concentration than the first-type dopant in the substrate **200**. In some embodiments, the deep implantation **902** comprises implanting a doping between about $1.5 \times 10^{15} \text{ cm}^{-3}$ to about $1.5 \times 10^{17} \text{ cm}^{-3}$ of a first dopant at an energy of between about 1.5 kiloelectron volts (keV) to about 2.5 keV. In some embodiments, the first dopant comprises at least one of nitrogen, phosphorus, or arsenic. In some embodiments, the first dopant comprises at least one of boron, aluminum, or gallium. In some embodiments, a first dopant of boron provides a junction depth between 2.0 μm and 3.0 μm with a depletion region depth greater than 2.5 μm . In some embodiments, a first dopant of boron provides a junction depth between 2.1 μm and 2.4 μm with a depletion region greater than 2.5 μm . In some embodiments, a depletion region greater than 2.5 μm provides a deep well depth

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D1_{dw} of the first well 519 greater than a deep storage well depth D2_{sw} of the first deep storage well section 516 of photodiode 500. In some embodiments, the first well 519 provides one of a reduction or an elimination of electrical cross-talk, where an electrical signal is produced from incident radiation that passes between adjacent photodiodes. Other processes and/or techniques for forming the first well 519 are within the scope of the present disclosure.

FIG. 10 illustrates the semiconductor arrangement 100 at a stage of fabrication with the isolation structure 518 comprising the first well 526, according to some embodiments. The first well 526 is formed in the substrate 200. The first well 526 is deeper than the first well 519 of FIG. 9 (illustrated herein for reference). In some embodiments, the first well 526 is formed via deep implantation 902 comprising the first-type dopant. In some embodiments, the first well 526 comprises the first-type dopant in a higher concentration than the substrate 200. In some embodiments, the deep implantation 902 comprises implanting a doping between about $1.5 \times 10^{15} \text{ cm}^{-3}$ to about $1.5 \times 10^{17} \text{ cm}^{-3}$ of a first dopant with a resistivity of about 10 ohm-cm at an energy of between about 3.0 keV to about 5.0 keV. In some embodiments, the first dopant comprises at least one of nitrogen, phosphorus, or arsenic. In some embodiments, the first dopant comprises at least one of boron, aluminum, or gallium. In some embodiments, a first dopant of boron provides a junction depth between 5.0 μm and 6.0 μm with a depletion region between 5.0 μm and 6.0 μm . In some embodiments, a depletion region greater than 5.0 μm provides a deep well depth D1_{dw} of the first well 526 greater than a deep storage well depth D2_{sw} of the first deep storage well section 516 of the photodiode 500. In some embodiments, the deep implantation 902 comprises implanting a doping between about $1.0 \times 10^{14} \text{ cm}^{-3}$ with a resistivity of about 100 ohm-cm at an energy of between about 3.0 keV to about 5.0 keV. In some embodiments, a first dopant of boron provides a junction depth greater than 6.0 μm with a depletion region greater than 6.0 μm . In some embodiments, a depletion region greater than 6.0 μm provides a deep well depth D1_{dw} of the first well 526 greater than a deep storage well depth D2_{sw} of the first deep storage well section 516 of the photodiode 500. In some embodiments, the first well 526 provides one of a reduction or an elimination of electrical cross-talk, where an electrical signal is produced from incident radiation that passes between adjacent photodiodes. Other processes and/or techniques for forming the first well 526 are within the scope of the present disclosure.

FIG. 11 illustrates the semiconductor arrangement 100 at a stage of fabrication after removal of the first patterned mask layer 802 and illustrated with a second mask layer 1102 formed over the substrate 200, according to some embodiments. The first patterned mask layer 802 is removed after at least one of the first well 519 or the first well 526 is formed. The first patterned mask layer 802 is removed by at least one of wafer back-grinding, chemical-mechanical polishing (CMP), planarization, a dry etching process, a wet etching process (i.e., a liquid etching process), an anisotropic etching process, an isotropic etching process, plasma etching, or other suitable techniques or processes. The etching process uses at least one of HF, diluted HF, HCl₂, H₂S, or other suitable material. In some embodiments, removal of the first patterned mask layer 802 exposes the first side 504 (e.g., a top surface) of the substrate 200.

The second mask layer 1102 at least one of overlies the substrate 200, is in direct contact with the substrate 200, or is in indirect contact with the substrate 200. An intermediate layer, such as at least one of a protection layer or a dielectric

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layer may be formed between the second mask layer 1102 and the substrate 200. The second mask layer 1102 is formed by at least one of PVD, sputtering, CVD, LPCVD, ALCVD, UHV-CVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. The second mask layer 1102 may be formed using a process akin to a process used for formation of the first mask layer 702 set forth above with reference to FIG. 7. A photoresist comprising a light-sensitive material, where properties, such as solubility, of the photoresist are affected by light may be used to form the second mask layer 1102. The photoresist may be a negative photoresist or a positive photoresist.

FIG. 12 illustrates the semiconductor arrangement 100 at a stage of fabrication with the second mask layer 1102 patterned to form a second patterned mask layer 1202 over the substrate 200, according to some embodiments. The second patterned mask layer 1202 may be formed using a process akin to a process used for formation of the first patterned mask layer 802 set forth above with reference to FIG. 8. The second patterned mask layer 1202 has openings exposing portions of the substrate 200. In some embodiments, the openings in the second patterned mask layer 1202 are between photodiodes of the photodiode array 104 or other components, such that the openings do not overlie or are laterally offset from the photodiodes or other components. In some embodiments, an opening in the second patterned mask layer 1202 is between two adjacent photodiodes of the photodiode array 104 or other components, such that the opening overlies a portion of the substrate 200 between a first photodiode and a second photodiode of the photodiode array 104. According to some embodiments, an opening in the second patterned mask layer 1202 overlies a portion of a photodiode 500, a portion of a photodiodes of the photodiode array 104, or other component.

FIG. 13 illustrates the semiconductor arrangement 100 at a stage of fabrication with a shallow trench 1302 formed using the second patterned mask layer 1202, according to some embodiments. The shallow trench 1302 extends through the substrate 200 and into the first wells 519. The shallow trench 1302 is at least one of laterally offset from photodiode 500 or between two photodiodes of the photodiode array 104. In some embodiments, the shallow trench 1302 is between two adjacent photodiodes of the photodiode array 104, a first portion 200a of the substrate 200 separates the shallow trench 1302 from a first photodiode of two adjacent photodiodes of the photodiode array 104, and a second portion 200b of the substrate 200 separates the shallow trench 1302 from a second photodiode of the two adjacent photodiodes of the photodiode array 104. In some embodiments, an etching process is performed to form the shallow trench 1302, where openings in the second patterned mask layer 1202 allow one or more etchants applied during the etching process to remove portions of the substrate 200 while the second patterned mask layer 1202 protects or shields portions of the substrate 200 that are covered by the second patterned mask layer 1202. The etching process is at least one of a dry etching process, a wet etching process, an anisotropic etching process, an isotropic etching process, or another suitable etching process. The etching process uses at least one of HF, diluted HF, HCl₂, H₂S, or other suitable material. Other processes and/or techniques for forming the shallow trench 1302 are within the scope of the present disclosure.

FIG. 14 illustrates the semiconductor arrangement 100 at a stage of fabrication after removal of the second patterned mask layer 1202, according to some embodiments. The second patterned mask layer 1202 is removed after the

shallow trench **1302** is formed. The second patterned mask layer **1202** is removed using a process akin to a process used for removal of the first patterned mask layer **802** set forth above with reference to FIG. **11**.

A portion of the substrate **200** defining the shallow trench **1302** has a first sidewall **1402** and a second sidewall **1404**. In some embodiments, at least one of at least some of the first sidewall **1402** is tapered or at least some of the second sidewall **1404** is tapered. With reference to the first side **504** of the substrate **200**, the first sidewall **1402** has a first slope, such as a negative slope, and the second sidewall **1404** has a second slope, such as a positive slope. In some embodiments, the second slope is opposite in polarity relative to the first slope. In some embodiments, a cross-sectional area of the shallow trench **1302** decreases along a direction opposite to the direction of projected radiation **201**, such that a width of an upper portion of the shallow trench **1302** is greater than a width of a lower portion of the shallow trench **1302**. In some embodiments, at least some of a sidewall defining the shallow trench **1302**, such as at least some of the first sidewall **1402** or at least some of the second sidewall **1404**, extends vertically, such as in a direction parallel to the direction of projected radiation **201**. The shallow trench **1302** has a first maximum width **W1st** and a first minimum width **W2st**. In some embodiments, the shallow trench **1302** is tapered from the first maximum width **W1st** to the first minimum width **W2st** in a direction opposite to the direction of projected radiation **201**. Other structures and/or configurations of the shallow trench **1302** relative to other elements, features, etc. are within the scope of the present disclosure.

In some embodiments, a lowermost portion of a shallow trench **1302** from the first side **504** of the substrate **200** is lower than an uppermost portion of a component such as, for example, at least one of the pinning layer **510**, the floating diffusion **514**, or the charge storage well **508**. According to some embodiments, the lowermost portion of the shallow trench **1302** from the first side **504** of the substrate **200** is higher than a lowermost portion of a component, such as, for example, the first deep storage well section **516** or the second deep storage well section **536**. According to some embodiments, the lowermost portion of the shallow trench **1302** is lower than a lowermost portion of a component, such as, for example, the charge storage well **508** or the floating diffusion **514**. According to some embodiments, the lowermost portion of the shallow trench **1302** is level or coplanar with the lowermost portion of the component. Other structures and/or configurations of the shallow trench **1302** relative to the photodiode **500** and/or components, other elements, features, etc. are within the scope of the present disclosure.

FIG. **15** illustrates the semiconductor arrangement **100** at a stage of fabrication with a first barrier layer **1502** formed over the substrate **200** in the shallow trench **1302**, according to some embodiments. In some embodiments, the first barrier layer **1502** is in direct contact with the first side **504** (e.g., the top surface) of the substrate **200**, such as the first sidewall **1402** and the second sidewall **1404** defining the shallow trench **1302**. In some embodiments, the first barrier layer **1502** is in indirect contact with the first side **504** (e.g., the top surface) of the substrate **200**. Other structures and/or configurations of the first barrier layer **1502** relative to other elements, features, etc. are within the scope of the present disclosure.

The first barrier layer **1502** comprises at least one of aluminum oxide (Al_2O_3), hafnium oxide, tantalum nitride, or other suitable material. The first barrier layer **1502** is formed by at least one of PVD, sputtering, CVD, LPCVD,

ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. In some embodiments, formation of the first barrier layer **1502** is followed by planarizing a top surface of the deposited material.

A first portion of the first barrier layer **1502** is in the shallow trench **1302**. The first portion of the first barrier layer **1502** has a third sidewall **1504** with which the first sidewall **1402** of the substrate **200** aligns. The first portion of the first barrier layer **1502** in the shallow trench **1302** has a fourth sidewall **1506** with which the second sidewall **1404** of the substrate **200** aligns. Other structures and/or configurations of the first barrier layer **1502** relative to other elements, features, etc. are within the scope of the present disclosure.

FIG. **16** illustrates the semiconductor arrangement **100** at a stage of fabrication with a first dielectric layer **1602** formed over the first barrier layer **1502** and in the shallow trench **1302**, according to some embodiments. In some embodiments, the first dielectric layer **1602** is in direct contact with the first barrier layer **1502**. In some embodiments, the first dielectric layer **1602** is in indirect contact with the first barrier layer **1502**. The first barrier layer **1502** is between the first dielectric layer **1602** and at least the substrate **200**. Other structures and/or configurations of the first dielectric layer **1602** relative to other elements, features, etc. are within the scope of the present disclosure.

The first dielectric layer **1602** comprises at least one of silicon dioxide, silicon nitride, silicon oxynitride, hafnium oxide, fluorinated silica glass (FSG), or other suitable material. The first dielectric layer **1602** is formed by at least one of PVD, sputtering, CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. In some embodiments, formation of the first dielectric layer **1602** is followed by planarizing a top surface of the deposited material. A coefficient of thermal expansion of the first dielectric layer **1602** is between about 1 to about 20 (such as between about 2.5 to about 16). Other structures and/or configurations of the first dielectric layer **1602** are within the scope of the present disclosure.

A first portion of the first dielectric layer **1602** is in the shallow trench **1302**. The first portion of the first dielectric layer **1602** has a fifth sidewall **1604** with which the third sidewall **1504** of the first portion of the first barrier layer **1502** aligns. The first portion of the first dielectric layer **1602** in the shallow trench **1302** has a sixth sidewall **1606** with which the fourth sidewall **1506** of the first portion of the first barrier layer **1502** aligns. Other structures and/or configurations of the first dielectric layer **1602** relative to other elements, features, etc. are within the scope of the present disclosure.

FIG. **17** illustrates the semiconductor arrangement **100** at a stage of fabrication with a shallow trench isolation **1702** formed over the first dielectric layer **1602**, over the first barrier layer **1502**, and in the shallow trench **1302**, according to some embodiments. In some embodiments, the shallow trench isolation **1702** is in direct contact with the first dielectric layer **1602**. In some embodiments, the shallow trench isolation **1702** is in indirect contact with the first dielectric layer **1602**. The first dielectric layer **1602** is between the shallow trench isolation **1702** and at least one of the first barrier layer **1502** or the substrate **200**. The first barrier layer is between the substrate **200** and at least one of the first dielectric layer **1602** or the shallow trench isolation **1702**. Other structures and/or configurations of the shallow trench isolation **1702** relative to other elements, features, etc. are within the scope of the present disclosure.

The shallow trench isolation **1702** comprises at least one of silicon dioxide, silicon nitride, silicon oxynitride, hafnium oxide, fluorinated silica glass (FSG), or other suitable material. The shallow trench isolation **1702** is formed by at least one of PVD, sputtering, CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. In some embodiments, formation of the shallow trench isolation **1702** is followed by planarizing a top surface of the deposited material. Other structures and/or configurations of the shallow trench isolation **1702** are within the scope of the present disclosure.

A first portion of the shallow trench isolation **1702** is in the shallow trench **1302**. The first portion of the shallow trench isolation **1702** has a seventh sidewall **1704** with which the fifth sidewall **1604** of the first portion of the first dielectric layer **1602** aligns. The first portion of the shallow trench isolation **1702** in the shallow trench **1302** has an eighth sidewall **1706** with which the sixth sidewall **1606** of the first portion of the first dielectric layer **1602** aligns. In some embodiments, at least one of at least some of the seventh sidewall **1704** is tapered or at least some of the eighth sidewall **1706** is tapered. The seventh sidewall **1704** has a first slope, such as a negative slope, and the eighth sidewall **1706** has a second slope, such as a positive slope. In some embodiments, the second slope is opposite in polarity relative to the first slope. In some embodiments, a cross-sectional area of the shallow trench isolation **1702** decreases along the direction of projected radiation **201**, such that a width $W1_{sti}$ of an upper portion of the shallow trench isolation **1702** is greater than a width $W2_{sti}$ of a lower portion of the shallow trench isolation **1702**. In some embodiments, at least some of a sidewall defining the shallow trench isolation **1702**, such as at least one of at least some of the seventh sidewall **1704** or at least some of the eighth sidewall **1706**, extends vertically, such as in a direction parallel to the direction of projected radiation **201**. In some embodiments, the shallow trench isolation **1702** has a first maximum width $W1_{sti}$ and a first minimum width $W1_{sti}$. In some embodiments, the shallow trench isolation **1702** is tapered from the first maximum width $W1_{sti}$ to the first minimum width $W2_{sti}$ in a direction opposite to the direction of projected radiation **201**. In some embodiments, the shallow trench isolation **1702** provides one of a reduction or an elimination of electrical cross-talk, where an electrical signal is produced from incident radiation that passes between adjacent photodiodes. Other structures and/or configurations of the shallow trench isolation **1702** relative to other elements, features, etc. are within the scope of the present disclosure.

FIG. **18** illustrates the semiconductor arrangement **100** at a stage of fabrication with a second dielectric layer **1802** formed over the shallow trench isolation **1702** and the substrate **200**, according to some embodiments. In some embodiments, the second dielectric layer **1802** is in direct contact with at least one of the shallow trench isolation **1702** or the substrate **200**. In some embodiments, the second dielectric layer **1802** is in indirect contact with at least one of the shallow trench isolation **1702** or the substrate **200**. In some embodiments, the second dielectric layer **1802** comprises a dielectric material such as at least one of oxide, nitride, or other suitable materials. The second dielectric layer **1802** is formed by at least one of PVD, sputtering, CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. Other structures and/or configurations of the second dielectric layer **1802** relative to other elements, features, etc. are within the scope of the present disclosure.

FIG. **19** illustrates the semiconductor arrangement **100** at a stage of fabrication with the transfer gate **512** formed over at least one of the second dielectric layer **1802** or the substrate **200**, according to some embodiments. In some embodiments, the transfer gate **512** is in direct contact with at least one of the second dielectric layer **1802** or the substrate **200**. In some embodiments, the transfer gate **512** is in indirect contact with at least one of the second dielectric layer **1802** or the substrate **200**.

The transfer gate **512** reads out a signal charge corresponding to detected signal strength of radiation impinging on the photodiode **500**. To read out signal charge from the charge storage well **508**, the transfer gate **512** is used to create a variable potential barrier for charge carriers in the junction of the charge storage well **508** and the substrate **200**. When a transfer gate voltage is applied to turn the transfer gate **512** "on" (e.g., a positive bias relative to the substrate **200** for an n-channel device where charge storage well **508** is the n-type) charge carriers in the charge storage well **508** transfer across a barrier in the substrate **200** to the floating diffusion **514** until a potential in the charge storage well **508** is approximately equal to the potential barrier. In the process of reading out the charge carriers, the charge storage well **508** is essentially reset to a higher potential. The transfer gate voltage is then removed to turn the transfer gate **512** "off" to a lower potential. New photoelectrons are collected in the charge storage well **508** during an integration period with an associated drop in potential in the charge storage well **508**. At the end of the integration period, the transfer gate **512** is again pulsed "on" to skim off the collected charge carriers.

The transfer gate **512** comprises a conductive material such as at least one of polysilicon, metal, or other suitable materials. The transfer gate **512** is formed by at least one of PVD, sputtering, CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. Other structures and/or configurations of the transfer gate **512** relative to other elements, features, etc. are within the scope of the present disclosure.

FIG. **20** illustrates the semiconductor arrangement **100** at a stage of fabrication with an interconnect structure **2002** formed over at least one of the transfer gate **512**, the second dielectric layer **1802**, or the substrate **200**, according to some embodiments. The interconnect structure **2002** at least one of overlies the transfer gate **512**, is in direct contact with the transfer gate **512**, or is in indirect contact with the transfer gate **512**. The interconnect structure **2002** at least one of overlies the second dielectric layer **1802**, is in direct contact with the second dielectric layer **1802**, or is in indirect contact with the second dielectric layer **1802**. The interconnect structure **2002** at least one of overlies the substrate **200**, is in direct contact with the substrate **200**, or is in indirect contact with the substrate **200**.

The interconnect structure **2002** comprises one or more interconnect layers, such as at least one of a first interconnect layer **2004**, a second interconnect layer **2006**, a third interconnect layer **2008**, or a fourth interconnect layer **2010**. The one or more interconnect layers of the interconnect structure **2002** comprise at least one of patterned dielectric layers or conductive layers that provide interconnections, such as wiring, between at least one of various doped features, circuitry, input/output, etc. of the semiconductor arrangement **100**. In some embodiments, the interconnect structure **2002** comprises an interlayer dielectric and multi-layer interconnect structures, such as at least one of contacts, vias, metal lines, or other types of structures. For purposes of illustration, the interconnect structure **2002** comprises

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conductive lines **2012**, where the positioning and configuration of conductive lines **2012** may vary depending upon design needs. Other structures and configurations of the interconnect structure **2002** are within the scope of the present disclosure.

FIG. **21** illustrates the semiconductor arrangement **100** at a stage of fabrication with a passivation layer **2102** formed over the interconnect structure **2002** and a first oxide layer **2104** formed over the passivation layer **2102**, according to some embodiments. A first wafer **2106** comprises the substrate **200**, the second dielectric layer **1802**, the transfer gate **512**, the interconnect structure **2002**, the passivation layer **2102**, and the first oxide layer **2104**. The passivation layer **2102** at least one of overlies the interconnect structure **2002**, is in direct contact with the interconnect structure **2002**, or is in indirect contact with the interconnect structure **2002**. The passivation layer **2102** at least one of overlies the substrate **200**, is in direct contact with the substrate **200**, or is in indirect contact with the substrate **200**. The first oxide layer **2104** at least one of overlies the passivation layer **2102**, is in direct contact with the passivation layer **2102**, or is in indirect contact with the passivation layer **2102**. The first oxide layer **2104** at least one of overlies the substrate **200**, is in direct contact with the substrate **200**, or is in indirect contact with the substrate **200**.

In some embodiments, the passivation layer **2102** protects at least some of the interconnect structure **2002** from undergoing corrosion or other alterations during subsequent formation processes of the semiconductor arrangement **100**. In some embodiments, the passivation layer **2102** includes at least one of aluminum nitride (AlN), aluminum oxide (Al₂O₃), SiO₂, silicon nitride (Si₃N₄), a chemically inert, corrosion-resistant dielectric material, or an organic compound having at least one of the n-type, the p-type, or other silicon-type molecular structure, or other suitable materials. Other configurations and/or compositions of the passivation layer **2102** are within the scope of the present disclosure.

In some embodiments, the passivation layer **2102** is formed by at least one of PVD, sputtering, CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, passivation, or other suitable techniques. A passivation process includes at least one of oxidation of a surface of an underlying material or complexing of the surface of the underlying material with an organic compound. The passivation layer **2102** includes a layer or layers of film covering an underlying material, such as a material of the interconnect structure **2002** or the substrate **200**. A chemically-stable material is used to produce the layer or layers of film. The layer or layers of film at least one of inhibits dissolution or reduces at least one of electrical or chemical reactivity with regard to the underlying material, at least one of during or after subsequent processes or treatments of the semiconductor arrangement **100**. Other processes of forming the passivation layer **2102** are within the scope of the present disclosure.

The passivation layer **2102** at least one of inhibits diffusion of at least one of charges, atoms, or ions into the underlying material, mitigates oxidation of the underlying material, protects the underlying material from environmental conditions, or acts as a diffusion barrier with regard to the underlying material. Other purposes and/or functions of the passivation layer **2102** are within the scope of the present disclosure.

In some embodiments, the first oxide layer **2104** is conformally formed over and across at least one of the passivation layer **2102**, the interconnect structure **2002**, or the substrate **200**. In some embodiments, the first oxide layer

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2104 comprises at least one of SiO₂, Si₃N₄, silicon oxynitride (SiON), dielectric material(s) with a high dielectric constant (high-k), or other suitable materials. The first oxide layer **2104** may be formed by at least one of CVD, PVD, ALD, high-density plasma CVD (HDPCVD), metal organic CVD (MOCVD), plasma enhanced CVD (PECVD), or other suitable techniques. In some embodiments, the oxide layer **2104** has a thickness of between 0.5 nm and 3 nm. If the oxide layer **2104** is too thick, such as greater than 3 nm, it may be difficult to remove in a following process. If the first oxide layer **2104** is too thin, such as less than 0.5 nm, the underlying structure may be damaged when bonding the first wafer **2106** to another wafer.

FIG. **22** illustrates the semiconductor arrangement **100** at a stage of fabrication with a second wafer **2202**, at times referred to as a carrier wafer, in a process to bond to the first wafer **2106**, according to some embodiments. The second wafer **2202** has a second substrate **2204** and a second wafer interface region **2206**. A layer of SiO₂ is formed in the second wafer interface region **2206** by exposing the second wafer interface region **2206** to **02**. The second wafer interface region **2206** at least one of overlies the second substrate **2204**, is in direct contact with the second substrate **2204**, or is in indirect contact with the second substrate **2204**. In some embodiments, the first wafer **2106** is treated with a process to create a first wafer interface region **2208** in the first oxide layer **2104**.

In some embodiments, the first wafer **2106** and the second wafer **2202** are subjected to a thermal process (heat treatment) for forming a thermal oxide layer in the first wafer interface region **2208** of the first wafer **2106** and the second wafer interface region **2206** of the second wafer **2202**. The thermal oxide layer is formed inside of a heated chamber, a furnace, or other suitable temperature-controlled environment. A thermal oxide layer is formed by subjecting the first wafer **2106** or second wafer **2202** to an environment of oxygen (e.g., O₂) and a specified temperature, such as 1000° C., for one hour. According to some embodiments, the thermal oxide layer is SiO₂ and is formed by a molecular reaction between the Si of the first wafer interface region **2208** of the first wafer **2106** or the second wafer interface region **2206** of the second wafer **2202** and the **02**. In some embodiments, a thickness of the thermal oxide layer of the first wafer interface region **2208** of the first wafer **2106** is selected to provide a good reaction with the second wafer interface region **2206** of the second wafer **2202** during a subsequent anneal process to form covalent bonds between the first wafer **2106** and the second wafer **2202**, such as a thickness of 1000 angstroms (Å).

In some embodiments, the first wafer **2106** is treated with plasma in a first plasma chamber (not shown) and the second wafer **2202** is treated with plasma in a second plasma chamber (not shown). In some embodiments, the first wafer **2106** and the second wafer **2202** are each treated with plasma in the same plasma chamber (not shown). Plasma treatment includes injecting a material into the first plasma chamber, the second plasma chamber, or the same plasma chamber and applying a voltage to the injected material to generate plasma from the injected material. The injected material may be argon (Ar) or other suitable materials, and the plasma may be Ar plasma or other suitable plasma. According to some embodiments, the plasma treatment includes subjecting the first wafer **2106** and the second wafer **2202** to an environment of silane (SiH₄) and nitrous oxide (N₂O), and applying a bias voltage directed at the first wafer interface region **2208** and the second wafer interface region **2206**. A plasma oxide layer of SiO₂ is formed by

subjecting the first wafer **2106** and the second wafer **2202** to an environment of SiH_4 and N_2O at a specified temperature, such as 300°C ., for a specified period of time, such as for one hour, and directing the voltage toward the first wafer interface region **2208** and the second wafer interface region **2206**. In some embodiments, a thickness of the plasma oxide layer of the first wafer interface region **2208** of the first wafer **2106** is selected to provide a good reaction with the second wafer interface region **2206** of the second wafer **2202** during a subsequent anneal process to form covalent bonds between the first wafer **2106** and the second wafer **2202**, such as a thickness of 1000 angstroms (Å).

The first wafer interface region **2208** and the second wafer interface region **2206** are aligned, such as to overlap or be vertically coincident with one another, and pressed together by applying at least one of a first pressure **P1** to the first wafer interface region **2208** in a direction toward the second wafer interface region **2206** or a second pressure **P2** to the second wafer interface region **2206** in a direction toward the first wafer interface region **2208**. Due to the force of at least one of the first pressure **P1** or the second pressure **P2**, the SiO_2 molecules at the second wafer interface region **2206** combine with the Si atoms at the first wafer interface region **2208** by way of Van der Waals interaction to form a Si— SiO_2 surface bond. In some embodiments, the anneal process includes subjecting an interface of the first wafer interface region **2208** and the second wafer interface region **2206** to a specified temperature for a specified amount of time, such as 350°C . for three hours. As the Si—O—Si covalent bonds form at the interface of the first wafer interface region **2208** and the second wafer interface region **2206**, the H atoms combine with O atoms to form H_2O as a byproduct. Other temperatures and/or anneal times for forming the Si—O—Si covalent bonds between the first wafer **2106** and the second wafer **2202** are within the scope of the disclosure.

FIG. **23** illustrates the semiconductor arrangement **100** at a stage of fabrication with the second wafer **2202** bonded to the first wafer **2106**, according to some embodiments. The second wafer **2202** is bonded to the first wafer **2106** after an anneal process for forming covalent bonds of Si—O—Si at the interface of the first wafer interface region **2208** and the second wafer interface region **2206**. Referring to FIG. **23**, the semiconductor arrangement **100** has been inverted.

FIG. **24** illustrates the semiconductor arrangement **100** at a stage of fabrication after removal of the second wafer **2202**, according to some embodiments. The first wafer **2106** is reduced by removing the second substrate **2204** and the second wafer interface region **2206**. According to some embodiments, the semiconductor arrangement **100** is rotated 180 degrees or inverted to facilitate removal of the second wafer **2202**. The second wafer **2202** is removed by at least one of wafer back-grinding, liquid etching, dry etching, plasma etching, chemical-mechanical polishing, planarization, or other suitable techniques.

FIG. **25** illustrates the semiconductor arrangement **100** at a stage of fabrication with a mask layer patterned to form a third patterned mask layer **2502** over the substrate **200** and isolation structure **518** comprising the first well **519** and the second well **528**, according to some embodiments. The third patterned mask layer **2502** may be formed using a process akin to a process used for formation of the first patterned mask layer **802** as set forth above with reference to FIG. **8**. In some embodiments, the second well **528** may be formed using a process akin to a process used for formation of the first well **519** in the substrate **200** as set forth above and as illustrated in FIG. **9**. In some embodiments, the second well

528 is formed via deep implantation **2504** and comprises the first-type dopant. In some embodiments, the second well **528** comprises the first-type dopant in a higher concentration than the first-type dopant in the substrate **200**. The deep implantation **2504** is formed using a process akin to a process used for formation of the deep implantation **902** set forth above with reference to FIG. **9**. In some embodiments, the first well **519** has a first side **519a** and a second side **519b**, and the second well **528** has a first side **528a** and a second side **528b**.

In some embodiments, the isolation structure **518** comprising a combination of the first well **519** and the second well **528** surrounds the photodiode **500**. In some embodiments, the first well **519** laterally surrounds the photodiode **500** and the second well **528** does not laterally surround the photodiode **500**. In some embodiments, the first well **519** laterally surrounds a portion of the photodiode **500** and the second well **528** laterally surrounds a portion of the photodiode **500** (e.g., when the second deep storage well section **536** of the photodiode **500** extends from the first side **504** of the substrate **200** beyond the first side **528a** of the second well **528**).

In some embodiments, the first well **519** is contiguous with the second well **528** (e.g., the second side **519b** of the first well **519** is contiguous with the first side **528a** of the second well **528**). In some embodiments, the first well **519** overlaps the second well **528** (e.g., the second side **519b** of the first well **519** overlaps the first side **528a** of the second well **528**). In some embodiments, the first well **519** and the second well **528** are contiguous and formed with the same dopant type. In some embodiments, the isolation structure **518** comprising a combination of the first well **519** and the second well **528** extends from the first side **504** of the substrate **200** to the second side **506** of the substrate **200**. In some embodiments, the isolation structure **518** comprising a combination of the first well **519** and the second well **528** extends from the first side **504** of the substrate **200** and does not extend to the second side **506** of the substrate **200**. In some embodiments, the isolation structure **518** comprising a combination of the first well **519** and the second well **528** does not extend from the first side **504** of the substrate **200** and does extend to the second side **506** of the substrate **200**. Relative widths and depths of the first well **519**, the second well **528**, and the photodiode **500** are set forth above with reference to FIG. **5C**. In some embodiments, the second well **528** provides one of a reduction or an elimination of electrical cross-talk, where an electrical signal is produced from incident radiation that passes between adjacent photodiodes. Other arrangements and/or configurations of the first well **519** and the second well **528**, and processes and/or techniques for forming the first well **519**, the second well **528**, and the third patterned mask layer **2502** are within the scope of the present disclosure.

FIG. **26** illustrates the semiconductor arrangement **100** at a stage of fabrication after removal of the third patterned mask layer **2502** and illustrates a mask layer patterned to form a fourth patterned mask layer **2602** over at least one of the substrate **200** or the isolation structure **518**, according to some embodiments. The third patterned mask layer **2502** is removed using a process akin to a process used for removal of the first patterned mask layer **802** set forth above with reference to FIG. **11**. In some embodiments, removal of the third patterned mask layer **2502** also removes at least some of the substrate **200**. The fourth patterned mask layer **2602** is formed using an etching process akin to an etching process used in the formation of the first patterned mask layer **802** as set forth above with reference to FIG. **8**. Other processes

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and/or techniques for forming the fourth patterned mask layer **2602** are within the scope of the present disclosure.

FIG. **27** illustrates the semiconductor arrangement **100** at a stage of fabrication with the fourth patterned mask layer **2602** used in the formation of recesses **2702** in the substrate **200**, according to some embodiments. In some embodiments, the recesses are formed using an etching process akin to an etching process used to form the shallow trench **1302** with reference to FIG. **13**. The recesses **2702** extend into the second side **506** of the substrate **200** and at least partially define an absorption enhancement structure. The recesses **2702** are at least one of vertically coincident with photodiode **500**, vertically coincident with a portion of the photodiode **500**, or vertically offset from the photodiode **500**. The recesses **2702** are at least one of surrounded by the isolation structure **518**, surrounded by a portion of the isolation structure **518**, laterally surrounded by the isolation structure **518**, or laterally surrounded by a portion of the isolation structure **518**. The recesses **2702** are at least one of surrounded by the second well **528**, surrounded by a portion of the second well **528**, laterally surrounded by the second well **528**, or laterally surrounded by a portion of the second well **528**. The recesses **2702** extend into the substrate **200** from the second side **506** by at least less than the second side **528b** of the second well **528**, coincident with the second side **528b** of the second well **528**, or greater than the second side **528b** of the second well **528**. Other arrangements of the recesses **2702** with respect to the isolation structure **518** and/or the second well **528** are within the scope of the present disclosure. The recesses **2702** are formed in accordance with the fourth patterned mask layer **2602**. The recesses **2702** are laterally adjacent and are at least one of laterally contiguous or laterally non-contiguous (e.g., separated by a distance in accordance with the fourth patterned mask layer **2602**). Other arrangements or configurations of the recesses **2702** are within the scope of the present disclosure.

The recesses **2702** define protrusions **2704**. The protrusions **2704** are configured to prevent incident radiation from reflecting away from the photodiode **500**. For example, the protrusions **2704** may have angled sidewalls to prevent such reflectance. By preventing incident radiation from reflecting away from the photodiode **500**, the protrusions **2704** increase an amount of radiation absorbed by the photodiode **500**, which may increase the sensitivity of the photodiode **500**. In some embodiments, the protrusions **2704** are in a periodic pattern or array. In some embodiments, the protrusions **2704** each have at least one of a cone shape, a pyramid shape, or some other shape. In some embodiments, the protrusions **2704** define a saw-toothed profile. In some embodiments, the protrusions **2704** have a pitch *P* of about 0.01 μm to 8.0 μm , of about 0.2 μm to 5.0 μm , of about 1.0 μm to 3.0 μm , or any combination of the foregoing. In some embodiments, the protrusions **2704** have a height *H* of about 0.2 μm to 20.0 μm , of about 1.0 μm to 15.0 μm , of about 5.0 μm to 10.0 μm , or any a combination of the foregoing. In some embodiments the second side **506** of the substrate **200** comprises a porous semiconductor material, and the protrusions **2704** are wholly or partially defined by the porous semiconductor material. In some embodiments, a porous semiconductor layer is formed on the second side **506** of the substrate **200** that comprises a porous semiconductor material, and the protrusions **2704** are wholly or partially defined by the porous semiconductor material. In some embodiments, the combination of the porous semiconductor material and the protrusions **2704** results in an absorption enhancement structure having a direct energy band gap that

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enhances the quantum efficiency of the photodiode **500**. Other arrangements or configurations of the protrusions **2704** are within the scope of the present disclosure.

FIG. **28** illustrates the semiconductor arrangement **100** at a stage of fabrication with removal of the fourth patterned mask layer **2602** and formation of a third dielectric layer **2802**, according to some embodiments. The fourth patterned mask layer **2602** is removed after the recesses **2702** are formed. The fourth patterned mask layer **2602** is removed using a process akin to a process used for removal of the first patterned mask layer **802** set forth above with reference to FIG. **11**. The third dielectric layer **2802** comprises at least one of Si_3N_4 , SiO_2 , SiO , MgO , Al_2O_3 , Yb_2O_3 , ZnO , Ta_2O_5 , ZrO_2 , HfO_2 , TeO_2 , TiO_2 , or other suitable material. In some embodiments, the third dielectric layer **2802** comprises a material that is substantially transparent to wavelengths of radiation to be detected by the photodiode array **104**, such as NIR wavelengths and optical wavelengths. The third dielectric layer **2802** is formed by at least one of PVD, sputtering, CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. In some embodiments, the third dielectric layer **2802** is in direct contact with at least one of the second side **506** of the substrate **200**, the recesses **2702** in the substrate **200**, or the protrusions **2704** defined by the recesses **2702**. In some embodiments, the third dielectric layer **2802** is in indirect contact with at least one of the second side **506** of the substrate **200**, the recesses **2702** in the substrate **200**, or the protrusions **2704** defined by the recesses **2702**. Other structures and/or configurations of the third dielectric layer **2802** are within the scope of the present disclosure.

FIG. **29** illustrates the semiconductor arrangement **100** at a stage of fabrication with the formation of a fourth patterned mask layer **2902** and formation of a deep trench **2904**, according to some embodiments. The formation of the fourth patterned mask layer **2902** is formed using a process akin to a process used for formation of the first patterned mask layer **802** set forth above with reference to FIG. **8**. The formation of the deep trench **2904** is formed using an etching process akin to an etching process for the formation of the shallow trench **1302** set forth above with reference to FIG. **13**. The etching process for the deep trench **2904** is at least one of a dry etching process, a wet etching process, an anisotropic etching process, an isotropic etching process, or another suitable etching process. The etching process uses at least one of an etching material of HF, diluted HF, HCl_2 , H_2S , or other suitable material. When compared to the formation of the shallow trench **1302**, the formation of the deep trench **2904** provides a deeper trench, and uses one or more of at least a longer duration of the etching process, a higher concentration of the etching material, a different type of etching material, a higher pressure of etching material, or a higher applied bias voltage. Other processes and/or techniques for forming the fourth patterned mask layer **2902** and the deep trench **2904** are within the scope of the present disclosure.

In some embodiments, the deep trench **2904** extends through the third dielectric layer **2802** and the substrate **200** and into the isolation structure **518**. According to some embodiments, the deep trench **2904** extends into at least one of the second well **528** or the first well **519**. In some embodiments, the deep trench **2904** extends past the second side **528b** of the second well **528**. In some embodiments, the deep trench **2904** extends past the second side **519b** of the second well **528**. In some embodiments, the deep trench **2904** extends into a portion of overlap between the first well

519 and the second well **528**. Other structures and/or configurations of the deep trench **2904** are within the scope of the present disclosure.

In some embodiments, the deep trench **2904** is at least one of laterally offset from the photodiode **500** or between two photodiodes of the photodiode array **104**. In some embodiments, the deep trench **2904** is between two adjacent photodiodes of the photodiode array **104**, a third portion **200c** of the substrate **200** separates the deep trench **2904** from a first photodiode of two adjacent photodiodes of the photodiode array **104**, and a fourth portion **200d** of the substrate **200** separates the deep trench **2904** from a second photodiode of the two adjacent photodiodes of the photodiode array **104**. In some embodiments, an etching process is performed to form the deep trench **2904**, where openings in the fourth patterned mask layer **2902** allow one or more etchants applied during the etching process to remove portions of the substrate **200** while the fourth patterned mask layer **2902** protects or shields portions of the substrate **200** that are covered by the fourth patterned mask layer **2902**. The etching process is at least one of an etching process for formation of the shallow trench **1302** set forth above with reference to FIG. **13**. Other structures and/or configurations for forming the deep trench **2904** are within the scope of the present disclosure.

A portion of the substrate **200** defining the deep trench **2904** has a ninth sidewall **2906** and a tenth sidewall **2908**. In some embodiments, at least one of at least some of the ninth sidewall **2906** is tapered or at least some of the tenth sidewall **2908** is tapered. With reference to the second side **506** of the substrate **505**, the ninth sidewall **2906** has a ninth slope, such as a negative slope, and the tenth sidewall **2908** has a tenth slope, such as a positive slope. In some embodiments, the tenth slope is opposite in polarity relative to the ninth slope. In some embodiments, a cross-sectional area of the deep trench **2904** decreases along the direction of projected radiation **201**, such that a width of an upper portion of the deep trench **2904** is greater than a width of a lower portion of the deep trench **2904**. In some embodiments, at least some of a sidewall defining the deep trench **2904**, such as at least some of the ninth sidewall **2906** or at least some of the tenth sidewall **2908**, extends vertically, such as in a direction parallel to the direction of projected radiation **201**. The deep trench **2904** has a minimum width $W1dt$ and a maximum width $W2dt$. In some embodiments, the deep trench **2904** is tapered from the maximum width $W2dt$ to the minimum width $W1dt$ in the direction of projected radiation **201**. Other structures and/or configurations of the deep trench **2904** are within the scope of the present disclosure.

FIG. **30** illustrates the semiconductor arrangement **100** at a stage of fabrication with formation of a second barrier layer **3002** formed in the deep trench **2904** of the substrate **200** and with removal of the fourth patterned mask layer **2902**, according to some embodiments. The second barrier layer **3002** is formed using a process akin to a process used for formation of the first barrier layer **1502** set forth above with reference to FIG. **15**. The fourth patterned mask layer **2902** is removed using a process akin to a process used for removal of the first patterned mask layer **802** set forth above with reference to FIG. **11**.

In some embodiments, the second barrier layer **3002** is in direct contact with at least one of the third dielectric layer **2802** or the second side **506** of the substrate **200**, such as the ninth sidewall **2906** or the tenth sidewall **2908** defining the deep trench **2904**. In some embodiments, the second barrier layer **3002** is in indirect contact with the surface along the second side **506** of the substrate **200**. In some embodiments,

formation of the second barrier layer **3002** is followed by planarizing a surface along the second side **506** of the substrate **200** of the deposited material. Other structures and/or configurations of the second barrier layer **3002** relative to other elements, features, etc. are within the scope of the present disclosure.

A first portion of the second barrier layer **3002** is in the deep trench **2904**. The first portion of the second barrier layer **3002** has an eleventh sidewall **3004** with which the ninth sidewall **2906** of the deep trench **2904** aligns. The first portion of the second barrier layer **3002** in the deep trench **2904** has a twelfth sidewall **3006** with which the tenth sidewall **2908** of the deep trench **2904** aligns. Other structures and/or configurations of the second barrier layer **3002** relative to other elements, features, etc. are within the scope of the present disclosure.

According to some embodiments, a fourth dielectric layer (not shown) is formed over the second barrier layer **3002** using a process akin to a process used for formation of the first dielectric layer **1602** set forth above with reference to FIG. **16**. In some embodiments, the fourth dielectric layer (not shown) is formed over the second barrier layer **3002** and in the deep trench **2904**. In some embodiments, the fourth dielectric layer (not shown) is in direct contact with the second barrier layer **3002**. In some embodiments, the fourth dielectric layer (not shown) is in indirect contact with the second barrier layer **3002**. The second barrier layer **3002** is between the fourth dielectric layer (not shown) and the substrate **200**. Other structures and/or configurations of the fourth dielectric layer (not shown) relative to other elements, features, etc. are within the scope of the present disclosure.

FIG. **31** illustrates the semiconductor arrangement **100** at a stage of fabrication with formation of the deep trench isolation **520**, according to some embodiments. The deep trench isolation **520** is formed over the second barrier layer **3002** and in the deep trench **2904**. In some embodiments, the deep trench isolation **520** is in direct contact with the second barrier layer **3002**. In some embodiments, the deep trench isolation **520** is in indirect contact with the second barrier layer **3002**, such as with contact with the fourth dielectric layer (not shown). The second barrier layer **3002** is between the deep trench isolation **520** and at least one of the deep trench isolation **520**, the fourth dielectric layer (not shown), or the substrate **200**. Other structures and/or configurations of the deep trench isolation **520** relative to other elements, features, etc. are within the scope of the present disclosure.

The deep trench isolation **520** comprises at least one of silicon dioxide, silicon nitride, silicon oxynitride, hafnium oxide, fluorinated silica glass (FSG), a metal such as aluminum or copper, a metal compound, a metal mixture, or other suitable material. The deep trench isolation **520** is formed by at least one of PVD, sputtering, CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. In some embodiments, formation of the deep trench isolation **520** is followed by planarizing a top surface of the deposited material. Other structures and/or configurations of the deep trench isolation **520** are within the scope of the present disclosure.

A first portion of the deep trench isolation **520** is in the deep trench **2904**. The first portion of the deep trench isolation **520** has a thirteenth sidewall **3102** with which the eleventh sidewall **3004** aligns. The first portion of the deep trench isolation **520** has a fourteenth sidewall **3104** with which the twelfth sidewall **3006** aligns. In some embodiments, at least one of at least some of the thirteenth sidewall **3102** is tapered or at least some of the fourteenth sidewall **3104** is tapered. The thirteenth sidewall **3102** has a thirteenth

slope, such as a negative slope, and the fourteenth sidewall **3104** has a fourteenth slope, such as a positive slope. In some embodiments, the fourteenth slope is opposite in polarity relative to the thirteenth slope. In some embodiments, a cross-sectional area of the deep trench isolation **520** decreases along the direction of projected radiation **201**, such that a width of an upper portion of the deep trench isolation **520** is greater than a width of a lower portion of the deep trench isolation **520**. In some embodiments, at least some of the sidewall defining the deep trench isolation **520**, such as at least some of at least one of the thirteenth sidewall **3102** or the fourteenth sidewall **3104**, extends vertically, such as in a direction parallel to the direction of projected radiation **201**. In some embodiments, the deep trench isolation **520** has a maximum width **W2dti** and a minimum width **W1dti**. In some embodiments, the deep trench isolation **520** is tapered from the maximum width **W2dti** to the minimum width **W1dti** in the direction of projected radiation **201**. In some embodiments, the deep trench isolation **520** provides one of a reduction or an elimination of optical cross-talk, where an optical signal is produced from incident radiation that passes between adjacent photodiodes. Other structures and/or configurations of the deep trench isolation **520** relative to other elements, features, etc. are within the scope of the present disclosure.

FIG. **32** illustrates the semiconductor arrangement **100** at a stage of fabrication with formation of a third barrier layer **3202** and formation of a metal grid **3204**, according to some embodiments. The third barrier layer **3202** is formed using a process akin to a process used for formation of the first barrier layer **1502** set forth above with reference to FIG. **15**. The third barrier layer **3202** defines openings for the formation of the metal grid **3204**. The metal grid **3204** is formed over the deep trench isolation **520** and over the deep trench **2904**. In some embodiments, the metal grid **3204** is in direct contact with at least one of the deep trench isolation **520**, the second barrier layer **3002**, the fourth dielectric layer (not shown), the substrate **200**, or the third dielectric layer **2802**. In some embodiments, the metal grid **3204** and the deep trench isolation **520** are contiguous or form the same structure. Other structures and/or configurations of the metal grid **3204** relative to other elements, features, etc. are within the scope of the present disclosure.

The metal grid **3204** is laterally arranged around and between photodiodes of the photodiode array **104** and defines openings for the transmission of incident radiation to the photodiode array **104**. The metal grid **3204** blocks incident radiation from passing between neighboring photodiodes of the photodiode array **104** near an upper surface of the substrate **200**, as well as above and below the upper surface, to help reduce cross talk. Therefore, the metal grid **3204** defines a substrate isolation grid or otherwise forms a part of a substrate isolation grid. In some embodiments, the metal grid **3204** is made of a reflective material such as a metal material. The metal grid **3204** may be, for example, aluminum, silver, copper, titanium, platinum, tungsten, tantalum, tantalum nitride, other suitable materials, or a combination thereof. In some embodiments, the metal grid **3204** is a reflective layer and is deposited using a suitable deposition process. The suitable deposition process includes, for example, a PVD process, an electroplating process, a CVD process, other applicable processes, or a combination thereof. Afterwards, the metal grid **3204** is patterned using, for example, a photolithography process and an etching process to form reflective elements. Other techniques for forming the metal grid **3204** are within the scope of the present disclosure.

FIG. **33** illustrates the semiconductor arrangement **100** at a stage of fabrication with formation of a second oxide layer **3302**, according to some embodiments. The second oxide layer **3302** is formed over at least one of the deep trench isolation **520**, the metal grid **3204**, the second barrier layer **3002**, the fourth dielectric layer (not shown), the third dielectric layer **2802**, or the substrate **200**. In some embodiments, the second oxide layer **3302** is in direct contact with at least one of the deep trench isolation **520**, the metal grid **3204**, the second barrier layer **3002**, the fourth dielectric layer (not shown), the third dielectric layer **2802**, or the substrate **200**. In some embodiments, the second oxide layer **3302** is at least one of contiguous with the third barrier layer **3202**, formed on the third barrier layer **3202**, or the same structure as the third barrier layer **3202**. In some embodiments, the second oxide layer **3302** is formed using a process akin to a process used for formation of the first oxide layer **2104** set forth above with reference to FIG. **21**. Other structures and/or configurations of the second oxide layer **3302** and the third barrier layer **3202** are within the scope of the present disclosure.

FIG. **34** illustrates the semiconductor arrangement **100** at a stage of fabrication with formation of the ARC layer **212**, the radiation filter layer **214**, and the micro-lens array **216**, according to some embodiments. The ARC layer **212** is arranged over the photodiode array **104**. The ARC layer **212** is formed over at least one of the second oxide layer **3302**, the metal grid **3204**, the deep trench isolation **520**, the second barrier layer **3002**, the fourth dielectric layer (not shown), the third dielectric layer **2802**, or the substrate **200**. In some embodiments, the ARC layer **212** is in direct contact with at least one of the second oxide layer **3302**, the metal grid **3204**, the deep trench isolation **520**, the second barrier layer **3002**, the fourth dielectric layer (not shown), the third dielectric layer **2802**, or the substrate **200**. In some embodiments, the ARC layer **212** is in indirect contact with at least one of the second oxide layer **3302**, the metal grid **3204**, the deep trench isolation **520**, the second barrier layer **3002**, the fourth dielectric layer (not shown), the third dielectric layer **2802**, or the substrate **200**. In some embodiments, the ARC layer **212** is at least one of contiguous with the second oxide layer **3302**, formed on the second oxide layer **3302**, or the same structure as the second oxide layer **3302**. In some embodiments, the ARC layer **212** is formed by at least one of PVD, sputtering, CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques.

According to some embodiments, the radiation filter layer **214** is arranged and formed over the ARC layer **212**. The radiation filter layer **214** comprises a radiation filter **3402** of the plurality of radiation pass filters **215**. The radiation filter **3402** corresponds to at least one of the NIR pass filter **218**, the red pass filter **220**, the green pass filter **222**, the blue pass filter **224**, or the white pass filter **226** set forth above with reference to FIG. **2**. The radiation filter layer **214** at least one of overlies the ARC layer **212**, is in direct contact with a top surface of the ARC layer **212**, or is in indirect contact with the top surface of the ARC layer **212**. The radiation filter **3402** comprises at least one of a pigment-dispersed color resist (PDCR) material, a photosensitive substance, a photoinitiator substance, a multifunctional monomer, one or more additives, a leveling agent, an adhesion promoter, a resin, a polymer soluble in alkaline solution, a color paste, a pigment, a dispersant, a solvent, or other suitable material. The radiation filter **3402** passes certain wavelengths of radiation. In some embodiments, different portions of the radiation filter **3402** have different material compositions to

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enable different wavelengths to be passed. The NIR pass filter **218** has an NIR pass material composition and passes NIR wavelengths, the red pass filter **220** has a red pass material composition and passes red wavelengths, the green pass filter **222** has a green pass material composition and passes green wavelengths, the blue pass filter **224** has a blue pass material composition and passes blue wavelengths, and the white pass filter **226** has a white pass material composition and passes white wavelengths. In some embodiments, the radiation filter **3402** comprises multiple radiation filter layers on top of each other. In some embodiments, a first radiation filter layer of the multiple radiation filter layers corresponds to the NIR pass filter **218**. In some embodiments, a second radiation filter layer of the multiple radiation filter layers is an optical pass filter that corresponds to at least one of the red pass filter **220**, the green pass filter **222**, the blue pass filter **224**, or the white pass filter **226**. In some embodiments, the second radiation filter layer overlies the first radiation filter layer. In some embodiments, the first radiation filter layer overlies the second radiation filter layer. The radiation filter layer **214** is formed by at least one of PVD, sputtering, CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. At least one of the multiple radiation filter layers, such as at least one of the first radiation filter layer or the second radiation filter layer, has a thickness of less than or equal to about 10,000 angstroms (such as less than or equal to about 5,000 angstroms). Other structures and/or configurations of the radiation filter layer **214** or the radiation filter **3402** are within the scope of the present disclosure.

According to some embodiments, the micro-lens array **216** is arranged over the radiation filter layer **214** and the radiation filter **3402** to steer radiation towards photodiodes of the photodiode array **104**. Lenses of the micro-lens array **216** are at least one of micro-lenses or other suitable lenses. In some embodiments, the micro-lens array **216** comprises at least one micro-lens of the plurality of micro-lenses **216a-216e**. In some embodiments, at least one micro-lens of the micro-lens array **216** is at least one of the micro-lens **216a** configured to transmit radiation through the NIR pass filter **218** to the photodiode **202a**, the micro-lens **216b** configured to transmit radiation through the red pass filter **220** to the photodiode **202b**, the micro-lens **216c** configured to transmit radiation through the green pass filter **222** to the photodiode **202c**, the micro-lens **216d** configured to transmit radiation through the blue pass filter **224** to the photodiode **202d**, or the micro-lens **216e** configured to transmit radiation through the white pass filter **226** to the photodiode **202e**. Other arrangements and/or configurations of the micro-lens array **216** are within the scope of the present disclosure.

In some embodiments, one or more lenses of the micro-lens array **216** overlie at least one of the radiation filter **3402**, the radiation filter layer **214**, the ARC layer **212**, the second oxide layer **3302**, the metal grid **3204**, the deep trench isolation **520**, the second barrier layer **3002**, the fourth dielectric layer (not shown), the third dielectric layer **2802**, or the substrate **200**. In some embodiments, one or more lenses of the micro-lens array **216** is in direct contact with at least one of the radiation filter **3402**, the radiation filter layer **214**, the ARC layer **212**, the second oxide layer **3302**, the metal grid **3204**, the deep trench isolation **520**, the second barrier layer **3002**, the fourth dielectric layer (not shown), the third dielectric layer **2802**, or the substrate **200**. In some embodiments, one or more lenses of the micro-lens array **216** is in indirect contact with at least one of the radiation filter **3402**, the radiation filter layer **214**, the ARC layer **212**, the second oxide layer **3302**, the metal grid **3204**,

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the deep trench isolation **520**, the second barrier layer **3002**, the fourth dielectric layer (not shown), the third dielectric layer **2802**, or the substrate **200**. Other arrangements and/or configurations of the micro-lens array **216** are within the scope of the present disclosure.

In some embodiments, the micro-lens array **216** comprises a lens substrate. The lens substrate comprises at least one of silicon, germanium, carbide, arsenide, gallium, arsenic, phosphide, indium, antimonide, SiGe, SiC, GaAs, GaN, GaP, InGaP, InP, InAs, InSb, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP, or other suitable material. The lens substrate is formed by at least one of PVD, sputtering, CVD, LPCVD, ALCVD, UHVCVD, RPCVD, ALD, MBE, LPE, spin on, growth, or other suitable techniques. In some embodiments, the micro-lens array **216** is formed by at least one of thermal reflow, microplastic embossing, microdroplet jetting, photolithography, reactive ion etching, machining, or other suitable methods. Other structures and/or configurations of the micro-lens array **216** are within the scope of the present disclosure.

FIGS. **35A** and **35B** illustrate a method **3500** of making a semiconductor arrangement, according to some embodiments. While the process of FIGS. **35A** and **35B** is applicable to many different systems, devices, components, arrangements, structures, and/or configurations, it is described with reference to FIGS. **1-34** for ease of understanding. The illustrated method **3500** is merely an example method. Therefore, it should be understood that any of a variety of operations may be omitted, re-sequenced, and/or added while remaining within the scope of the present disclosure. As illustrated in FIG. **35A**, the method **3500** comprises, at **3502**, forming a first photodiode with a first depth from a first side in a substrate. For example, in FIG. **5A** and FIG. **6**, the photodiode **500** can be in the form of the first photodiode **108f** of FIG. **1E**. The photodiode **500** can be formed in various examples with at least one of the charge storage well **508**, the first deep storage well section **516**, or the second deep storage well section **536**. With respect to the first side **504** of the substrate **200**, the charge storage well **508** has the storage well depth **D1_{sw}**, the first deep storage well section **516** has the deep storage well depth **D2_{sw}**, and the second deep storage well section **536** has the deep storage well depth **D3_{sw}**.

At **3504**, the method **3500** comprises forming a first isolation structure laterally surrounding the first photodiode, wherein the first isolation structure comprises a first well extending to a second depth from the first side in the substrate greater than the first depth. For example, in FIG. **5A** and FIG. **9**, the isolation structure **518** (e.g., including the first well **519**) can be in the form of the first isolation structure **116f** of FIG. **1E** and can laterally surround the photodiode **500**. With respect to the first side **504** of the substrate **200** at least one of, the deep well depth **D1_{dw}** of the first well **519** can be greater than the storage well depth **D1_{sw}** of the charge storage well **508**, the deep storage well depth **D2_{sw}** of the first deep storage well section **516**, or the deep storage well depth **D3_{sw}** of the second deep storage well section **536**.

At **3506**, the method **3500** comprises forming a deep trench from a third depth from the first side in the substrate less than the second depth to the second side of the substrate, and wherein at least a portion of the deep trench underlies the first well. For example, in FIG. **29**, the deep trench **2904** can be formed using an etching process to etch into the second side **506** of the substrate **200** opposite to the first side **504** of the substrate **200**. In FIG. **5A** and FIG. **29**, with respect to the first side **504** of the substrate **200**, the deep

trench **2904** can extend into the substrate **200** from the deep trench isolation depth **D1dti**, which can be less than the deep well depth **D1dw** of the first well **519**.

At **3508**, the method **3500** comprises forming a deep trench isolation in the deep trench. For example, in FIG. **31**, the deep trench isolation **520** can be formed over the second barrier layer **3002** and in the deep trench **2904**. The deep trench isolation **520** can be in the form of the first isolation structure **116f** of first subset of isolation structures **114**.

At **3510**, the method **3500** comprises forming a near-infrared pass filter overlying the second side of the substrate and the first photodiode, wherein the near-infrared pass filter is configured to allow a first range of wavelengths to pass through and be detected by the first photodiode. For example, in FIG. **34**, the radiation filter layer **214** can be arranged and formed over the ARC layer **212**. The radiation filter layer **214** can include the radiation filter **3402** of the plurality of radiation pass filters **215**. The radiation filter **3402** corresponds, for example, to the NIR pass filter **218** illustrated in FIG. **2**. With reference to FIG. **3**, an NIR range of wavelengths may be from about 750 nm to about 1400 nm or may be from about 780 nm to about 2500 nm. The NIR range of wavelengths may pass through and be detected by the photodiode **500** in the form of the photodiode **202a** of FIG. **2**.

Referring to FIG. **35B**, at **3512**, the method **3500** comprising forming the first isolation structure with a first lateral cross-sectional shape. For example, in FIG. **1E** and FIG. **9**, the isolation structure **518** (e.g., including the first well **519**) can be in the form of the first isolation structure **116f** having the octagon shape.

At **3514**, the method **3500** comprises forming a second photodiode in the first side of the substrate. For example, in FIG. **5A** and FIG. **6**, the photodiode **500** can be in the form of the second photodiode **112d** of FIG. **1J** and formed in the first side **504** of the substrate **200**.

At **3516**, the method **3500** comprises forming a second isolation structure laterally surrounding the second photodiode, wherein the second isolation structure comprises a second well extending to the second depth from the first side in the substrate and has a second lateral cross-sectional shape. For example, in FIG. **5A** and FIG. **9**, the isolation structure **518** (e.g., including the first well **519**) can be in the form of the second isolation structure **124d** of FIG. **1J** and can laterally surround the photodiode **500**. With respect to the first side **504** of the substrate **200**, the first well **519** can have the deep well depth **D1dw**. In FIG. **1J** and FIG. **9**, the isolation structure **518** (e.g., including the first well **519**) can be in the form of the second isolation structure **124d** having the diamond shape.

At **3518**, the method **3500** comprises performing the etch into the second side of the substrate to form a second deep trench, wherein the second deep trench extends, with reference to the first side, from the third depth in the substrate to the second side of the substrate, and wherein at least a portion of the second deep trench underlies second well. For example, in FIG. **29**, the deep trench **2904** can be formed using an etching process into the second side **506** of the substrate **200** opposite to the first side **504** of the substrate **200**. In FIG. **5A** and FIG. **29**, with respect to the first side **504** of the substrate **200**, the deep trench **2904** can extend into the substrate **200** from the deep trench isolation depth **D2dti** to the second side **506** of the substrate **200**.

At **3520**, the method **3500** comprises forming a second deep trench isolation in the second deep trench. For example, in FIG. **31**, the deep trench isolation **520** can be formed over the second barrier layer **3002** and in the deep

trench **2904**. The deep trench isolation **520** can be in the form of the second isolation structure **124d** of the second subset of isolation structures **122**.

At **3522**, the method **3500** comprises forming a radiation pass filter overlying the second side of the substrate and the second photodiode, the radiation pass filter comprising one of a first radiation pass filter, a second radiation pass filter, or a third radiation pass filter. For example, the radiation pass filter can include at least one of an NIR pass filter (e.g., the NIR pass filter **218**) or an optical pass filter (e.g., the red pass filter **220**, the green pass filter **222**, the blue pass filter **224**, or the white pass filter **226**).

FIG. **36** illustrates an implementation **3600** of an exemplary computer-readable medium, according to some embodiments. One or more embodiments involve a computer-readable medium comprising processor-executable instructions configured to implement one or more of the techniques presented herein. An exemplary computer-readable medium is illustrated in FIG. **36** wherein the implementation **3600** comprises a computer-readable medium **3606** (e.g., a CD-R, DVD-R, flash drive, a platter of a hard disk drive, etc.), on which is encoded computer-readable data **3604**. This computer-readable data **3604** in turn comprises a set of processor-executable computer instructions **3602** that when executed are configured to facilitate operations according to one or more of the principles set forth herein. In some embodiments, the processor-executable computer instructions **3602** are configured to facilitate performance of a method **3601**, such as at least some of the aforementioned method(s). In some embodiments, the processor-executable computer instructions **3602** are configured to facilitate implementation of a system, such as at least some of the one or more aforementioned system(s). Many such computer-readable media may be devised by those of ordinary skill in the art that are configured to operate in accordance with the techniques presented herein.

By implementing the semiconductor arrangement **100** set forth above, in some embodiments, the first subset of photodiodes **106** may be resized with respect to the second subset of photodiodes **110**. In an example, when the first subset of photodiodes **106** are optical image sensors and the second subset of photodiodes **110** are NIR image sensors, the optical image sensors may be resized to provide enhanced quality and/or resolution for each pixel of the optical image. In this case, the NIR image sensors may detect an NIR image corresponding to the optical image, and the NIR image may be used to enhance the optical image. In another example, the NIR image may convey more important information than the optical image. The NIR image may detect, for example, that a subject, such as produce, contains a defect, such as bruising or spoilage, that relates to quality of the subject. In this case, the NIR image may provide greater value, and the NIR image sensors may be resized to provide enhanced quality and/or resolution for each pixel of the NIR image. In another example, when the first subset of photodiodes **106** are NIR image sensors and the second subset of photodiodes **110** are optical image sensors, the first subset of photodiodes **106** may be larger than the second subset of photodiodes **110** to provide an enhanced NIR image.

By implementing the semiconductor arrangement **100** set forth above, in some embodiments, the first subset of photodiodes **106** are laterally surrounded by the first subset of isolation structures **114**, such that the dimensions of the first subset of isolation structures **114** define the size of the first subset of photodiodes **106**. Likewise, the second subset of photodiodes **110** are laterally surrounded by the second

subset of isolation structures 122 such that the dimensions of the first subset of isolation structures 114 define the size of the first subset of photodiodes 106. The first subset of isolation structures 114 may be resized with respect to the second subset of isolation structures 122 such that the each photodiode of the first subset of photodiodes 106 has a greater dimension than each photodiodes of the second subset of photodiodes 110 to provide enhanced quality and/or resolution for each pixel of the first subset of photodiodes 106.

By implementing the semiconductor arrangement 100 set forth above, in some embodiments, the first subset of isolation structures 114 may be resized with a predetermined shape, such as an octagon shape, and the second subset of isolation structures 122 may be resized with a predetermined shape, such as a diamond shape. The octagon shape and the diamond shape permit greater precision in resizing of the first subset of isolation structures 114 and the second subset of isolation structures 122 when compared to a rectangular grid array of isolation structures. The first subset of isolation structures 114 may be resized with respect to the second subset of isolation structures 122 such that the each photodiode of the first subset of photodiodes 106 has a greater dimension than each photodiodes of the second subset of photodiodes 110 to provide enhanced quality and/or resolution for each pixel of the first subset of photodiodes 106.

In some embodiments, a semiconductor arrangement is provided. The semiconductor arrangement includes a photodiode extending to a first depth from a first side in a substrate. An isolation structure laterally surrounds the photodiode, wherein the isolation structure includes a first well extending to a second depth from the first side in the substrate greater than the first depth. A deep trench isolation extends from a third depth from the first side in the substrate to a fourth depth from the first side in the substrate, wherein at least a portion of the deep trench isolation underlies the first well.

In some embodiments, a semiconductor arrangement is provided. A photodiode array is provided over a substrate, wherein the photodiode array includes a first photodiode, a second photodiode, and a third photodiode. An isolation structure is disposed between the first photodiode and the second photodiode, wherein the isolation structure surrounds the third photodiode, and the isolation structure includes a first well and a shallow trench isolation at least partially surrounded by the first well. A near-infrared pass filter overlies the second photodiode and is configured to allow a first range of wavelengths to pass through and be detected by the second photodiode.

In some embodiments, a method of making a semiconductor arrangement is provided. The method includes forming a first photodiode with a first depth from a first side in a substrate and forming a first isolation structure laterally surrounding the first photodiode, wherein the first isolation structure comprises a first well extending to a second depth from the first side in the substrate greater than the first depth. The method also includes performing an etch into a second side in the substrate opposite the first side to form a deep trench, wherein the deep trench extends from a third depth from the first side in the substrate less than the second depth to the second side of the substrate, and wherein at least a portion of the deep trench underlies the first well. The method also includes forming a deep trench isolation in the deep trench and forming a near-infrared pass filter overlying the second side of the substrate and the first photodiode,

wherein the near-infrared pass filter is configured to allow a first range of wavelengths to pass through and be detected by the first photodiode.

In some embodiments, the method of making the semiconductor arrangement further includes forming the first isolation structure with a first lateral cross-sectional shape, and forming a second photodiode from the first side of the substrate. The method further includes forming a second isolation structure laterally surrounding the second photodiode, wherein the second isolation structure comprises a second well extending to the second depth from the first side in the substrate and has a second lateral cross-sectional shape. The method further includes forming a radiation pass filter overlying the second side of the substrate and the second photodiode. The radiation pass filter includes one of a first radiation pass filter configured to allow a first radiation pass range of wavelengths to pass through and be detected by the second photodiode, a second radiation pass filter configured to allow a second radiation pass range of wavelengths to pass through and be detected by the second photodiode, or a third radiation pass filter configured to allow a third radiation pass range of wavelengths to pass through and be detected by the second photodiode.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Although the subject matter has been described in language specific to structural features or methodological acts, it is to be understood that the subject matter of the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing at least some of the claims.

Various operations of embodiments are provided herein. The order in which some or all of the operations are described should not be construed to imply that these operations are necessarily order dependent. Alternative ordering will be appreciated having the benefit of this description. Further, it will be understood that not all operations are necessarily present in each embodiment provided herein. Also, it will be understood that not all operations are necessary in some embodiments.

It will be appreciated that layers, features, elements, etc. depicted herein are illustrated with particular dimensions relative to one another, such as structural dimensions or orientations, for example, for purposes of simplicity and ease of understanding and that actual dimensions of the same differ substantially from that illustrated herein, in some embodiments. Additionally, a variety of techniques exist for forming the layers, regions, features, elements, etc. mentioned herein, such as at least one of etching techniques, planarization techniques, implanting techniques, doping techniques, spin-on techniques, sputtering techniques, growth techniques, or deposition techniques such as CVD, for example.

Moreover, "exemplary" is used herein to mean serving as an example, instance, illustration, etc., and not necessarily as

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advantageous. As used in this application, “or” is intended to mean an inclusive “or” rather than an exclusive “or”. In addition, “a” and “an” as used in this application and the appended claims are generally to be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form. Also, at least one of A and B and/or the like generally means A or B or both A and B. Furthermore, to the extent that “includes”, “having”, “has”, “with”, or variants thereof are used, such terms are intended to be inclusive in a manner similar to the term “comprising”. Also, unless specified otherwise, “first,” “second,” or the like are not intended to imply a temporal aspect, a spatial aspect, an ordering, etc. Rather, such terms are merely used as identifiers, names, etc. for features, elements, items, etc. For example, a first element and a second element generally correspond to element A and element B or two different or two identical elements or the same element.

Also, although the disclosure has been shown and described with respect to one or more implementations, equivalent alterations and modifications will occur to others of ordinary skill in the art based upon a reading and understanding of this specification and the annexed drawings. The disclosure comprises all such modifications and alterations and is limited only by the scope of the following claims. In particular regard to the various functions performed by the above described components (e.g., elements, resources, etc.), the terms used to describe such components are intended to correspond, unless otherwise indicated, to any component which performs the specified function of the described component (e.g., that is functionally equivalent), even though not structurally equivalent to the disclosed structure. In addition, while a particular feature of the disclosure may have been disclosed with respect to only one of several implementations, such feature may be combined with one or more other features of the other implementations as may be desired and advantageous for any given or particular application.

What is claimed is:

1. A semiconductor arrangement, comprising:
 - a first photodiode;
 - a second photodiode;
 - an isolation structure; and
 - a near-infrared pass filter overlying the second photodiode, wherein:
 - the isolation structure laterally surrounds the first photodiode to separate the first photodiode from the second photodiode,
 - the isolation structure comprises a first well and a shallow trench isolation at least partially surrounded by the first well,
 - the near-infrared pass filter allows merely wavelengths in a range of 750 nm and 2500 nm to pass through and be detected by the second photodiode,
 - the first photodiode has a first lateral cross-sectional diameter,
 - the second photodiode has a second lateral cross-sectional diameter different than the first lateral cross-sectional diameter,
 - the second photodiode extends to a first depth from a first side in a substrate, and
 - the first well extends to a second depth from the first side in the substrate greater than the first depth.
2. The semiconductor arrangement of claim 1, comprising:
 - a third photodiode, wherein the isolation structure is disposed between the second photodiode and the third photodiode.

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3. The semiconductor arrangement of claim 1, wherein:
 - the first photodiode has a first lateral cross-sectional area, and
 - the second photodiode has a second lateral cross-sectional area less than the first lateral cross-sectional area.
4. The semiconductor arrangement of claim 1, wherein:
 - the first photodiode has a first lateral cross-sectional area, and
 - the second photodiode has a second lateral cross-sectional area greater than the first lateral cross-sectional area.
5. The semiconductor arrangement of claim 1, comprising:
 - a radiation pass filter overlying the first photodiode and configured to allow a first radiation pass range of wavelengths to pass through and be detected by the first photodiode, the first radiation pass range of wavelengths corresponding to wavelengths within at least a portion of a visible light spectrum.
6. The semiconductor arrangement of claim 1, wherein the semiconductor arrangement comprises a deep trench isolation extending from a second side in the substrate, wherein at least a portion of the deep trench isolation overlaps with the first well.
7. The semiconductor arrangement of claim 6, wherein at least a second portion of the deep trench isolation underlies the first well.
8. The semiconductor arrangement of claim 1, comprising:
 - a deep trench isolation, wherein at least a portion of the deep trench isolation underlies the first well.
9. The semiconductor arrangement of claim 1, comprising:
 - a deep trench isolation extending from a third depth from the first side in the substrate to a fourth depth from the first side in the substrate, wherein at least a portion of the deep trench isolation underlies the first well.
10. The semiconductor arrangement of claim 9, wherein the third depth is less than the second depth and the first well laterally surrounds the deep trench isolation.
11. A semiconductor arrangement, comprising:
 - a first photodiode;
 - a near-infrared pass filter overlying the first photodiode, wherein the near-infrared pass filter allows merely wavelengths in a range of 750 nm and 2500 nm to pass through and be detected by the first photodiode;
 - an isolation structure laterally surrounding the first photodiode and comprising a first well, a shallow trench isolation, and a second well; and
 - a deep trench isolation, wherein:
 - a width of the second well is greater than a width of the first well,
 - at least a first portion of the deep trench isolation underlies the first well and is surrounded by the second well,
 - the deep trench isolation is separated from the shallow trench isolation by the first well,
 - the first well laterally surrounds at least a second portion of the deep trench isolation,
 - the first photodiode extends to a first depth from a first side in a substrate, and
 - the first well extends to a second depth from the first side in the substrate greater than the first depth.
12. The semiconductor arrangement according to claim 11, comprising:
 - a second photodiode, wherein the isolation structure and the deep trench isolation are between the first photodiode and the second photodiode.

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13. The semiconductor arrangement according to claim 11, wherein the deep trench isolation extends from a third depth from the first side in the substrate to a fourth depth from the first side in the substrate.

14. The semiconductor arrangement of claim 13, wherein the third depth is less than the second depth.

15. A semiconductor arrangement, comprising:

a first photodiode;

a near-infrared pass filter overlying the first photodiode, wherein the near-infrared pass filter allows merely wavelengths in a range of 750 nm and 2500 nm to pass through and be detected by the first photodiode;

an isolation structure laterally surrounding the first photodiode and comprising a shallow trench isolation, a first well, and a second well, wherein:

a first portion of the second well overlaps a first portion of the first well,

a plane parallel to a top surface of the first photodiode intersects the shallow trench isolation and the first photodiode,

the first photodiode extends to a first depth from a first side in a substrate, and

the first well extends to a second depth from the first side in the substrate greater than the first depth; and

a deep trench isolation, wherein a first portion of the deep trench isolation overlaps the first portion of the first well such that a sidewall and a top surface of the deep trench isolation contact the first well and overlap the

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first portion of the second well such that the first portion of the first well, the first portion of the second well, and the first portion of the deep trench isolation are laterally coincident.

16. The semiconductor arrangement of claim 15, wherein the first photodiode comprises:

a charge storage well; and

a deep storage well section below the charge storage well, wherein a width of the deep storage well section is greater than a width of the charge storage well where the charge storage well interfaces with the deep storage well section.

17. The semiconductor arrangement of claim 15, comprising:

a transfer gate between the first photodiode and the isolation structure.

18. The semiconductor arrangement of claim 15, wherein the isolation structure comprises a dielectric layer between the shallow trench isolation and the first well.

19. The semiconductor arrangement of claim 15, wherein the sidewall of the deep trench isolation contacts second well.

20. The semiconductor arrangement of claim 15, wherein a width of the second well is greater than a width of the first well such that a first sidewall and a second sidewall of the first portion of the first well contact the first portion of the second well.

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